Miscategorization as an Information Friction? Activating School-Based Agricultural Education in Liberia with Parental Engagement

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Abstract

Many African countries have vastly under-staffed agricultural extension systems, while rural households face stark tradeoffs between children's schooling and farm labor. This paper studies a school-based agricultural extension (SBAE) program in Liberia that integrates (i) experiential science instruction and student-centered extra-curricular activities; (ii) school farms as demonstration plots; and (iii) student home farming projects to tackle the dual challenges of diffusing agricultural technologies and motivating school attendance. I randomly assign SBAE across 197 junior high schools to estimate its intent-to-treat effects on voluntary participants. To tackle weak parent-school linkages, in a random subset of SBAE schools, I engage parents through a promotional video highlighting the complementarities between agriculture and education under SBAE, an annual farmer field day, or both. By the third year, SBAE successfully trains teachers and students but fails to improve technology adoption on any household parcels—and increases student absenteeism by 31%. Conversely, SBAE augmented with both types of engagement increases adopted techniques by 26% on parents' parcels and 53% on students' parcels. It also reduces student dropouts by 4-5 percentage points and absenteeism by 18%-28% over consecutive years, and increases the proportion of students saving for university by 6 percentage points. Parental engagement is pivotal because it activates muted mechanisms—(i) parents increase (otherwise reduced) school visits, learn techniques, and observe harvests; (ii) students manage household farms, use techniques, show parents improved yields, and increase (otherwise reduced) studying hours. Consistent with parents' confusion of SBAE with school gardens, the video clarifying SBAE's student-centered training changes parents' beliefs and plays a dominant role in diffusing technologies through mechanism (ii) and reducing dropouts. The augmented program is more cost-effective than most extension programs in Africa and over twice as cost-effective as conditional cash transfers in raising attendance. The findings underscore the promise of positioning schools as rural development hubs that break the cycle of stagnant agricultural productivity and low levels of human capital, and exemplify how strengthening parent-school linkages reveals school programs' hidden potential.

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1. Introduction

The majority of sub-Saharan Africa's population depends on agriculture for their livelihoods, yet the region has experienced persistently low productivity growth since the 1960s (The World Bank, 2008), which is a stark contrast to the widespread diffusion of Green Revolution technologies and subsequent productivity growth in Asia and Latin America (Gollin et al., 2021). This stagnation is particularly puzzling given decades of international development efforts targeting African agriculture.[1] The persistence of low productivity reflects a vicious cycle of mutually reinforcing constraints that trap rural households in poverty. Increasing economic and population pressures degrade soil quality and lower agricultural yields (Jayne et al., 2014; Sheahan & Barrett, 2017), while severely understaffed agricultural extension systems—with ratios often approaching 1 extension officer to 5000 farm households (Anderson & Feder, 2007; Moore, 2017)—limit the diffusion of improved techniques that could help households manage soils more effectively. Low agricultural productivity in turn forces binding constraints on rural household livelihoods, leading parents to employ children in farm labor and reduce their school attendance (Beegle et al., 2006; Edmonds, 2005), while dampening aspirations for children to pursue higher education and non-agricultural careers (Bernard et al., 2014; Macours & Vakis, 2014). The resulting lack of human capital undermines both the capacity of farming households to experiment with productivity-enhancing technologies (Foster & Rosenzweig, 1996; Wossen et al., 2017) and the supply of educated workers needed to staff effective extension systems and other rural institutions (Juma, 2011; Bold et al., 2017), thereby closing the cycle and perpetuating agricultural stagnation.

Breaking this cycle requires interventions that address both agricultural productivity and human capital constraints simultaneously. This paper studies an approach that repositions schools from passive educational institutions into active catalysts for agricultural development. By using schools as platforms for technology diffusion, the intervention reaches farming households through their children while strengthening educational engagement, thereby tackling the dual challenges of diffusing agricultural technologies and increasing school attendance. Specifically, I evaluate a school-based agricultural extension (SBAE) program that integrates innovations in both education and agricultural extension. The education component trains teachers in experiential science instruction and engages students in extracurricular leadership and agribusiness training. The extension component creates two complementary mechanisms for technology diffusion. First, school farms serve as centralized demonstration plots for parents and nearby farmers. Second, students implement home projects that function as decentralized demonstration plots on their family farms. This dual approach aligns with established findings on technology adoption: farmers prefer learning from socioeconomic peers (BenYishay & Mobarak, 2019), observing technologies under familiar conditions (Munshi, 2004), and gathering multiple observations before adopting (Beaman et al., 2021). These mechanisms are especially valuable in sub-Saharan Africa, where agricultural technique performance varies greatly across small distances (Suri & Udry, 2022), limiting the effectiveness of centralized extension.

Despite its promise, implementing SBAE in the fragile education systems typical of developing countries raises important concerns. First, high teacher turnover and student dropout rates at the targeted upper primary and junior high levels dissipate training investments as trained individuals leave the program. Second, teachers may resist adopting experiential instruction methods and monitoring student activities,

[1] Examples include the establishment of international agricultural research centers in the 1970s (Evenson & Gollin, 2003), more recent initiatives such as the Alliance for a Green Revolution in Africa (Denning et al., 2009), and substantial increases in aid flows to the sector (IFAD and UNEP, 2013).

viewing these as diversions from standard curricula and exam preparation. Agriculture and science teachers in rural areas possess the technical knowledge to function as extension agents, but may lack incentives to invest the additional effort required. Third, students may be reluctant to participate if school farms are poorly maintained and offer limited tangible benefits. These implementation challenges could make SBAE less viable than traditional village-based extension, which reaches households through dedicated agents rather than relying on the education system.

Moreover, even successful teacher and student training may fail to translate into changes in household agricultural practices and educational investments if parent-school linkages are weak. A large literature documents that parents in developing countries hold inaccurate beliefs about what and how much their children learn in school.[2] These information gaps persist because parent-school engagement is typically limited: schools rarely take initiative to involve parents in school activities, parents with low educational aspirations see little value in visiting schools, and poor road infrastructure makes travel to schools costly in time and effort. As a result, SBAE's dual diffusion mechanisms—school farms as demonstration plots and student home farming projects—could fail, as both require parental awareness of the promoted farming techniques and SBAE's student-centered training.

Weak parent-school linkages pose particular risks for students' education. Critical for rural households' decisions in choosing students' farm work and schooling, it is important to recognize that SBAE provides both opportunities for (i) improving households' production through learning and adopting promoted techniques; and (ii) student-centered training on agribusiness, improved education, and a national network of mentors. For this reason, SBAE could impose negative effects on students' education if parents are unaware of (ii). In fact, focus group discussions reveal that many parents confuse SBAE with typical school gardens that use students as assistants. Without understanding SBAE's educational objectives, parents may view students primarily as more skilled and obedient farm assistants, intensifying child labor demands and reducing school attendance while maintaining low educational aspirations. As I show in a companion paper (JMP2, Lee, 2025), weak parent-school linkages can amplify students' uncertainty about parents' preferences and beliefs, preventing even motivated students from acting: students hesitate to propose changes in farm management as they are uncertain about whether parents are open to such changes in resource control; and, as a result, fail to make timely earnings that could help them finance school fees.

This paper evaluates whether school-based agricultural extension (SBAE) can overcome these barriers and provides proof of concept that developing countries can create a cost-effective SBAE system that both diffuses yield-improving technologies to students' households and encourages school attendance. I partner with non-governmental organizations and government ministries to randomly assign an SBAE program across 197 elementary and junior high schools in rural Liberia—a country where only 3.4% of farm households receive extension services (Liberia Institute of Statistics and Geo-Information Services (LISGIS), 2017) and 24% of students in targeted grades drop out within two school years.

Adapted from the century-old 4-H model that brought practical agricultural education to rural America, the program designs training to be both relevant and empowering, encouraging active student participation. First, unlike typical school gardens in sub-Saharan Africa where student labor is compulsory, participation is voluntary. Rather than following teacher directives, students make major decisions on school

[2] For example, Dizon-Ross (2019) shows that parents in Malawi hold wildly inaccurate beliefs about their children's academic performance even when report cards are available, with larger inaccuracies among higher socioeconomic status households.

farms—including what crops to grow and how to use farm proceeds. Second, the program enables students to earn income by pairing yield-improving technologies with agribusiness training and supporting students to manage their own farms under teacher guidance.

The program is also designed to strengthen school attendance and educational aspirations. It integrates agricultural training with core academic subjects, training teachers to demonstrate connections between agriculture, science, and mathematics through experiential learning on school farms. This approach is particularly valuable in resource-constrained environments where school farms function as open-air laboratories that partially address the lack of textbooks. The program also builds students' sense of belonging through a national network that organizes annual agriculture fairs where students compete, showcase their agricultural knowledge and agribusiness success, and meet role models for future careers.

These design features address student and teacher engagement, but parental engagement remains a critical bottleneck. Only 36% of program school parents attend parent-teacher meetings in any given school year. Over the three-year trial period, only 44% of parents report visiting school farms. To address this bottleneck, I implement parental engagement interventions in a random subset of SBAE schools. Specifically, I randomly assign SBAE schools to receive: (i) a one-time promotional video session in which a parent of each sampled student watches a video explaining how agricultural and education components interact under SBAE, with testominials from past participants and authority figures highlighting SBAE's transformative effects on students' abilities to manage their own farms, livelihoods, and aspirations; (ii) an annual farmer field day inviting parents to observe, discuss, and experiment with promoted technologies on school farms; (iii) both interventions; or (iv) no parental engagement. School-level randomization enables clean identification of SBAE's effects with and without parental engagement, accounting for potential spillovers within communities. Crucially, these interventions are light-touch and cost-effective: together, they account for only 10% of average per-school implementation costs, or 30% if the video is extended to all cohorts of students in the three-year trial period.

Mirroring the layered randomization design, the program evaluation is separated into two stages: (i) a first follow-up survey of students in the first rainy season and before parental engagement treatments were implemented; and (ii) midline surveys of students and endline surveys of students and parents, which were conducted in the second and the third rainy seasons. I track a total of 3757 households, sampled from a pool of students with farming experiences, which is highly representative of students in sampled schools, surveying one parent per student. I then estimate the intent-to-treat effects of the program.

The first follow-up survey confirms that the program has strong training effects on students early on — students increase usage of and knowledge about promoted techniques, and improve their beliefs about the number of cassava tubers that they could harvest from a plant, consistent with agronomic trials. Furthermore, the program increases the fraction of students who report an intention to look after farms by 9 percentage points. Meanwhile, there are early signs of pitfalls — students spend 5% less hours per week studying.

Results in the subsequent midline and endline surveys suggest that the program completely fails to achieve its goals in the absence of parental engagement. By the third rainy season (endline), the program fails to improve adoption of the main yield-improving farming technique on both students' and parents' parcels. Worse, the program, on its own, increases student absenteeism, measured by the number of days missed out of the most recent five, by 31% and reduces weekly studying hours by 20%.

By contrast, parental engagement interventions transform program outcomes. When SBAE is augmented

with both the promotional video and farmer field days, technology adoption increases substantially: the main yield-improving technique is adopted on 18 percentage points more students' parcels and 11 percentage points more parents' parcels (from baseline rates of 29% and 74%, respectively). Considering the full package of soil management techniques—which have synergistic effects on yields—adoption on parents' parcels and students' parcels increases by 26% (0.38 standard deviations) and 53% (0.48 standard deviations), which are large effects relative to traditional village-based extension programs in sub-Saharan Africa that routinely fail to increase adoption among farmers who are not directly trained (Suri & Udry, 2022). The augmented program also dramatically improves educational outcomes over two consecutive school years: student dropouts decrease by 4-5 percentage points (from 22% at midline and 33% at endline), absenteeism falls by 18% and 28%, and weekly studying hours increase by 7% and 22%.

Beyond agricultural technology adoption and school attendance, the augmented SBAE program shows promising effects on students' economic well-being and educational aspirations without generating adverse side effects. While students' spending and food consumption remain unchanged, the program increases students' savings: 10 percentage points more students report positive savings, though the amount of annual savings among students who save remain unchanged. Estimates from Poisson regressions indicate that average annual savings rise by 21% from approximately $20. These gains are consistent with increased farm sales and parental transfers. More importantly, students show signs of aspiring for more education under SBAE: the program increases the share of students taking agriculture as an elective by 18 percentage points and naming agricultural careers among their top three choices by 12 percentage points; while also increasing the fraction of students saving for university by 6 percentage points. Students reallocate time away from working as farm assistants while maintaining total hours in farming—contrary to concerns that SBAE would increase child labor. Future surveys will track longer-term outcomes including crop yields on both students' and parents' parcels, as well as spillovers of farming practices to non-parent farmers.

So what drives the pivotal effects of parental engagement? While we have confirmed that the program itself has strong training effects on students in the first follow-up survey, it could still be the case that parental engagement substantially increases students' responses to training. This is, however, not true. First, parental engagement has no effects on students' participation in the program by the endline, which already reaches 81% in the absence of engagement. Second, by the third year, the program on its own has large positive effects on students' agricultural knowledge and beliefs about attainable crop yields, whereas adding parental engagement has close to zero and statistically insignificant impact on these outcomes. For improved responses to training to explain the dramatic effects of parental engagement — large differentials on technology adoption and reversed effects on education — we would expect positive marginal effects of parental engagement on key intermediate outcomes and effects that are different from pure program effects.

Further inspection suggests that parental engagement enables SBAE's dual diffusion mechanisms, which are otherwise completely muted. First, parents increase visits to school farms and attendance at parent-teacher meetings. In the absence of parental engagement, despite 81% of students in program schools have joined the program by the endline, only 44% of parents report visits to school farms. Strikingly, program school parents are not more likely to visit school farms in the second rainy season than control school parents. With parental engagement, 67% of parents visit school farms. Parents double their visits to school farms in the second rainy season. Relative to counterparts in schools with SBAE only, parents report learning five times more practices from schools and 23 percentage point more parents report having observed harvest on school farms. Parents' increased exposure translates into both better knowledge (0.2 standard deviations) and

beliefs about promoted techniques — there is a 10 percentage point increase in parents who believe students can produce more cassava tubers per plant than typical adults in the community. In line with reversals on students' school attendance, adding parental engagement also overturns SBAE effects on parent-teacher meeting attendance from a 39% decrease to a 76% increase.

Second, students manage their own farms as well as use and show improved yields from promoted techniques to parents. In the absence of parental engagement, the program does not increase the fraction of students who are involved in the choice of farming methods on students' parcels, which remains at 23%. This reflects the absence of student home projects and mirrors the lack of adoption of promoted techniques. With parental engagement, this fraction increases by 15.4 percentage points. This change aligns with the large increases in students' technology adoption. Correspondingly, the number of promoted techniques that students have shown their parents how to make increases by a dramatic 94%. Most importantly, there is a 18 percentage-point increase in parents who report having observed improved yields on students' parcels.

Contrasting the marginal effects of the farmer field day and the promotional video sheds further light on what drives households' technology adoption and reversal in educational decisions. Both interventions provide information to parents, but of different types. The farmer field day offers technical training on promoted techniques, which parents can employ while retaining students as assistants on farms. By contrast, the video highlights complementarities between students' agricultural work and education under SBAE, emphasizing students' ability to manage their own farms and pursue further education that strengthens their skills as farmers. Consistent with these differences, the video plays a dominant role in activating the second diffusion mechanism of SBAE: only the video, not the farmer field day, increases the fraction of students who look after farms at midline and endline. Students are also three times more likely to show parents how to use promoted techniques, and ten times more likely to advise parents to use them, when parents receive the video rather than the farmer field day. In turn, the fraction of parents who report observing improved cassava yields on students' parcels is three times higher with the video than with the farmer field day. In line with the importance of activating both diffusion from school farms and student home projects (BenYishay & Mobarak, 2019; Munshi, 2004; Beaman et al., 2021; Suri & Udry, 2022), both treatments increase technology adoption on students' parcels, but only the video produces a strong enough effect to increase adoption on parents' parcels, which tend to generate much more farm sales for households. Similarly, consistent with the video changing parents' aspirations for their children's education, the video accounts for the entire reduction in student dropouts.[3] Consistent with the video tackling a key information friction, parents who watch the video anticipate growth in students' farm management abilities, and their beliefs about students' cassava yields remain significantly more optimistic at midline and endline than those of parents in schools without video sessions. Moreover, parents' beliefs measured immediately after watching videos strongly predict their subsequent visits to schools, ability to name field officers, and attendance at parent-teacher meetings.

When the video and the farmer field day are included, the SBAE program is highly effective in achieving multiple goals: increasing adoption of yield-improving technologies in rural households, school attendance, and students' savings. But at $40 per student——the total cost per beneficiary across three cohorts, covering field officer and teacher training, school farm setup, school monitoring, and national events for extracurricular clubs—do the benefits justify the investment? I conduct a back-of-the-envelope calculation with two sets of

[3] Disentangling the effects of different versions of the promotional video and subsequent engagement with students, Lee (2025) (JMP2) provides additional evidence that an interaction between parents' acceptance and students' requests for timely harvests of root crops—helping finance school fees—is crucial in explaining the video's effects on reduced dropouts under SBAE.

conservative assumptions. First, in calculating the increase in parents' net production value due to technology adoption, I consider only increases in the number of parents who adopt the main yield-improving technique but not other complementary techniques; parents achieve only half of the yield gains in agronomic trials; adoption takes place only in 1/4 of the cultivated area and only in one season. Second, in calculating the increase in students' discounted lifetime earnings, I assume a 8/200% return on each day of schooling, which implies a 8% annual return that reflects findings on secondary school returns in sub-Saharan Africa.

There are three takeaways from this exercise. First, benefits outweigh costs at a ratio of 1.33 in the three-year trial period—this ratio is close to one even if parents' benefits from technology adoption are dropped. Second, despite its broad scope, the augmented SBAE program outperforms major alternatives in cost-effectiveness when comparing impact per $100 spent with other randomized trials in Africa. For technology adoption, it induces more households to adopt new techniques than all other rigorously evaluated village-based extension programs, with the exception of BenYishay & Mobarak (2019), which provides financial incentives for trained farmers to promote new technologies to other farmers; relative to this study, the augmented SBAE program performs better against one of their techniques but worse against the other. For school attendance, the program is more than twice as cost-effective as conditional cash transfers in Malawi (Baird et al., 2011) in increasing total days students remain enrolled and attend school. Third, these favorable economic returns extend well beyond the trial period: once capacity is built at the school level, dramatically slashed monitoring costs after the first three years enable the augmented program to achieve a benefit-cost ratio of 1.53 over the first ten years even if annual benefits halve after the first three years. This robustness strongly justifies long-term policy commitment to capitalize on the institutional capacity and teacher training already established, which can deliver sustained returns at a fraction of initial costs.

Related Literature. This paper provides proof of concept that schools in developing countries—particularly low-income African nations—can serve as cost-effective hubs for rural development by (i) diffusing improved agricultural technologies to students' households while (ii) increasing school attendance. In doing so, it deviates from a large literature proposing ways to strengthen existing agricultural extension systems that target adults as contact farmers, including providing incentives for contact farmers to communicate with peers (BenYishay & Mobarak, 2019), selecting contact farmers using social network data (Beaman et al., 2021), leveraging mobile phone advice to reach farmers directly (Aker, 2011; Fafchamps & Minten, 2012; Cole & Fernando, 2021; Fabregas et al., 2025), and targeting private input suppliers to disseminate information about new technologies (Dar et al., 2024). Most closely related are two papers examining the transmission of school-taught agricultural knowledge—through computer classes—from students to parents, using class-level randomization in Peru (Nakasone & Torero, 2016) and Uganda (Fernández et al., 2024). The Ugandan study, more relevant to the sub-Saharan African context, finds substantial gains in parents' agricultural knowledge but only modest increases in adoption of promoted techniques. Despite this promising evidence of student-to-parent knowledge transmission, important questions remain about the cost-effectiveness of promoting agricultural technology adoption in fragile education systems characterized by high teacher turnover and student dropout. This study addresses these gaps in four key respects. First, unlike the other two studies that rely solely on computer-based instruction, SBAE in the current study employs multiple channels for technology diffusion—including experiential science instruction, extracurricular activities, school demonstration farms, and student home projects. This multi-channel approach better addresses the barriers to social learning in African contexts, where vast heterogeneity in technology appropriateness poses unique challenges. Second, randomization occurs at the cluster-of-school

rather than class level, with intent-to-treat effects accounting for voluntary student participation and potential dropout among both students and teachers. Third, we demonstrate that this integrative model is nore cost-effective at increasing technology adoption than traditional extension approaches. Fourth, we show that school-based agricultural extension can be highly effective in regions where computer and internet infrastructure is absent—which represent the majority of rural areas in Africa.

This paper also contributes a new perspective to a large literature on the interconnections between students' agricultural work and education. Agricultural households in developing countries face a well-documented tradeoff between children's immediate value in production and long-term human capital investments.[4] At the same time, despite the fact that most students in rural Africa are projected to find employment in agriculture (Filmer & Fox, 2014), many are increasingly reluctant to pursue agricultural livelihoods due to perceived low returns and the view of farming as low-status work (Yeboah, 2018; Blimpo & Pugatch, 2021). This context leads to skepticism about school-based agricultural training programs for two reasons. First, such programs might intensify child labor and student absenteeism if they lead parents to view students as more obedient and effective farm assistants. Second, student participation in these programs might be low, and students might not take the training seriously if they aspire to non-agricultural careers. Against this backdrop, this paper documents a school-based agricultural training program that simultaneously (i) trains students effectively and increases their aspirations for agricultural careers; (ii) improves students' school attendance via reduced dropouts and absenteeism, as well as aspirations for progressing to universities; and (iii) increases students' experiences in managing agribusinesses without changing their overall working hours. More specifically, I find evidence that informational interventions persuading parents that students' agricultural work and education are compatible are crucial for (i), (ii), and (iii) to simultaneously happen.

Beyond immediate impacts, school-based agricultural extension can serve as a long-term development strategy. While the current RCT focuses on establishing proof-of-concept for technology adoption and attendance effects, the SBAE model's core components—student-centered training, agribusiness skills, and monthly student data collection—create opportunities for three synergistic interventions. These include: (i) providing experiential, student-centered pedagogy training for teachers who lack exposure to methods beyond traditional rote memorization (Nourani et al., 2025); (ii) enhancing agribusiness training by connecting school farm production to home-grown school feeding initiatives; and (iii) using high-frequency student outcome data to evaluate and refine which agricultural practices are most effective for dissemination. Together, these interventions transform SBAE into an enhanced platform that diffuses appropriate agricultural technologies while equipping students with generalizable skills beyond agriculture. This dual focus offers low-income countries a development strategy that simultaneously promotes rural livelihoods and prepares youth for structural transformation towards high-value jobs and sectors. I elaborate on these synergies, partnerships, and broader implications in the conclusion.

Lastly, this paper exemplifies how the hidden potential of school programs can be revealed through joint evaluations with household interventions. A large literature shows substantial room for improving educational outcomes through informational interventions that correct parental misperceptions, including beliefs about returns to education (Jensen, 2010), students' academic performance (Dizon-Ross, 2019), and study effort (Bergman, 2021). While these papers hint at weak parent-school linkages in developing countries, there have been no attempts to exploit this insight to enhance school program effectiveness.

[4] For example, households trade off children's farm labor with school attendance in response to enrollment subsidies (Ravallion & Wodon, 2000) and conditional cash transfers (De Janvry et al., 2006; Edmonds & Schady, 2012).

This paper, along with complementary findings in JMP2 (Lee, 2025), demonstrates that strengthening parent-school linkages delivers exceptional value for both agricultural technology diffusion and rural education through SBAE. Remarkably, while parental engagement proves pivotal to program success, it costs only less than 30% of SBAE's total budget. Furthermore, the strong evidence that parental engagement activates SBAE's mechanisms suggests that evaluations of school programs should routinely test parental engagement components to avoid underestimating program impacts. This insight has broader implications for program evaluation, as demonstrated success is typically necessary to generate buy-in for subsequent trials and scaling efforts, potentially biasing meta-analyses toward programs evaluated without key complementary components.

Outline. The rest of this paper is structured as follows. Section 2 provides the study background and introduces the school-based agricultural extension program in Liberia. Section 3 explains the experimental design. Section 4 discusses the main empirical results. Section 5 discusses mechanisms that help interpret the results. Section 6 provides cost-benefit and cost-effectiveness analyses of the SBAE program when the promotional video and the farmer field day are included. Section 7 concludes.

2. Study background

2.1. Agriculture and education in Liberia

This study is conducted in Liberia where 72% of households live in absolute poverty (Benfica et al., 2018). Agriculture is the primary source of livelihood for about 80% of the population (African Development Bank, 2008). Smallholder farmers in this tropical region traditionally employ a fallow system under collective ownership of the land.[5] This system operates smoothly with minimal inputs to the soil — no tillage nor use of fertilizers — if the fallow period is sufficiently long. However, as in many other sub-Saharan African countries, farmers are forced to shorten the fallow period under economic and population pressures, leading to commonly observed soil degradation and declining agricultural output.

Agricultural intensification — increasing output per unit of land by increasing inputs to the soil — is the common response to soil degradation. Managing the shift to agricultural intensification, however, requires farmers to acquire soil management techniques that are fundamentally different from traditional techniques. In Liberia, this progress is hindered by low levels of coverage by extension services (3.4%) and low levels of education — the average household head has four years of schooling. As a result, Liberia has very low levels of yields in major staples even by the standard of sub-Saharan Africa — FAO (2023) data suggests that rice and cassava yields are only 54% and 62% of the regional average.

Besides low agricultural productivity, Liberia also has an education system with lower school attendance than those in most sub-Saharan African nations. Liberia has a net primary enrollment rate of only 44% in 2017, compared to the regional average of about 80% in sub-Saharan Africa (UNESCO Institute for Statistics, 2020). Overage enrollment is particularly serious — over 40% of primary school students are above 12 years old; and about 50% of junior high students are 18-20 years old (Liberian Ministry of Education, 2017).

[5] I use the term collective ownership here to refer to communal land tenure or lineage-based land tenure.

Another reason is the high annual rates of student dropouts, which stand at 10-15% in upper primary and junior high grades.

Other than socioeconomic factors, such as poverty and needs for farm labor, a lack of quality instruction and educational resources are important reasons discouraging school attendance. Only 51% of all teachers are qualified to teach — this drops to 33% in junior high. Lacking the required certificates, unqualified teachers also do not receive in-service teacher training from the government. A lack of essential educational resources, such as textbooks, further worsen teacher instruction. Many students are forced to spend a significant portion of class time on copying from the blackboard. These factors are particularly discouraging for rural students who anticipate an agricultural career, as the way knowledge is taught hardly feels relevant to them.

2.2. School-based agricultural extension as an integrative model

School-based agricultural extension (SBAE) has a century-old history in rural America, dating back to the Smith-Hughes Act of 1917. At its core is an integrative model that blends the promotion of modern farming practices with student-centered learning experiences. According to Croom (2008), this model has three interrelated components. Through formal instruction, students learn the scientific basis of modern farming practices through lectures and demonstration. But more important are the other two components that ensure personalized learning experiences for rural students. Through participation in extra-curricular agricultural clubs, students develop their sense of belonging in a national network, meet mentors and role models, and gain leadership experiences and recognition in public services and competitions. Through supervised agricultural experiences, students can develop home projects tailored to their own pace and needs. Rufus Stimson, an early advocate of the method, argues in particular that home projects allow students to profit from farm work and complement school projects that are subject to overcrowding amongst students (Stimson, 1919).

In the sub-Saharan African context, SBAE is an important new model in the delivery of agricultural extension and rural education.[6] In particular, SBAE should be distinguished from typical school gardens due to its emphasis on student-centered learning experiences. Typical school gardens are directed by teachers without integration with scientific instruction of modern farming practices, home projects, and a national network of agricultural clubs. By contrast, students participate in SBAE voluntarily, make major decisions on school farms (e.g. choice of crops and use of proceeds), and are empowered to make profits from home projects and gain social recognition from clubs. These distinguishing features of SBAE make it particularly suited to tackle intertwined challenges in agricultural extension and school attendance.

2.3. Components of the 4-H Liberia program

This study examines an SBAE program in Liberia that is conducted by the local NGO 4-H Liberia in collaboration with the ministries of agriculture and education. Focusing on the RCT period from January 2021 to December 2023, this program implements the three-component model mainly through in-service teacher training and regular extension visits by extension officers.

[6] As in the case of agricultural education in early US history, separate provisions of agricultural training and agricultural youth clubs can be found in some sub-Saharan African countries — for example, agricultural youth clubs affiliated with the international 4-H network can be found in 15 African countries. But the integrative SBAE model has only been introduced by the US-based NGO AgriCorps to Liberia and Ghana since the 2010s.

Teacher training is offered twice a year to two teachers per school. These teachers are selected by the participating schools with the requirement that they teach either science or agriculture. The duration of each training workshop is 20 hours that are spread across several days, with roughly equal time covering (i) the integrative SBAE model; and (ii) the promoted yield-improving agricultural technologies. To ensure proper implementation of the SBAE model and diffusion of technologies, teacher training is reinforced by monthly extension visits. During these visits, extension officers monitor progress on school farms, collect information from student participants, and provide lessons on agriculture.

The 4-H Liberia program implements the three-component SBAE model in the following ways. First, formal instruction of agriculture and science is offered through a combination of extension visits and experiential instruction by teachers. Students attend lessons provided by extension officers, while teachers are given instructional ideas that they can adopt during classes. In particular, teachers are given structured lesson plans to illustrate scientific concepts that are associated with promoted farming practices, such as (i) using seed germination tests to illustrate how to conduct a scientific experiment; (ii) using farm site selection to illustrate what a soil profile is; and (iii) using the practice of composting to illustrate the concept of a chemical compound. Besides work on the school farm, teachers are also encouraged to use other experiential methods such as field trips, role plays, and dramas to pique students' interest in scientific knowledge.

Then, extra-curricular 4-H Clubs provide a series of learning experiences. In each school year, every participating school recruits 25-30 club members. These members then participate in, on average, two hours of farm work per week after school. They also meet, on average, on a bi-weekly basis to discuss club affairs. The most important issues are about the choice of crops in a growing season, ways to guard the school farm against thieves and pests, and ways to use the proceeds from farm. By enrolling in the club, students also earn the opportunity to represent their schools in annual national events. The most important event is the national agriculture fair, during which students compete in displaying their knowledge of agriculture to judges and share stories of successes in applying modern farming practices.

Lastly, the program aims to amplify the diffusion of agricultural technologies to rural communities through (i) using the school farm as a demonstration plot; and (ii) encouraging students to start home projects. The school farm is instructed to be open to visitors, with posters encouraging learning of promoted techniques by parents and farmers. To enable students to manage their own farms and make profits from the adoption of promoted techniques, extension officers provide lessons on agribusiness, including farm management and record keeping practices. Teachers are also encouraged to supervise students through home and farm visits.

2.4. Promoted agricultural technologies

Agriculture is an inherently complex activity that requires flexible practices with respect to local environments. The 4-H Liberia program thus provides a wide range of techniques that extension officers and teachers can choose to implement on the school farm. These techniques are grouped under the categories "soil management" (e.g. bed preparation), "crop protection" (e.g. weeding), "improved input use" (e.g. improved seeds or crop varieties), "improved planting practices" (e.g. transplanting), "improved nursery practices", "harvesting practices", and "record keeping practices".

Under the advice of the Liberian Central Agricultural Research Institute, we focus on four soil management techniques that are most widely adopted and expected to improve crop yields, based on agronomic trials

and findings from agricultural development programs in Liberia. The four specific techniques are (i) bed preparation methods; (ii) composting; (iii) regular spacing; and (iv) row planting.

These four techniques are promoted to improve agricultural productivity amidst soil degradation in a bush fallow system. Farmers employing the system typically clear land by cutting and burning vegetation. Because of the fertile soils created, farmers do not need extra tillage and instead sow seeds directly into the ground. To reduce the risk of single crop failures and pests, farmers often plant several crops, such as rice and cassava, in proximity. While these practices quickly drain soil nutrients, they are sustainable with traditional fallow periods, which could be as long as 10-15 years. With reduced fallow periods, however, farmers are increasingly unable to cope with agronomic problems such as soil erosion, soil compaction, and waterlogging.

Bed preparation methods, which include mounds or ridges for root crops and raised beds for vegetables, are the main yield-improving technique in this context. These methods help loosen the soil and enhance drainage, thus facilitating root penetration while avoiding waterlogging and topsoil erosion. Other promoted techniques tend to reinforce the effects of beds — with larger yield gains when adopted together. Composting provides a low-cost alternative to chemical fertilizers for improving soil fertility. Row planting and regular crop spacing help optimize the growth of individual plants by avoiding inter-plant competition for nutrients.

Agronomic trials in the tropical humid region in general recommend the promotion of these soil management techniques. In particular, several agronomic trials by the International Institute of Tropical Agriculture (IITA) suggest that the use of mounds and ridges can significantly increase root crop yields — cassava by 31-46% (Odjugo, 2008) and sweet potatoes by 70% (Agbede & Adekiya, 2009). These trials have two important features. First, the yield gains are found under conditions that mimic the traditional cultivation style with minimal soil inputs and random spacing. Second, consistent with anecdotes from program participants, the differences are easily visible, as indicated in the stem height and leaf size within 3 months of planting, and the number and size of tubers per plant at harvest.

These techniques are widely applicable for root crops and vegetables, which are grown by over 80% and 60% of households respectively (Liberia Institute of Statistics and Geo-Information Services (LISGIS), 2017). Farmers are also generally aware of these techniques — for example, 80% of control school parents are able to describe mounds or ridges as a way of planting root crops and 74% apply bed preparation methods to some parts of their farms in our endline survey.

However, a mixture of risks and increased labor can deter adoption by rural households. Though beneficial on average, these techniques tend to perform differently under heterogeneous growing conditions. Unsure about the benefits, farmers might not incur the additional labor for bed preparation — in Section 6, I estimate that the labor takes about 150 hours per hectare in a season. In addition, while these techniques are particularly beneficial for root crops, farmers might be reluctant to change their practices due to to the important role root crops play in maintaining subsistence. For example, cassava tubers can be stored underground for up to 24 months while its leaves are consumed. One drawback of using bed preparation methods, which loosen the soil, is that farmers might be forced to harvest all produce together at once.

Because of these concerns, farmers seldom apply these techniques at a large scale and rarely apply all techniques together. For example, farmers commonly construct mounds opportunistically (e.g. where ashes are concentrated or where soils are depleted) and plant multiple cassava stakes, along with other crops, in and along the edges of each mound — minimizing labor and insuring against sprouting failures while missing

the benefits from consistent spacing. The proportion of control school parents who adopt bed preparation jointly with one of the other techniques is only 34% in the endline survey. Nor do farmers drastically modify traditional cultivation practices (e.g. intercropping rice with root crops at close distances and planting a large number of crops per parcel) that help maintain subsistence and optimize overall production rather than single crop yields — only 50% of parents use beds while deviating from these traditional practices. These limitations in full-scale adoption, together with a lack of market access, substantially lower the total benefits that farmers can tap from these yield-improving techniques.

While technical training (e.g. management of weeds and pests under new ways of planting) is involved, in this context extension serves mainly to encourage agricultural intensification by demonstrating the possible extent of yield gains. In particular, school farms bear the risks that farmers cannot afford when a bundle of promoted techniques are applied together. By the same token, small-scale projects undertaken by students, who have less obligations to feed the family, can help farmers learn about the returns to a systemic shift to promoted techniques under familiar growing conditions.

The actual yields could, of course, differ from those in agronomic trials.[7] I nonetheless anticipate a net benefit from adoption for the average targeted farmer. In Section 6, assuming only half of the yield gains from trials — under the traditional cultivation environment — and accounting for increased labor costs, I estimate a $65 increase in net production value from using beds on a quarter of the cultivated area in one season — roughly 13% of a typical farmer's annual income. This measure omits changes at the intensive margin — the joint adoption of bed preparation with other techniques and under deviations from subsistence practices. Given the lack of trial estimates on the magnitude of yield gains under joint adoption, I maintain an agnostic view of its net effects on welfare but track changes in Section 4.3.1, noting its associations with increases in farm sales and agricultural intensification.

3. Experimental design

3.1. Sampling of schools and randomization of treatments

Figure 1 presents the experimental design. This randomized evaluation of SBAE draws from a sampling frame of 304 eligible junior high schools across five major food-producing counties in Liberia: Bong, Lofa, Nimba, Gbarpolu, and Margibi. School eligibility was determined by two criteria: available land and water sources for establishing school farms. These 304 schools were organized into 206 geographical clusters, with schools located within two kilometers of each other grouped together. This clustering strategy ensured a minimum distance of two kilometers between schools in different clusters, limiting spillovers between SBAE and control schools.

The randomization proceeded as follows. Geographical clusters were assigned to SBAE and control conditions within pairs matched on extension officer regions and average school enrollment in 2015/16. All schools within a given cluster received the same treatment assignment. Resource constraints limited recruitment to 100

[7] Suri (2011) provides evidence in Kenya that an agricultural technology that offers large average returns in a population can nonetheless have much lower returns for non-adopters. Focusing on common farming techniques promoted in agricultural intensification, Udry (2024) discusses recent evidence on the drastic difference in yields between agronomic trials and general population in Northern Ghana.

SBAE schools. The randomization yielded 100 SBAE schools (78 clusters) matched to 97 control schools (77 clusters). All schools assigned to SBAE treatment accepted the program. Figure 2 shows the geographical distribution of sampled schools relative to other junior high schools in Liberia and distinguishes between SBAE and control schools.

Following the school-level randomization, SBAE schools were further randomized into parental engagement conditions through a two-step process.[8] 50 SBAE schools were selected to receive a program promotional session for every sampled parent during the first survey in the rainy season of 2021.[9] Second, we randomized among 85 schools that have established school farms by August 2021, stratified by extension officer region. 44 SBAE schools were selected to host an annual farmer field day (FFD) in 2021 and 2022.[10]

Panel A of Table A1 shows how the sampled schools compare to others in the targeted counties. The 197 sampled schools were selected through a three-stage process: first, 447 out of 1794 schools were identified as having a functional parent-teacher association (PTA) and water source according to a Water, Sanitation, and Hygiene (WASH) dataset; then, field inspections narrowed this to 304 schools with available land and water; lastly, 197 schools were randomly selected from the 304 schools. As a result, eligible schools tend to be better than county averages in several dimensions: more students, more teachers, and smaller student-teacher ratios.

3.2. Details of treatments

3.2.1. The SBAE treatment

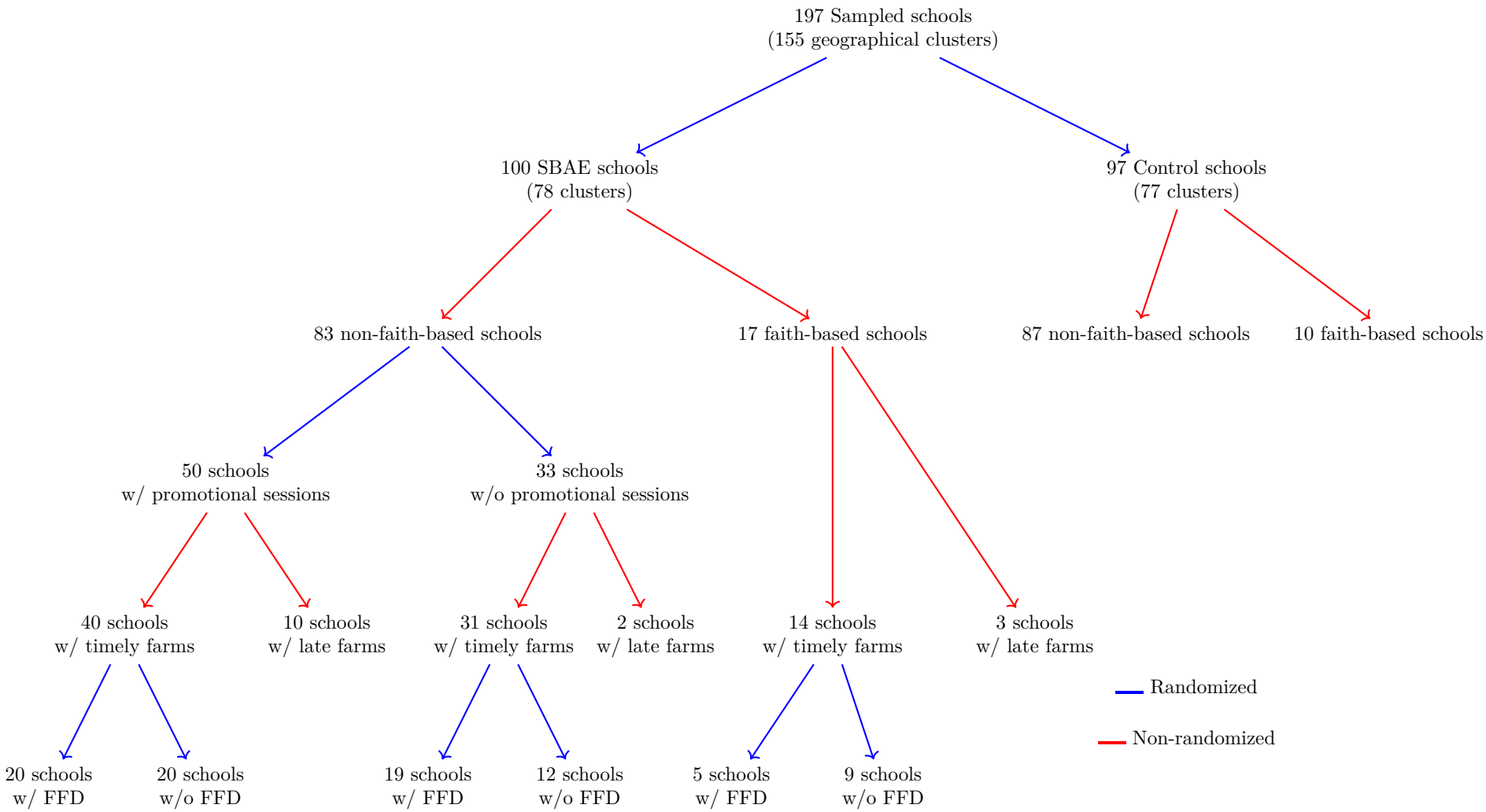
All schools assigned to the SBAE treatment receive a package of interventions as stated in Section 2.3: (i) bi-annual training of two teachers per school on the integrative SBAE model and promoted yield-improving technologies; and (ii) monthly visits from extension officers. Teachers and extension officers then implement the three-component SBAE model: (i) formal instruction of agriculture and science through lessons during extension visits and experiential instruction during classes; (ii) extra-curricular activities including weekly farm work, bi-weekly club meetings, and annual national events; (iii) technology diffusion to rural households through using the school farm as a demonstration plot and encouraging students to start home projects.

Crucial for our interpretation of the intent-to-treat effects, implementation of the SBAE program was far from perfect. By the end of the first rainy season, only 70 out of 100 SBAE schools have started their school farms. This implementation shortfall was documented in my pre-analysis plan before the midline survey.

[8] Faith-based schools were excluded from the parental engagement intervention due to the high cost of reaching parents across long distances. There is an imbalance in the percentage of faith-based schools between SBAE and control schools (see Figure 1). This imbalance arose because randomization occurred at the cluster level and faith-based schools tend to be geographically clustered together. Among the 83 clusters, the average cluster contains 1.3 schools and 17% contain multiple schools. By chance, SBAE was assigned more multi-school clusters (14 with more than 1 school; 5 with more than 2 schools) compared to control (13 with more than 1 school; 3 with more than 2 schools). Because faith-based schools cluster geographically, these multi-school clusters in SBAE were more likely to contain faith-based schools, resulting in the observed imbalance.

[9] A higher number of SBAE schools were assigned to the arm with program promotional sessions than the arm without because we needed more subjects in the former for a companion paper (JMP2, Lee, 2025).

[10] The experimental design resulted in unbalanced numbers of schools across combinations of promotional sessions and FFD (see Figure 1). This is due to (i) independent randomization of promotional sessions and FFD; and (ii) an unexpectedly lower rates of timely school farm establishment among SBAE schools with promotional sessions (80%) than SBAE schools without (94%).



Notes: This figure presents the randomization of three treatments. (1) The SBAE treatment: 100 schools (78 clusters) are assigned to the SBAE arm and 97 schools (77 clusters) to the control arm. (2) Among the 83 non-faith-based SBAE schools, 50 are assigned to host promotional sessions, and 33 are not. (3) Among 85 schools with timely farm establishment by August 2021, 44 are assigned to host a farmer field day (FFD). Blue arrows indicate randomized assignments; red arrows indicate non-randomized selection (faith-based status and farm establishment timing).

Figure 1: Experimental Design of the Study

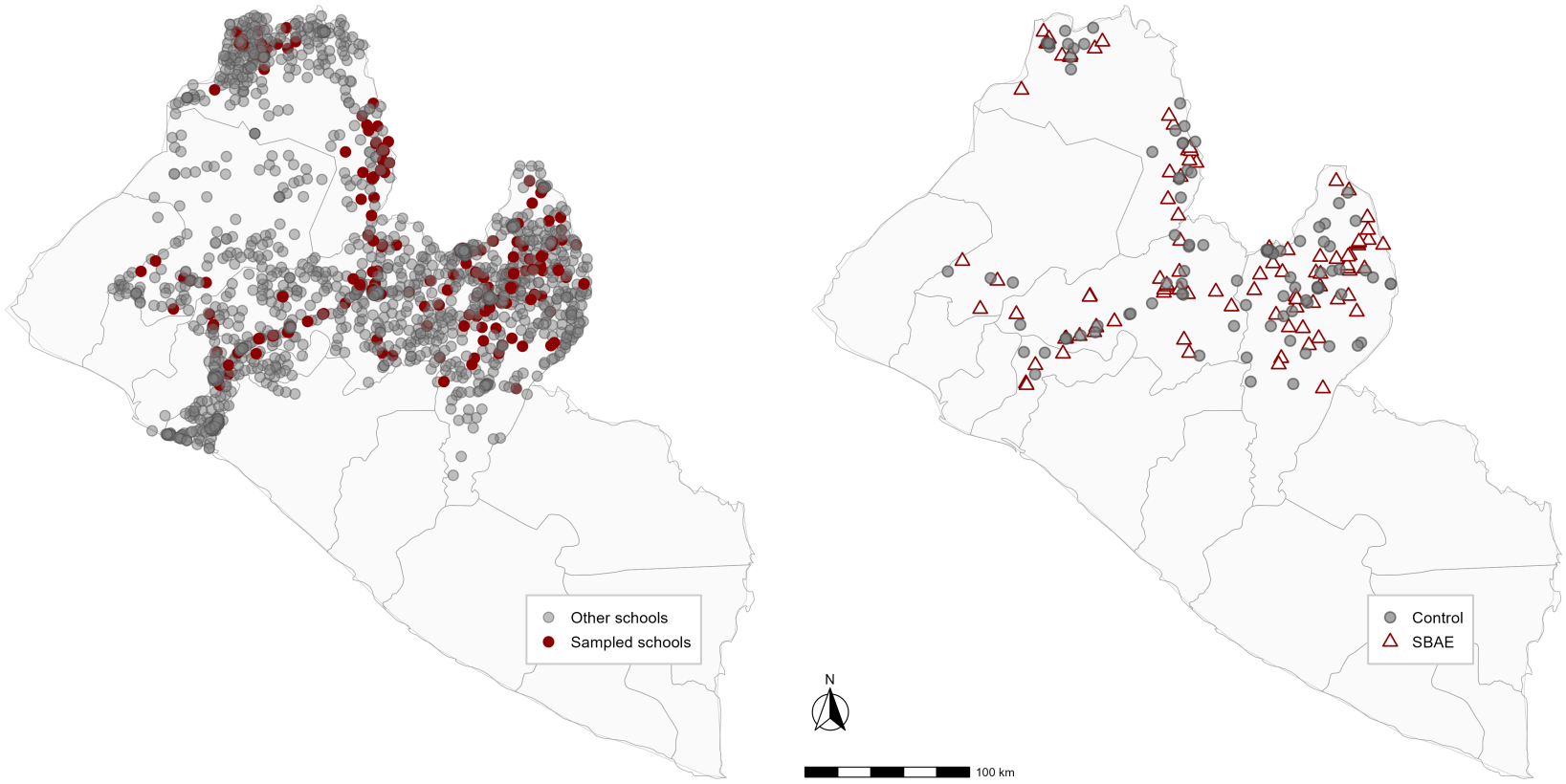


Figure 2: Geographical distribution of sampled schools

The common reasons for failed implementation include teacher turnovers, teacher non-compliance to start school farms, and damage to the school farm due to depradation and theft.

3.2.2. Parental engagement treatments

The parental engagement treatments are intentionally designed to be light-touch, aiming to strengthen weak parent-school linkages by providing parents with program promotional information. A primary goal of this paper is, therefore, to assess whether these treatments are jointly sufficient to activate SBAE's capacity to address the dual challenges of agricultural technology diffusion and school attendance.

While there are overlaps, each treatment targets distinct informational gaps. The program promotional session is inspirational in nature—it provides no training and no direct technical information about the promoted soil management techniques. Instead, through the means of a video, it uses promotional messaging and testimonials to help parents envision transformative possibilities for students, parents, and their communities under the SBAE program. This treatment addresses information constraints and corrects misperceptions, but imparts no technical skills. In contrast, the farmer field day is a training intervention. It focuses exclusively on one program component—the use of school farms for demonstration—and delivers direct instruction on the recommended use of soil management techniques. Rather than inspiration and vision, it exposes parents to the technical curriculum of SBAE.

While the FFD is common in agricultural extension programs, it has significant limitations in the SBAE context. Specifically, it does not help parents visualize the interconnections between the four program components: experiential science instruction, extra-curricular activities, school demonstration farms, and student home projects, nor the nationwide SBAE network that aids students' progression. Without this holistic understanding, parents may emphasize only the immediate agricultural productivity gains without reconceiving (i) students' roles within household production and (ii) the value of avoiding dropouts. In fact, my companion paper (JMP2, Lee, 2025) demonstrates that the FFD alone leads parents to employ students as assistants in subsistence farming rather than as managers of sales-oriented parcels, while targeted information nudges signaling parents' openness to delegate farms to students can eliminate this unintended effect.

3.2.2.1. The program promotional session treatment

Parents in my sample live, on average, one hour's walk from SBAE schools. At endline, only 30% of parents in SBAE schools without parental engagement had attended parent-teacher meetings during the 2022-23 school year. Even more concerning, just 44% had visited school farms over the three-year RCT period. Given this limited exposure, it is unsurprising that many parents remain unaware of the program entirely. Moreover, focus group discussions revealed that many parents cannot distinguish the integrative SBAE model from typical school gardens in the Liberian context. In the latter, students receive no systematic agricultural training and no yield-improving technologies are employed. This misperception may explain parents' limited engagement in supporting both the program and their children's education more broadly.

The program promotional session treatment was conducted during the first rainy season, approximately 6 months into the program, coinciding with the first student follow-up survey (see Section 3.3). In 50 SBAE schools, one parent per sampled student was invited to an information session hosted by enumerators outside

the school. During 20- to 30-minute sessions, parents watched a promotional video in groups of four to five and completed survey questions before and after viewing.

To understand the barriers hindering parental engagement with the program, we randomized which video parents watched. Half of the parents, randomly selected, watched the *"basic video"*, which described SBAE as an innovative education model that integrates experiential science instruction, extracurricular activities on school farms, and student home projects to advance technology diffusion, educational outcomes, and youth empowerment. The other half watched the *"persuasive video"*, which included all content from the basic video and added testimonials from past participants and locally relevant stakeholders—town chiefs, extension officers, teachers, students, and parents. These persuasive elements highlighted SBAE's effectiveness in empowering students to manage farms, grow more food, and use proceeds productively (e.g., "I paid my own school fees"), demonstrating how the program transforms lives through improved agricultural techniques (e.g., "the cassava I got was big, big").

Focusing on parents' anticipated improvements for students under SBAE over one year, my companion paper (JMP2, Lee, 2025) provides strong evidence that both versions of the promotional video had immediate positive effects on parents' beliefs about students' farm management abilities—a key factor determining their openness to delegating farms to students. Specifically, 97% of parents predict improvement on at least one out of seven belief measures, and 89% expect higher cassava yields from students. The share anticipating students will surpass typical adults in the community rises dramatically from 31% before viewing to 70% after. While both videos generate similar levels of optimism, parents who view the persuasive version anticipate stronger harvest improvements. Moreover, beliefs about cassava yields remain elevated at midline and endline compared to parents in schools without video sessions. Conditional on the type of video viewed, parents' beliefs about students' harvests strongly predict their visits to school farms and attendance at parent-teacher meetings one year later.[11]

Overall speaking, the program promotional session treatment effectively tackles two beliefs of parents: (i) the returns to promoted soil management techniques (and thus the need to visit school farms); and (ii) the value of students' learning under SBAE (and thus the need to delegate farms to students and support their education). My companion paper (JMP2, Lee, 2025) provides a theoretical model linking how these beliefs guide intrahousehold production decisions, such as whether students switch from assisting on subsistence parcels to managing sales-oriented parcels and whether students harvest root crops early to support timely school fee payment. Crucially, the promotional video does not directly mention any specific promoted techniques, so any treatment effects on household farming practices must operate through parents' interactions with students and the SBAE program.

3.2.2.2. The annual farmer field day treatment

While SBAE is an integrative extension model, the introduction of improved soil management techniques is one of its key elements and an important aspect that sets it apart from typical school gardens. As such, the program can potentially benefit from creating an explicit opportunity for parents to learn and observe the promoted farming techniques. To this end, farmer field days—a common practice in agricultural extension—offer a valuable mechanism. Under this approach, a group of "seed farmers" are first trained on

[11] These belief measures are incentivized and carefully elicited, reminding subjects of the stated anticipated changes (including comparison of stated beliefs about students' harvests and typical adults' harvests) and allows revisions if needed.

farming techniques, and then the extension program engages neighboring farmers in a meeting where the techniques are discussed and their performance is observed.[12]

The annual farmer field day treatment was conducted each September during the first two years of the three-year program. This timing allowed extension officers to harvest some of the root crops to illustrate the benefits of the main yield-improving technique—bed preparation methods. To ensure broad community participation, 4-H extension officers informed nearby town chiefs and asked them to select 25-30 farming group leaders per school to participate alongside parents. Each event lasted approximately six hours. While parents' attendance was not directly measured, Section 5 provides evidence that the annual farmer field day significantly improves the likelihood that parents have observed harvests on the school farms by the endline, as well as their beliefs about the number of cassava tubers per plant that students can harvest.

3.3. Design of data collection

3.3.1. Timeline of activities

Figure 3 shows a timeline of SBAE activities and randomized treatments.

The SBAE program began in January 2021, which was the beginning of the school year 2020/21,[13] and ended in December 2023. The program covered three school years and three rainy seasons. We conducted a major survey with subjects in the midst of each rainy season. This timing was chosen for two reasons. First, it coincided roughly with the conclusion of a school year, which provided a natural point to track program impact. Second, it coincided with the periods where most Liberian farmers conduct land preparation for their cultivation, which allowed us to ask questions about adoption of promoted soil management techniques.

Mirroring the layered randomization design, the program evaluation was separated into two stages. In the first year, I tracked the program's initial impact in the first follow-up survey with students and implemented the two parental engagement treatments. Then, in the second and third years, I conducted the midline and endline surveys with teachers, students, and parents to assess the primary outcomes. We have also planned a future survey potentially taking place in 2026. We will track parents' livelihoods, households' production indicators (e.g. yields), and technology adoption among non-parent households in nearby communities.

[12] According to Emerick & Dar (2021), farmer field days differ from informal communication among farmers in that (i) farmers can physically verify the benefits of techniques; (ii) the attributes of the techniques are discussed explicitly; (iii) farmers can hear the experiences of (multiple) demonstrators; (iv) farmers can interact with extension officers. In the current context, parents can also indirectly acquire information about what students learn in school.

[13] The commencement of the school year was delayed by 3 months due to school closures during the COVID-19 pandemic.

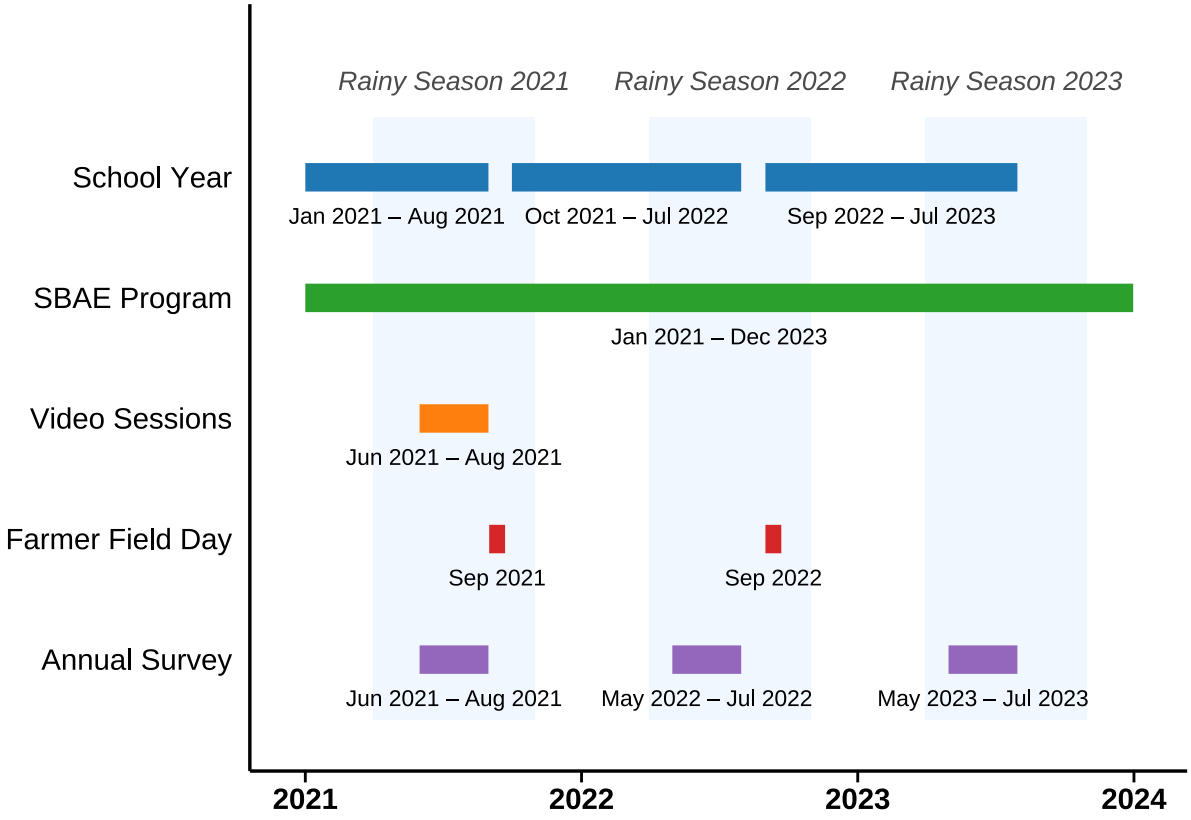


Figure 3: Timeline of SBAE program and data collection activities

3.3.2. Sampling of subjects

Sample of students

The SBAE program targeted upper primary and junior high grades in its recruitment of students. To increase the likelihood of successful follow-up after three school years, I focused on grades 4, 5, 7, and 8, excluding 6th and 9th grade students who would soon transition to different schools.

My sample was drawn from the pool of intended beneficiaries — students who farmed outside school and had parents who farmed. To identify this pool, enumerators conducted screening surveys in these targeted grades during the 2020/21 school year. To maximize the likelihood of capturing actual program participants, enumerators employed a hierarchical sampling strategy within each school to select approximately 20 students.[14] Students were prioritized in the following order: (1) students who both farmed before 2021 and joined cooperative farming groups (kuu) before 2021; (2) students who joined kuu before 2021; (3) students who farmed before 2021; and (4) other eligible students. Enumerators selected students sequentially down this priority list until reaching the target sample size.[15] Despite this purposive approach, sampled students

[14] Enumerators were instructed to oversample in larger schools and undersample in smaller schools to ensure adequate representation while maintaining feasibility.

[15] Based on a pilot survey, students with prior farming experience and prior kuu participation were substantially more likely to participate in the SBAE program.

closely resemble the broader population of eligible students in targeted grades. As shown in Panel B of Table A1, sampled students have only slightly higher rates of prior farming experience (99.4% vs. 95.0%) and prior kuu participation (82.1% vs. 77.1%) compared to all eligible students, suggesting that the sampling strategy successfully captured a representative sample of the target population while increasing the likelihood of observing program participation.

A total of 3,757 students were sampled across 197 schools, averaging approximately 19 students per school and representing 30% of students in targeted grades. The sample was stratified by school, section (upper primary or junior high), and gender. As only one student per household was sampled, the total number of sampled households was also 3,757.

Sample of parents

For every sampled student, enumerators invited one parent or guardian (henceforth, parent) who was planting root crops, vegetables, or both in 2021.[16] In households with both male and female parents meeting this requirement, I randomized which parent was invited to maintain gender balance in the sample. In households with only one eligible parent, that parent was invited. A total of 3,757 parents were invited, corresponding to the 3,757 sampled students. In 95% of cases, the invited parent participated in the survey; in the remaining 5%, a substitute household member participated instead.

Sample of teachers

To assess the SBAE program's impact on school instruction, I collected repeated cross-sections of science and agriculture teachers in 2022 and 2023. Rather than tracking individual teachers over time, different teachers were sampled in each survey round to measure school-level instructional changes.[17] In each survey round, enumerators sampled all science and agriculture teachers in each school. A total of 452 teachers were sampled in 2022 and 472 in 2023.

3.3.3. Surveys

Due to a tight budget and funds arriving per round, survey coverage varied across the three annual rounds. In the first follow-up survey (2021), I tracked and interviewed only 2520 students — all sampled students in the 50 SBAE schools assigned to the promotional session treatment, and a random half in the other 147 schools.

We placed a stronger focus on tracking students than parents overall, as there were more primary outcomes involving students. In the midline survey (2022), all students were tracked but not parents. In the endline survey (2023), all students were tracked, together with 45% of parents. All sampled teachers were interviewed in the midline and endline surveys.

3.3.4. Farm survey modules

Due to budget constraints, I randomly selected a subset of subjects for a module of parcel-level farm questions in the midline student survey and endline parent survey. In the midline student survey, 1388

[16] This requirement ensured parents were engaged in agricultural activities relevant to the SBAE curriculum. During student sampling, households were pre-screened to ensure at least one parent met this criterion.

[17] Teachers were not tracked from baseline due to high teacher turnover in rural Liberian schools and because the study focused on school-level instructional changes rather than individual teacher development.

out of 2425 were selected for the farm survey module. In the endline parent survey, 1363 out of 1543 parents were selected for the farm survey module. For questions about agricultural technology adoption and farm management, questions are specifically asked about a "student parcel" (midline and endline) and a "parent parcel" (endline). These parcels are randomly chosen up to two criteria about priority: (i) parcels that are looked after by the subject are prioritized; (ii) parcels that involve the planting of root crops and vegetables are prioritized.

3.4. Measurement of outcomes

3.4.1. Primary outcomes

Agricultural technology adoption. I measure the adoption of promoted soil management techniques on students' parcels (midline and endline) and parents' parcels (endline). For each subject-round, there are two primary outcomes: (i) whether bed preparation is used for root crops or vegetables; and (ii) the total number of techniques (out of the four outlined in Section 2.4) that are used for root crops or vegetables.

School attendance. I measure students' school attendance (midline and endline) by three measures: (i) whether students have dropped out of schools by the time of survey, according to school records and tracking to the household (in the case students have moved to other schools); (ii) absenteeism in the last five school days; and (iii) the imputed total days of school attendance in 2021/22 to 2022/23.[18]

Students' studying efforts. I use the number of hours students spend on studying in the week preceding surveys (midline and endline) as a primary outcome. As supplementary information, I also measure the number of hours students spend on farming (inside and outside schools) and doing household chores.

Students' livelihoods. I track three dimensions of students' livelihoods. First, their annual incomes (reported retrospectively in the endline) - disaggregated by farm and non-farm incomes. Second, their annual savings (reported retrospectively in the endline) - which is a better measure than incomes in this context because many students farm for their households and may receive transfers instead of incomes. Third, their food consumption and non-food spending. Food consumption is captured (inversely) by the number of meals students missed in the week preceding surveys (midline and endline). Non-food spending during the past 6 months (midline and endline) is measured on three categories: agricultural inputs, spending on school fees and supplies, and consumption goods (e.g. clothes and footwear, musical instruments and sports accessories). All variables with monetary values are log-transformed and winsorized at the top and bottom 2.5%.

Students' aspirations. I measure students' aspirations in two ways. First, I measure whether students list an agricultural career (including farmer, agronomist, and extension officer) as one of their top three jobs without being given options in the midline survey. Second, I measure whether students report saving for attending university, as a proxy for their educational aspirations, in the endline survey.

[18] This imputed measure is based on students' enrollment status, rate of absence from school, and a total number of 200 school days per school year. A student who drop outs from school is counted as having zero school days. A student who switches schools is counted as having the same number of school days as control school average.

3.4.2. Secondary outcomes

Beliefs and experiences about attainable crop yields. Both anecdotes and agronomic trials (see Section 2.4) suggest quick visible signs that yields improve under bed preparation — the main promoted technique, and more so when used jointly with other techniques — especially for root crops such as cassava. Beliefs about the number of attainable cassava tubers per plant thus provide good proxies for subjects' exposure to the SBAE program and, accordingly, beliefs about the returns to promoted soil management techniques. To evaluate whether the promoted techniques are beneficial to sampled households, in the endline survey we further ask students and parents whether they have experienced an improvement in the yields (as in the number and size of tubers) or quality of produce over the past two years.

Knowledge about promoted soil management techniques. In each annual survey, I design a test based on the SBAE curriculum to evaluate whether subjects know how, when, and why to use the promoted techniques.

Teachers' pedagogy. Teachers' pedagogy is reported by both students and teachers during midline and endline. I use student reports of teachers' pedagogy, unless otherwise stated. I measure teachers' adoption of SBAE pedagogy in three ways. First, whether teachers have used agricultural projects or field visits for instruction in the last five school days. Second, the number of scientific concepts (out of six from the science curriculum) that have ever been illustrated with farm work. Third, the number of home visits (last month) and farm visits (last rainy season). As supplementary information on whether SBAE crowd out other types of teaching efforts, I also measure (i) teachers' use of student-centered instruction that is not related to agriculture, such as student presentations and group projects; and (ii) whether teachers have provided after-school classes in the last five school days.

Parents' visits to school farms. During the endline survey, I ask parents (i) whether they have ever visited the school farm and (ii) the number of times they visited the school farm in the preceding rainy season. I also ask parents whether they have ever worked and observed harvests on the school farm.

Students' management of farms. During the midline and endline surveys, I ask students who is involved in choosing the crops and planting methods on the "student parcels", to which students can answer (i) they choose on their own; (ii) they choose after discussing with parents; (iii) they choose what parents would prefer; (iv) parents choose; and (v) others choose. As primary measures, I define the variables "student chooses crops" and "student chooses methods" to be one if students choose on their own and zero otherwise. As supplementary information, I also ask (i) whether students have ever looked after a farm without guidance from older persons; (ii) whether students look after a farm in the current season; and (iii) the number of parcels they look after; and (iv) the number of parcels on which they assist household members.

Technology diffusion within households. During the midline survey, I ask students whether they have ever demonstrated, in four specific ways, the promoted techniques they have applied on the "student parcels" to household members, and to whom.[19] This enables me to define two measures of demonstration efforts that are analogous to measures of technology adoption: (i) whether students have demonstrated bed preparation to parents; and (ii) the number of promoted techniques students have demonstrated to parents.

Parents' support for students' education. During the endline survey, I collect two proxies for parents' support by (i) the number of parent-teacher meetings attended; and (ii) whether students have ever been absent from

[19] The four specific ways are:

school because of household work. As supplementary information, I also ask subjects (i) whether parents pay for students' school fees on time in the current semester; and (ii) the amount parents spend for students' agricultural inputs, school fees and supplies, and non-food consumption.

Measures of joint technology adoption. My primary measures of agricultural technology adoption capture the standalone adoption of promoted techniques. As discussed in Section 2.4, there are synergies from joint adoption of bed preparation with other techniques. I use two measures to capture the degree to which farmers deepen their agricultural intensification. First, whether bed preparation is applied jointly with at least one of the three other soil management techniques. Second, whether bed preparation is applied jointly with at least one deviation out of three traditional subsistence practices (fallowing in previous rainy season; cultivation on rice fields; and intercropping with rice).

3.5. Balance of baseline characteristics

Table A2 shows that baseline characteristics of sampled schools, students, and parents are balanced across treatment arms at baseline, that is, at the beginning of the school year 2020/21. Mirroring our main regressions in Section 4.3, I regress baseline characteristics on an indicator of SBAE, an interaction term between SBAE and program promotional session, and an interaction term between SBAE and annual farmer field day, with pair-of-school-clusters fixed effects and clustering at the school-cluster level. None of the F-tests of treatment coefficients across baseline characteristics are significant at the 10% significance level.

Panel A focuses on school characteristics. It shows that schools are balanced in terms of students, teachers, student-teacher ratio, classrooms, and the number of students who are sampled. Importantly, schools are also balanced in variables that potentially explain the success of school farms — whether water source is available nearby, and whether breakfasts or lunches are provided. In addition, schools are balanced in school management — the number of visits by government district education officers; school ownership (public, private, or faith-based); whether an enrollment list is kept; and the number of parent-teacher association meetings and school management committee meetings.

Panel B focuses on characteristics of sampled students. Students are balanced in characteristics used for stratified sampling — students have an average age of 15.7, a grade of 5.7, and 52% of students are male. In addition, students are balanced in having 7 years of farming experience.

Panel C focuses on characteristics of sampled parents. As in the case of students, parents are balanced in demographics — the average is 43.2, 47% are male, 74% are parents (as opposed to guardians), and 63% are household heads. The fact that we did not survey parents in the first survey does not have a substantive impact on the experiment. 98% of parents who ultimately attended surveys are the ones invited based on student reports and all sampled parents farm. As is typical in rural Liberia, only 59% of parents can read and write English, with 54% and 39% having completed elementary and junior high school respectively.

3.6. Attrition

3.6.1. Levels and correlates of attrition

Table A3 shows the attrition levels of students in the midline and endline surveys, as well the attrition levels of parents in the endline survey.

Attrition among students is a serious concern in this study. Panel A shows that student attrition rate is 35% by the midline survey and 48% by the endline survey. Panel B shows that this high rate of attrition is mainly due to student dropouts from schools (18% by midline; 24% by endline) or switching to other schools (16% by midline; 21% by endline). The high rate of switching to other schools reflects the fact that many students are progressing to another school section (junior high or senior high); and in addition, students who seek better opportunities in schools closer to urban areas.

Besides the high levels of attrition, Panel A of Table A3 also shows that student attrition is imbalanced across treatment arms. Both the SBAE treatment and the annual farmer field day treatment do not create imbalanced student attrition relative to control schools. However, the program promotional session strongly predicts student attrition, lowering the midline rate by 7.3 percentage points (from 36%) and the endline rate by 9.2 percentage points (from 48%). Panel C shows that this differential attrition can be explained entirely by the promotional session reducing dropouts or switching of schools.

Relative to students, the attrition among parents is much lower. Panel A shows that parent attrition rate is only 12% by the endline survey. Panel B shows that the main reasons for parent attrition are due to travelling (4.6%) or relocation (4.3%).

As in the case of students, endline parent attrition is also imbalanced across treatment arms. Specifically, Panel A shows that the SBAE treatment increases parent attrition by 4.2 percentage points and the annual farmer field day reduces attrition by 4 percentage points; whereas the program promotional session has no statistically significant effects.

In view of the high and imbalanced attrition rates, Table A4 examines (i) whether student and parent attrition is selective in terms of baseline characteristics; and (ii) whether selective attrition is differential across treatment arms.

Panel A shows that students are selected into the endline sample mainly based on age and grade, but not based on gender or years of farming experiences. An increase of students' age (grade) by one increases attrition probability by 2.62% (1.96%). The attritors are, on average, 0.21 standard deviations older (or in 0.12 standard deviations higher grade) than non-attritors. This is a natural reflection of the student sample — upper-grade students are closer to the point where they need to progress to another school section (junior high or senior high). Meanwhile, Wald tests for the interactions of treatments with baseline student characteristics indicate that none of the selective attrition is differential across treatments.

Panel B shows that parents are selected into the endline sample based on two factors. An increase in parents' age by one reduces attrition probability by 0.23%; while parents are 4.27% less likely to attrit than non-parent guardians. Attritors are, on average, 0.17 standard deviations (0.18 standard deviations) younger (less likely to be parents). Meanwhile, Wald tests again indicate that selective attrition is not differential across treatment arms.

3.6.2. Summary and countering strategies

In Section 4.3), I will present the main intent-to-treat (ITT) effects from the midline and endline samples. The current section shows that non-random attrition, especially among students assigned to the program promotional session treatment, poses a threat to internal validity. However, since students and parents are selected into the endline samples in the same manner across treatment arms, addressing differential attrition rates is sufficient to preserve the comparability between groups. Accordingly, I will present Lee lower and upper bounds (Lee, 2009) to account for the worse-case and best-case effects of differential attrition.

Given selective attrition, one might still worry that the ITT estimates from the midline and endline samples are not representative of the full sample. I offer three remarks to support the validity of my estimates. First, students' dropouts are a natural and expected feature in this context, which provides a useful setting to study the robustness of SBAE effects on parents' agricultural technology adoption. Second, as shown in Table A4, students' selective attrition is largely driven by mechanical grade progression. In this case, the ITT effects on school-specific outcomes — such as absenteeism — are estimated from the appropriate sample: students who remain and could benefit from SBAE. Third, I assess the sensitivity of main results to sample composition. For example, I demonstrate consistency in treatment effects on key outcomes across all three rounds (Section 4.2 and Section 4.3). In the Online Appendix, I will also show that treatment effects are similar for upper-grade (Grades 5 and 8) and lower-grade students (Grades 4 and 7).

4. Experimental results

4.1. SBAE impact on teachers

To show that the program is effective in training teachers, we begin by documenting SBAE's impact on teachers, which was measured during the midline and endline surveys. I run the following regression:

$$Y_{tscp} = \alpha + \beta_1 \text{SBAE}_p + \gamma_p + \epsilon_{tscp} \tag{1}$$

where Y_{tscp} is the outcome of interest for teacher t, school s, geographical cluster c, and pair-of-clusters p; SBAE_p indicates whether pair p is randomly assigned to the SBAE arm; γ_p denotes randomization pair fixed effects; ϵ_{tscp} is an error term. Standard errors are clustered at the geographical cluster level, which is the highest level at which random assignment took place. β_1 is the coefficient of interest, capturing the ITT effects of SBAE on teachers.

Table 1 presents the results. Columns (1) to (4) indicate consistently positive ITT effects of SBAE on teachers' knowledge about promoted techniques and beliefs about students' attainable yields, in both the midline and endline surveys. Teachers' scores on knowledge tests improve by 12% (0.4σ) and 14% (0.44σ) respectively. Teachers' beliefs about students' attainable cassava tubers in 9-10 months improve by 24% (0.53σ) and 20% (0.42σ) respectively. The large magnitude of these ITT effects validates both (i) the strong receptiveness among teachers and (ii) the novelty of SBAE's agricultural curriculum, considering that 67 out of 97 control schools also have school gardens and most teachers farm outside schools. Columns (5) and (6)

Table 1: Midline and endline intent-to-treat effects on teachers

	Dependent variables					
	Midline		Endline		Endline	
	Score: test on techniques (0-10)	Beliefs: students' tubers from 4 cassava plants	Score: test on techniques (0-12)	Beliefs: students' tubers from 4 cassava plants	Pedagogy: used ag projects/ field visits (0/1)	Pedagogy: # concepts demonstrated (0-6)
	(1)	(2)	(3)	(4)	(5)	(6)
$SBAE_p$	0.729***	4.940***	0.671***	4.536***	0.126***	0.395***
Standard errors	(0.122)	(0.917)	(0.142)	(1.244)	(0.024)	(0.070)
Control Mean	6.02	20.93	4.85	22.40	0.35	1.57
Control SD	1.81	9.30	1.53	10.84	0.48	1.73
Observations	450	452	445	445	1970	1970
Clusters	150	150	149	149	155	155

Notes: This table reports intent-to-treat effects of the SBAE treatment on agriculture and science teachers in the midline and endline surveys. Columns (1) and (3) refer to teachers' scores on a test of promoted agricultural techniques. Columns (2) and (4) refer to teachers' beliefs about the number of tubers that students can harvest (in 9-10 months) from 4 cassava plants. Column (5) refers to whether teachers have used agricultural projects or field visits in science instruction in the past five days, according to student reports. Column (6)refers to the number of scientific concepts that teachers have demonstrated using farm work, according to student reports. All regressions include pair-of-clusters-of-schools fixed effects. Standard errors are clustered at the cluster-of-schools level.

indicate also positive ITT effects of SBAE on teachers' pedagogy in the endline survey.[20] 12.6 percentage points more students report that their teachers have used agricultural projects or field visits in instruction in the last five days. Students also report a 25% (0.23σ) increase in the number of scientific concepts that teachers have ever demonstrated with farm work. Taken together, Table 1 thus indicates that the SBAE program has strong effects on teachers as intended.

4.2. Early training effects on students

Now we analyze SBAE's impact on students. We first discuss the results of the first follow-up survey, which captured training effects by 5-7 months, and importantly, before parental engagement treatments were implemented. I run the following regression:

$$Y_{hscp} = \alpha + \beta_1 \text{SBAE}_p + \gamma_p + \epsilon_{hscp} \tag{2}$$

where Y_{hscp} is the outcome of interest for household h, school s, geographical cluster c, and pair-of-clusters p; SBAE_p indicates whether pair p is randomly assigned to the SBAE arm; γ_p denotes randomization pair fixed effects; ϵ_{hscp} is an error term. Standard errors are clustered at the geographical cluster level, which is the highest level at which random assignment took place. β_1 is the coefficient of interest, capturing the ITT effects of SBAE on students.

Table 2 presents the results. Columns (1) to (4) indicate consistently positive ITT effects of SBAE on students' exposure to promoted farming techniques. The fractions of students who have ever used bed preparation and composting improve by 4 and 6 percentage points respectively. Reflecting SBAE's training on

[20] In the Online Appendix, I show similar positive ITT effects on teachers' pedagogy in the midline survey. The results are also robust to using teacher reports.

Table 2: Intent-to-treat effects on students in the first follow-up survey

	Dependent variables							
	Ever used: beds (0/1)	Ever used: compost (0/1)	Score: site selection (0–2)	Beliefs: tubers from 4 cassava plants		Intend to look after farms (0/1)	Hours studying (weekly)	Hours farming (weekly)
				(student)	(typical farmer)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$SBAE_p$	0.042***	0.063***	0.042**	0.844**	0.242	0.090***	-0.366*	0.316
Standard errors	(0.010)	(0.017)	(0.020)	(0.411)	(0.334)	(0.016)	(0.198)	(0.379)
Control Mean	0.90	0.30	1.63	19.86	20.30	0.60	7.25	13.23
Control SD	0.30	0.46	0.56	9.12	8.84	0.49	4.42	7.15
Observations	2520	2520	2520	2520	2520	2520	2520	2520
Clusters	155	155	155	155	155	155	155	155

Notes: This table reports intent-to-treat effects of the SBAE treatment on students in the first follow-up survey. Columns (1) and (2) refer to student reports about whether they have ever used bed preparation and composting. Column (3) refers to students' scores on two questions about site selection. Columns (4) and (5) refer to refer to students' beliefs about the number of tubers that can be harvested (in 9-10 months) from 4 cassava plants. Column (4) refers to students' beliefs about their own harvest. Column (5) refers to students' beliefs about the harvest of a typical farmer in their community. Column (6) refers to whether students report an intention to look after farms in that rainy season. Columns (7) and (8) refer to students' reported hours spent on studying and farming in the week preceding the survey. All regressions include pair-of-clusters-of-schools fixed effects. Standard errors are clustered at the cluster-of-schools level.

farm management skills, students' knowledge about site selection improves by 3% (0.08σ). Most importantly, students' beliefs about students' attainable cassava tubers in 9-10 months improve by 4% (0.09σ) — these early improvements in beliefs are consistent with descriptions in agronomic trials that report early, visible differences in the height of stems and the size of leaves when bed preparation is applied. To verify that these changes in beliefs are due to promoted techniques in the SBAE program, column (5) shows that there are no changes in students' beliefs about cassava tubers attained by typical farmers in their communities. Columns (6) to (8) indicate ITT effects on students' behavior outside schools. The fraction of students who report intending to look after a farm outside school in the first rainy season improves by 9 percentage points from the control mean of 60%. Meanwhile, there are early signs of pitfalls as discussed in the Introduction — students' weekly studying hours drop by 5% and farming hours increase by 2%, though the latter is not statistically significant at conventional levels. Taken together, Table 2 thus indicates that before parental engagement, the SBAE program had already increased students' intentions to start home projects and apply promoted techniques outside schools, with some diversion of students' efforts from academic studies.

4.3. SBAE impact on primary outcomes, with and without parental engagement

Having analyzed SBAE's training effects on teachers and students in Section 4.1 to Section 4.2, I now analyze SBAE's impact on primary outcomes — agricultural technology adoption, school attendance, students'

livelihoods, and students' aspirations. I run the following regressions to estimate ITT effects of interest:

$$Y_{hscp} = \alpha_1 + \beta_1 \text{SBAE}_p + \gamma_p + \epsilon_{hscp} \tag{3a}$$

$$Y_{hscp} = \alpha_2 + \beta_1 \text{SBAE}_p + \beta_2 \text{SBAE}_p \times \text{FFD}_s + \beta_3 \text{SBAE}_p \times \text{Session}_s + \lambda \text{Faith-based}_s + \gamma_p + \epsilon_{hscp} \tag{3b}$$

where Y_{hscp} is the outcome of interest for household h, school s, geographical cluster c, and pair-of-clusters p; SBAE_p indicates whether pair p is randomly assigned to the SBAE arm; FFD_s indicates whether school s is randomly assigned to the FFD arm; Session_s indicates whether school s hosts program promotional sessions;Faith-based_s indicates whether school s is funded by faith-based institutions, which are excluded from the randomization of video sessions; γ_p denotes randomization pair fixed effects; ϵ_{hscp} is an error term. Standard errors are clustered at the geographical cluster level, which is the highest level at which random assignment took place.

(3a) estimates the overall SBAE effect, disregarding differences in parental treatment conditions among SBAE schools. (3b) estimates the distinct effect of each treatment combination. I present results from both regressions for the following reason. Regression (3a) contains the overall SBAE treatment effect, β_1, which is a weighted average of the effects in SBAE schools assigned to different parental engagement conditions as captured by β's in (3b). On the one hand, if we treat SBAE schools asssigned to different conditions to be the same, then β_1 in (3a) provides a more precise estimate of SBAE's effects. On the other hand, SBAE schools assigned to different conditions can have varying effects — in this case we will want to see which combination of SBAE and parental engagement is sufficient to improve outcomes, thus the need of (3b).

To be specific, we will discuss the following estimates of interest: (i) overall SBAE effect (β_1 in (3a)); (ii) pure SBAE effect (β_1 in (3b)); (iii) "SBAE + FFD" effect ($\beta_1 + \beta_3$ in (3b)); (iv) "SBAE + Session" effect ($\beta_1 + \beta_2$ in (3b)); and (v) "SBAE + Session + FFD" effect ($\beta_1 + \beta_2 + \beta_3$ in (3b)).

To maximize statistical power, (3b) uses what Muralidharan et al. (2025) refer to as the "short model" in factorial design. β_2 and β_3 give the marginal effects of the promotional session and the annual FFD in a sample in which (roughly) half the households are offered the other treatment. We are underpowered to detect interaction effects. In the Online Appendix, I show that the main point estimates are similar when using the "long model" that controls for an interaction term between the two parental engagement treatments. Following Kaur et al. (2025), In the Online Appendix I also show that the effects of the promotional session (FFD) are qualitatively similar in the sub-samples with and without FFD (promotional session).

4.3.1. Agricultural technology adoption and agricultural intensification

We begin by presenting the results on households' technology adoption in Table 3. Panel A shows treatment effects on the student parcel. Columns (1), (3), (5), and (7) indicate that considering all SBAE schools together, treatment effects on both midline and endline adoption are positive. Columns (2), (4), (6), and (8) provide a drastically different view by separating the effects of parental engagement from the SBAE program. In both the midline and endline surveys, pure SBAE effects on both the extensive margin (a binary indicator of bed preparation) and the intensive margin (the number of promoted techniques used) are close to zero and statistically insignificant. By contrast, the program promotional session increases midline and endline

bed preparation by 10.8 and 10.4 percentage points respectively (from 50% and 45% in control schools), with similar effects on the intensive margin. The annual farmer field day also has positive effects on all outcomes, but the effects are not statistically significant at the conventional levels.

Panel B shows endline treatment effects on the parent parcel. Again, columns (9) and (11) indicate that considering all SBAE schools together, treatment effects are positive for adoption on the parent parcel, whereas columns (10) and (12) indicate that the effects tend to be driven by parental engagement. SBAE alone has exactly zero effects on the use of bed preparation and increases the number of techniques used by 0.2 (14% from 1.38 in control schools). The program promotional session raises both — increasing bed preparation by 8.8 percentage points (from 74% in control schools) and the number of techniques by 0.24 (17% from 1.38 in control schools). Surprisingly, even though the annual farmer field day targets the parents, it increases technology adoption on the student parcel (Panel A) but fails to do so on the parent parcel (Panel B) — this likely reflects parents' tendency to let students experiment with promoted techniques before applying on "parent parcels" that are often larger and more central to livelihoods.

The bottom half of Table 3 presents the linear combinations of estimates along with Lee bounds, allowing us to examine the combined effects of different treatments. These estimates provide two messages. First, there is little change to the null pure SBAE effects on both technology adoption on the student and parent parcels, even if the Lee upper bounds are considered. Second, when we add together the treatment effects of the SBAE program, the program promotional session, and the annual farmer field day, the increases in technology adoption are even more impressive. In particular, SBAE augmented with both forms of parental engagement raise students' use of bed preparation by 18.2 percentage points (from 45% in control schools) and number of techniques used by 0.39 (53% from 0.73 in control schools). It raises parents' use of bed preparation by 10.9 percentage points (from 74% in control schools) and number of techniques used by 0.36 (26% from 1.38 in control schools). All these strong effects are preserved when considering Lee lower bounds. Alternatively, the augmented SBAE program raises the number of techniques used by parents by 0.38 standard deviations — a sizable effect considering that parents are not directly trained in the program and that households do not receive other forms of interventions (e.g. provision of seeds or credit) than provision of knowledge and skills. By comparison, a large literature on sub-Saharan Africa (Suri & Udry, 2022) suggests that many village-based agricultural extension fails to create statistically significant impacts on farmers who are not directly trained, unless social network targeting (Beaman et al., 2021) or incentives for trained farmers to communicate with others (BenYishay & Mobarak, 2019) are built into such extension programs. Table 3 thus provides strong proof-of-concept that school-based agricultural extension can be a highly effective method in increasing technology diffusion.

While Table 3 focuses on the primary measures of adoption, Table A5 indicates the augmented SBAE program is deepening agricultural intensification in a comprehensive way. I consider two secondary measures of agricultural intensification: first, whether the main yield-improving technique, bed preparation, is applied jointly with at least one of the three other soil management techniques, noting the potential of such pairing to increaes yields; second, whether bed preparation is applied jointly with at least one deviation from three traditional cultivation practices (fallowing in previous rainy season; cultivation of roots or vegetables on rice fields; and intercropping with rice), noting the potential of such pairing to increase sales.

Whereas Table 3 suggests that the SBAE program alone is entirely ineffective in increasing the use of bed preparation, Table A5 provides a slightly different picture. Columns (6) and (10) indicate that the SBAE

program, on its own, increases both students' and parents' joint use of bed preparation with other promoted techniques — by 8.8 percentage points for students (from 17%) and by 7.6 percentage points for parents (from 34%). By contrast, the SBAE program, on its own, has close to zero and statistically insignificant effects on whether students and parents use bed preparation along with deviations from traditional cultivation practices (columns (8) and (12)). While the program promotional session intensifies students' and parents use of bed preparation along with other techniques (columns (6) and (10)), its main effect is to dramatically increase the use of bed preparation in settings deviating from traditional cultivation practices — by 12.1 percentage points for students (from 25%; column (8)) and 9.8 percentage points for parents (from 52%; column (12)). As in the case of primary measures of technology adoption, the annual farmer field day does not affect these secondary measures of technology adoption. In summary, Table A5 suggests that besides pivotally increasing the use of promoted techniques, the program promotional session also dramatically increases the use of yield-improving techniques in settings that are more amendable to sales.

4.3.2. School attendance and time use

Table 4 presents results on students' school attendance. Columns (1), (3), (5), and (7) indicate that considering all SBAE schools together, there is a slight increase in sampled students who remain enrolled in school but close to no change in students' rate of absenteeism. Columns (2), (4), (6), and (8) reveal drastistically different effects of the SBAE program and each type of parental engagement. The SBAE program itself has no effects on enrollment and increases absenteeism in the midline and endline surveys by 26.2% (an increase of 0.18 from 0.69 out of 5 days) and 31.2% (an increase of 0.2 from 0.65 out of 5 days). Increases in students' enrollment are driven by the program promotional session treatment, but not the annual farmer field day — with a marginal effect of 5 percentage points by the endline (from 67% in control schools). Both the program promotional session and the annual farmer field day contribute to a reversal of the SBAE-induced increase in absenteeism — while some marginal effects are not statistically significant, the joint estimates strongly reject the null that applying both types of parental engagement has no effects on absenteeism relative to SBAE-only schools.

The bottom half of Table 4 presents the combined effects of treatments with Lee bounds. These estimates provide three messages. First, the Lee lower bounds re-affirm us that the SBAE program on its own increases student absenteeism. Second, including both types of parental engagement in the SBAE program leads to an increase in enrollment by 3.8 to 5 percentage points — alternatively, a reduction in dropouts by 3.8 to 5 percentage points (from 22% in midline and 33% in endline).[21] Third, including both types of parental engagement is sufficient to overturn the pure SBAE effect on absenteeism — reducing absenteeism by 18% in the midline (a drop of 0.12 from 0.69 out of 5 days) and by 27.8% (a drop of 0.18 from 0.65 out of 5 days).

The Lee bounds in general confirm the direction of effects for the augmented program including both types of parental engagement. However, in some cases (e.g. midline enrollment and midline absenteeism), precision is not preserved. Columns (9) and (10) use a single imputed measure of school attendance to account for both the effects on students' school enrollment and absenteeism. This measure counts the total number of

[21] There is a drop in the magnitude and precision of point estimates when moving from "SBAE + Session" to "SBAE + Session + FFD". This is caused by a negative but imprecise effect of the annual farmer field day. In view of this, I take the effect of "SBAE + Session + FFD" (3.8 percentage points in the midline and 3.9 percentage points in the endline) to be the lower bound and the effect of "SBAE + Session" (4.5 percentage points in the midline and 5 percentage points in the endline) to be the upper bound of the true effect of the augmented SBAE program.

school days students attend in the school year 2021/22 (year of the midline) to 2022/23 (year of the endline) with three imputation rules: first, days of absence are subtracted from a school year of 200 school days, assuming that the rate of absenteeism is constant throughout the year; second, students who have dropped out by the midline and endline surveys are counted as having zero school days in the corresponding school year; third, students who have switched schools are counted as having the same days of absence as the average among control school students who respond to surveys. Column (10) shows that both the SBAE program and the annual farmer field day have statistically insignificant effects on overall school attendance, whereas the program promotional (which reduces both dropouts and absenteeism) increases overall school attendance by 19.2 days per student. The combined estimate of the SBAE program with both types of parental engagement indicates an increase in overall school attendance by 17.6 days per student (from 229.1 days in control schools). I will use this number in cost-benefit and cost-effectiveness analyses in Section 6.

Table 5 presents results on students' time use, with treatment effects on studying hours that mirroring those on absenteeism. Columns (1) and (5) indicate close to zero and statistically insignificant effects on students' studying hours when all SBAE schools are considered together. Columns (2) and (6) indicate that the SBAE program and parental engagement have opposite effects on students' studying hours. The SBAE program on its own reduces students' weekly studying hours by 10.6% (a drop of 0.6 hours from 5.69 hours per week) in the midline and by 19.9% (a drop of 1.07 hours from 5.39 hours per week) — these effects are consistent with the drop in students' studying hours in the first follow-up survey. By contrast, the program promotional session increases students' weekly studying hours by 12.5% (an increase of 0.71 hours from 5.69 hours per week) in the midline and by 20.8% (an increase of 1.12 hours from 5.39 hours per week). Likewise, the annual farmer field day increases students' weekly studying hours by 5.4% (an increase of 0.31 hours from 5.69 hours per week) in the midline and by 20.8% (an increase of 1.12 hours from 5.39 hours per week).

The bottom half of Table 5 presents the combined effects of treatments with Lee bounds. These estimates provide two messages. First, the Lee upper bounds re-affirm us that the SBAE program on its own reduces students' studying hours. Second, including both types of parental engagement is sufficient to overturn the pure SBAE effect — increasing students' studying hours by 21.7% (an increase of 1.17 hours from 5.39 hours per week) in the endline.

To complete the description of students' time use, columns (3), (4), (7), and (8) of Table 5 provide estimates on students' weekly hours spent on farming (inclusive of hours spent on school farms). These results suggest that when including both types of parental engagement in the SBAE program, students' farming hours do not increase significantly relative to control schools — rebutting concerns that SBAE might increase child labor.

4.3.3. Students' livelihoods and aspirations

Table 6 presents results on students' farm sales, non-farm income, total income, and savings in 2022, retrospectively reported in the endline survey. For each outcome, I estimate effects at the extensive margin (whether students have positive values), the intensive margin (log values among students with positive outcomes), and overall percentage changes using Poisson regression. All monetary values are expressed in 2022 Liberian dollars and winsorized at the 2.5 and 97.5 percentiles.

Considering all SBAE schools together, the program increases farm sales primarily at the extensive margin—6

percentage points more students report positive farm sales (from a control mean of 70%)—with an overall increase of 11% in the Poisson specification (from 3407 Liberian dollars). Conversely, the program reduces non-farm income: 5 percentage points fewer students engage in non-farm work (from a control mean of 47%), though the overall effect on non-farm income is not statistically significant in the Poisson specification. These offsetting changes result in modest and statistically insignificant effects on total income. However, the program increases savings both at the extensive margin (5 percentage points more students report positive savings, from a control mean of 55%) and overall, though the Poisson estimate is imprecise.

Parental engagement interventions substantially amplify these effects. The program promotional session combined with SBAE increases farm sales by 12 percentage points at the extensive margin and by 14% overall (Poisson specification), while reducing non-farm income by 11 percentage points at the extensive margin and by 27% overall. The farmer field day alone shows no significant effects on income sources, consistent with its limited impact on students' household farm management (discussed in Section 5). When both engagement interventions are combined with SBAE, the effects on savings become substantial: 11 percentage points more students report positive savings, with an overall increase of 21% (from a baseline of approximately $20 annually).

While students' increases in farm sales tend to be offset by decreases in non-farm income, our measurement of total income likely understates actual gains from the program, since many students work as assistants to household members and do not directly sell farm produce. Savings provide a better measure of students' economic gains in this context, as they reflect transfers from household members in addition to direct earnings. The combined parental engagement interventions significantly increase savings beyond the SBAE-only effect (p=0.04 for the joint test). Table **??** confirms that this increase in savings stems from higher income rather than reduced consumption—the SBAE program has close to zero and statistically insignificant effects on students' spending and meal consumption, with or without parental engagement.

Turning to the last category of primary outcomes, Table 7 shows results on students' aspirations in the midline and endline surveys. Columns (1), (3), and (5) indicate that considering all SBAE schools together, the program has positive effects on whether students study agriculture as an elective subject, whether students name agricultural jobs in their top three career choices (without being given options), and whether students report having saved for attending university. Column (2) indicates that parental engagement plays a pivotal role in increasing students' study of agriculture, leading to an aggregate increase of 18.4 percentage points (from 50% in control schools) when both types of parental engagement are included. Column (4) indicates that SBAE alone increases students' naming of agricultural careers by 7.8 percentage points (from 43% in control schools), while including both types of parental engagement intensifies this effect to 11.5 percentage points over control schools. Column (6) shows that both the SBAE program and the program promotional session have no significant effects on students' saving for university, whereas the annual farmer field day increases this by 7.5 percentage points (from 17% in control schools).

4.3.4. Summary and robustness to multiple hypothesis testing

In summary, Tables 3 to 7 all suggest the pivotal role of parental engagement in SBAE's effectiveness. SBAE, on its own, (i) fails to increase technology adoption by students and parents; (ii) increases students' absenteeism and reduces their studying hours; (iii) fails to improve students' livelihoods and aspirations

(with the exception of students' naming agricultural careers). By contrast, SBAE augmented with both types of parental engagement is sufficient to achieve all desired goals: (i) increasing parents' adoption of promoted techniques by 0.38 standard deviations; (ii) reducing dropouts by 3.8 to 5 percentage points and absenteeism by 27.8%; (iii) increasing students' weekly studying hours by 21.7%; (iv) increasing students' annual savings by 92% without sacrificing their spending and meals consumption; and (v) increasing the fraction of students who find an agricultural career favorable and save for attending university.

The program promotional session, which is experimentally designed to reach every parent, plays an especially important role in parental engagement. This can be seen in several ways: (i) both the promotional session and the annual farmer field day increase technology adoption on the student parcel in the endline, but only the former increases technology adoption on the parent parcel and students' farm sales; (ii) only the promotional session reduces dropouts and increases students' studying of agriculture as an elective subject.

While the annual farmer field day, which is organized at the school level, has quantitatively smaller effects on technology adoption and school attendance, it is nonetheless an important supplement in making SBAE complete. This can be seen in several ways: (i) it speeds up students' adoption of technologies; (ii) it is crucial in ensuring the overturn of SBAE effects on absenteeism and studying hours; (iii) it ensures that students' total farming hours do not increase; (iv) it is crucial in ensuring positive effects on students' savings; (iv) only it increases students who save for attending university.

To economize on space, Tables 3 to 7 have not shown robustness checks with respect to multiple hypothesis testing. Figure 4 addresses this issue by plotting the naive p-values and adjusted q-values based on the procedure of Anderson (2008), which controls the false discovery rate across multiple tests. To be specific, I conduct two types of adjustments: (i) pooling all 18 outcomes together and adjusting for the four linear combination coefficients tested (SBAE, SBAE + Session, SBAE + FFD, and SBAE + Session + FFD) — thus 72 tests in total, disregarding potential correlations between outcomes, such as between students' and parents' adoption; (ii) pooling only outcomes within pre-specified categories: testing 6 outcomes for 4 coefficients under "technology adoption"; testing 3 outcomes for 4 coefficients under "school attendance and learning"; testing 2 outcomes for 4 coefficients under "income and savings"; testing 4 outcomes for 4 coefficients under "spending and food security"; and testing 3 outcomes for 4 coefficients under "aspirations".

Figure 4 presents the robustness of ten main results for the augmented SBAE program with program promotional session and the annual farmer field day. Eight out of ten results have p-values below 0.01 before correction, and adjusted q-values remain below 0.05. For the two results with p-values around 0.05 (total days of attendance and farm sales), the adjusted q-values are around 0.1. Hence my main results are, on the whole, robust to multiple hypothesis testing.

Table 3: Midline and endline intent-to-treat effects on agricultural technology adoption

	Panel A: `Student parcel`							
	Midline		Midline		Endline		Endline	
	Bed preparation (0/1)		# techniques		Bed preparation (0/1)		# techniques	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$SBAE_p$	0.066*	-0.013	0.143*	-0.100	0.107***	0.011	0.255***	0.086
	(0.034)	(0.050)	(0.080)	(0.141)	(0.030)	(0.046)	(0.054)	(0.083)
$SBAE_p \times FFD_s$		0.055		0.205*		0.066		0.128
		(0.046)		(0.119)		(0.048)		(0.085)
$SBAE_p \times Session_s$		0.108**		0.280*		0.104**		0.175**
		(0.042)		(0.143)		(0.044)		(0.082)
p-value: $(\beta_2 + \beta_3 = 0)$		0.02		0.04		0.01		0.01
Summary: Combined effects of SBAE and informational treatments								
SBAE only (β_1)	0.066*	-0.013	0.143*	-0.100	0.107***	0.011	0.255***	0.086
	(0.034)	(0.050)	(0.080)	(0.141)	(0.030)	(0.046)	(0.054)	(0.083)
Lee bounds	[0.057*	[-0.025	[0.097	[-0.146	[0.107***	[0.002	[0.254***	[0.071
	(0.034),	(0.050),	(0.080),	(0.140),	(0.030),	(0.047),	(0.054),	(0.083),
	0.076**	-0.001	0.172**	-0.043	0.108***	0.028	0.257***	0.139*
	(0.034)]	(0.949)]	(0.080)]	(0.759)]	(0.030)]	(0.046)]	(0.054)]	(0.079)]
SBAE + FFD ($\beta_1 + \beta_2$)		0.042		0.105		0.077**		0.214***
		(0.043)		(0.093)		(0.037)		(0.069)
Lee bounds		[0.028		[0.016		[0.066*		[0.186**
		(0.043),		(0.098),		(0.038),		(0.069),
		0.057		0.150		0.100**		0.288***
		(0.043)]		(0.095)]		(0.037)]		(0.062)]
SBAE + Session ($\beta_1 + \beta_3$)		0.095**		0.180*		0.115**		0.261***
		(0.041)		(0.104)		(0.044)		(0.082)
Lee bounds		[0.082*		[0.117		[0.109**		[0.240***
		(0.042),		(0.101),		(0.045),		(0.082),
		0.109**		0.216**		0.119**		0.271***
		(0.042)]		(0.104)]		(0.045)]		(0.081)]
SBAE + Session + FFD ($\beta_1 + \beta_2 + \beta_3$)		0.150***		0.386***		0.182***		0.388***
		(0.045)		(0.135)		(0.037)		(0.068)
Lee bounds		[0.131***		[0.304**		[0.159***		[0.300***
		(0.046),		(0.139),		(0.038),		(0.071),
		[0.170***		[0.450***		[0.199***		[0.430***
		(0.045)]		(0.134)]		(0.038)]		(0.069)]
Summary Statistics for Control Schools								
Control Mean	0.45	0.45	1.43	1.43	0.29	0.29	0.73	0.73
Control SD	0.50	0.50	1.23	1.23	0.45	0.45	0.85	0.85
Observations	1384	1384	1384	1384	1488	1488	1488	1488
Number of clusters	155	155	155	155	155	155	155	155

Notes: This table reports treatment effects on the adoption of the promoted techniques on the designated `student parcel` (midline and endline; Panel A) and the designated `student parcel` (endline; Panel B). The top portion of each panel reports the estimates under each specification; the second portion then reports the combined effects of the SBAE program and informational treatments. These effects are computed using linear combinations of estimates. All regression specifications include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 3: Midline and endline intent-to-treat effects on agricultural technology adoption *(continued)*

	Panel B: `Parent parcel`			
	Endline		Endline	
	Bed preparation (0/1)		# techniques	
	(9)	(10)	(11)	(12)
$SBAE_p$	0.058**	0.001	0.294***	0.200**
	(0.024)	(0.037)	(0.060)	(0.095)
$SBAE_p \times FFD_s$		0.020		-0.080
		(0.037)		(0.096)
$SBAE_p \times Session_s$		0.088**		0.242**
		(0.037)		(0.097)
p-value: $(\beta_2 + \beta_3 = 0)$		0.07		0.30
Summary: Combined effects of SBAE and informational treatments				
SBAE only (β_1)	0.058**	0.001	0.294***	0.200**
	(0.024)	(0.037)	(0.060)	(0.095)
Lee bounds	[0.056**	[-0.001	[0.274***	[0.193**
	(0.024),	(0.037),	(0.060),	(0.095),
	0.067**	0.001	0.315***	0.207**
	(0.024)]	(0.038)]	(0.060)]	(0.095)]
SBAE + FFD ($\beta_1 + \beta_2$)		0.021		0.120
		(0.034)		(0.084)
Lee bounds		[0.006		[0.102
		(0.034),		(0.084),
		0.024		0.180**
		(0.034)]		(0.082)]
SBAE + Session ($\beta_1 + \beta_3$)		0.089***		0.442***
		(0.031)		(0.078)
Lee bounds		[0.087**		[0.416***
		(0.031),		(0.077),
		0.100***		0.465***
		(0.031)]		(0.076)]
SBAE + Session + FFD ($\beta_1 + \beta_2 + \beta_3$)		0.109***		0.362***
		(0.038)		(0.101)
Lee bounds		[0.104**		[0.319***
		(0.038),		(0.099),
		[0.127***		[0.408***
		(0.039)]		(0.106)]
Summary Statistics for Control Schools				
Control Mean	0.74	0.74	1.38	1.38
Control SD	0.44	0.44	0.96	0.96
Observations	1363	1363	1363	1363
Number of clusters	155	155	155	155

Notes: This table reports treatment effects on the adoption of the promoted techniques on the designated `student parcel` (midline and endline; Panel A) and the designated `student parcel` (endline; Panel B). The top portion of each panel reports the estimates under each specification; the second portion then reports the combined effects of the SBAE program and informational treatments. These effects are computed using linear combinations of estimates. All regression specifications include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 4: Midline and endline intent-to-treat effects on school attendance

	Midline Enrolled in school (0/1)		Midline Days absent (0-5)		Endline Enrolled in school (0/1)		Endline Days absent (0-5)		Imputed Total days in attendance	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$SBAE_p$	0.023*	0.008	0.014	0.181*	0.018	-0.001	-0.004	0.203**	6.734	-3.572
	(0.014)	(0.023)	(0.060)	(0.092)	(0.014)	(0.028)	(0.050)	(0.081)	(4.499)	(7.964)
$SBAE_p \times FFD_s$		-0.007		-0.162		-0.011		-0.275**		1.957
		(0.023)		(0.099)		(0.031)		(0.107)		(9.505)
$SBAE_p \times Session_s$		0.037		-0.143		0.050*		-0.109		19.198**
		(0.025)		(0.103)		(0.028)		(0.121)		(9.075)
p-value: $(\beta_2 + \beta_3 = 0)$		0.46		0.06		0.41		0.01		0.14
Summary: Combined effects of SBAE and informational treatments										
SBAE only (β_1)	0.023*	0.008	0.014	0.181*	0.018	-0.001	-0.004	0.203**	6.734	-3.572
	(0.014)	(0.023)	(0.060)	(0.092)	(0.014)	(0.028)	(0.050)	(0.081)	(4.499)	(7.964)
Lee bounds	[0.018	[-0.010	[-0.089	[0.159*	[0.018	[-0.019	[-0.005	[0.191**	[6.602	[-8.998
	(0.014),	(0.023),	(0.054),	(0.091),	(0.014),	(0.027),	(0.050),	(0.082),	(4.503),	(7.951)
	0.040***	0.013	0.024	0.222***	0.018	0.008	0.000	0.280***	6.820	0.732
	(0.013)]	(0.023)]	(0.061)]	(0.073)]	(0.014)]	(0.028)]	(0.050)]	(0.078)]	(4.496)]	(7.876)]
SBAE + FFD ($\beta_1 + \beta_2$)		0.001		0.018		-0.011		-0.072		-1.615
		(0.019)		(0.100)		(0.020)		(0.116)		(7.296)
Lee bounds		[-0.005		[-0.105		[-0.036*		[-0.095		[-9.725
		(0.020),		(0.078),		(0.021),		(0.118),		(7.273),
		0.025		0.040		-0.000		-0.053		5.458
		(0.019)]		(0.102)]		(0.021)]		(0.116)]		(7.505)]
SBAE + Session ($\beta_1 + \beta_3$)		0.045**		0.038		0.050**		0.094		15.626**
		(0.019)		(0.073)		(0.024)		(0.089)		(7.571)
Lee bounds		[0.040**		[-0.054		[0.047*		[0.060		[14.306*
		(0.019),		(0.066),		(0.024),		(0.083),		(7.562),
		0.065***		0.047		0.055**		0.100		17.217**
		(0.019)]		(0.076)]		(0.024)]		(0.091)]		(7.575)]
SBAE + Session + FFD ($\beta_1 + \beta_2 + \beta_3$)		0.038		-0.124		0.039		-0.181**		17.583**
		(0.026)		(0.109)		(0.028)		(0.088)		(8.776)
Lee bounds		[0.032		[-0.221**		[0.027		[-0.344***		[11.694
		(0.026),		(0.106),		(0.028),		(0.074),		(8.731),
		0.067***		-0.103		0.068**		-0.159*		26.611***
		(0.025)]		(0.108)]		(0.028)]		(0.090)]		(8.768)]
Summary Statistics for Control Schools										
Control Mean	0.78	0.78	0.69	0.69	0.67	0.67	0.65	0.65	229.09	229.09
Control SD	0.42	0.42	1.13	1.13	0.47	0.47	1.25	1.25	144.02	144.02
Observations	3728	3728	2418	2418	3115	3115	1970	1970	3728	3728
Number of clusters	155	155	155	155	155	155	155	155	155	155

Notes: This table reports treatment effects on students' school attendance in the midline and endline surveys. The top portion of each panel reports the estimates under each specification; the second portion then reports the combined effects of the SBAE program and informational treatments. These effects are computed using linear combinations of estimates. All regression specifications include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 5: Midline and endline intent-to-treat effects on time use

	Midline Hours studying (weekly)		Midline Hours farming (weekly)		Endline Hours studying (weekly)		Endline Hours farming (weekly)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$SBAE_p$	-0.060	-0.602**	0.395	0.555	0.117	-1.070***	0.155	0.407
	(0.166)	(0.262)	(0.311)	(0.609)	(0.218)	(0.373)	(0.488)	(0.772)
$SBAE_p \times FFD_s$		0.306		-0.827		1.119***		-0.823
		(0.301)		(0.644)		(0.367)		(0.707)
$SBAE_p \times Session_s$		0.706***		0.658		1.117**		0.455
		(0.260)		(0.567)		(0.432)		(0.674)
p-value: $(\beta_2 + \beta_3 = 0)$		0.01		0.86		0.00		0.72
Summary: Combined effects of SBAE and informational treatments								
SBAE only (β_1)	-0.060	-0.602**	0.395	0.555	0.117	-1.070***	0.155	0.407
	(0.166)	(0.262)	(0.311)	(0.609)	(0.218)	(0.373)	(0.488)	(0.772)
Lee bounds	[-0.393**	[-0.757***	[-0.120	[0.356	[0.112	[-1.194***	[0.145	[0.252
	(0.163),	(0.258),	(0.300),	(0.607),	(0.217),	(0.368),	(0.488),	(0.767),
	0.067	-0.226	0.565*	0.821	0.148	-0.758**	0.240	0.765
	(0.165)]	(0.228)]	(0.310)]	(0.617)]	(0.218)]	(0.353)]	(0.488)]	(0.768)]
SBAE + FFD $(\beta_1 + \beta_2)$		-0.296		-0.273		0.050		-0.417
		(0.229)		(0.505)		(0.423)		(0.731)
Lee bounds		[-0.666***		[-0.868*		[-0.136		[-0.700
		(0.218),		(0.470),		(0.414),		(0.728),
		-0.121		-0.048		0.222		-0.223
		(0.236)]		(0.521)]		(0.427)]		(0.737)]
SBAE + Session $(\beta_1 + \beta_3)$		0.103		1.212***		0.047		0.862
		(0.273)		(0.439)		(0.310)		(0.602)
Lee bounds		[-0.299		[0.626		[-0.108		[0.637
		(0.243),		(0.406),		(0.295),		(0.600),
		0.245		1.439***		0.098		0.956
		(0.276)]		(0.436)]		(0.306)]		(0.604)]
SBAE + Session + FFD $(\beta_1 + \beta_2 + \beta_3)$		0.409		0.385		1.167***		0.038
		(0.255)		(0.523)		(0.319)		(0.638)
Lee bounds		[-0.042		[-0.332		[0.603**		[-0.777
		(0.265),		(0.551),		(0.265),		(0.600),
		[0.650**		[0.747		[1.414***		[0.379
		(0.250)]		(0.509)]		(0.318)]		(0.642)]
Summary Statistics for Control Schools								
Control Mean	5.69	5.69	9.46	9.46	5.39	5.39	9.94	9.94
Control SD	4.33	4.33	7.26	7.26	4.15	4.15	8.12	8.12
Observations	2425	2425	2425	2425	1843	1843	1646	1646
Number of clusters	155	155	155	155	155	155	155	155

Notes: This table reports intent-to-treat effects on students' time use in the midline and endline surveys. The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. Again, these effects are computed using linear combinations of estimates. The reference regression specifications include pair-of-clusters fixed effects and standard errors are clustered at the cluster-of-schools level. The main regression specification includes school-grade section-gender fixed effects and robust standard errors are presented. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 6: Endline effects on students' income and savings

	Y=Farm sales (Liberian dollars)					
	OLS extensive margin: Pr(Y>0)		OLS intensive margin: Log(Y)|Y>0		Poisson regression: Proportion change in Y	
	(1)	(2)	(3)	(4)	(5)	(6)
$SBAE_p$	0.060***	0.039	0.027	-0.009	0.106*	0.092
	(0.021)	(0.043)	(0.048)	(0.084)	(0.061)	(0.113)
$SBAE_p \times FFD_s$		-0.027		0.054		-0.000
		(0.056)		(0.100)		(0.133)
$SBAE_p \times Session_s$		0.084*		0.034		0.040
		(0.043)		(0.109)		(0.124)
p-value: $(\beta_2 + \beta_3 = 0)$		0.45		0.56		0.82
Summary: Combined effects of SBAE and informational treatments						
SBAE only (β_1)	0.060***	0.039	0.027	-0.009	0.106*	0.092
	(0.021)	(0.043)	(0.048)	(0.084)	(0.061)	(0.113)
Lee bounds	[0.052**	[0.012	[0.021	[-0.089	[0.104*	[0.061
	(0.021),	(0.043),	(0.049),	(0.086),	(0.061),	(0.111),
	0.053***	0.036	0.029	0.019	0.111*	0.219*
	(0.020)]	(0.044)]	(0.048)]	(0.084)]	(0.061)]	(0.125)]
SBAE + FFD ($\beta_1 + \beta_2$)		0.012		0.045		0.091
		(0.039)		(0.091)		(0.120)
Lee bounds		[-0.018		[-0.074		[0.054
		(0.040),		(0.094),		(0.115),
		0.034		0.136*		0.123
		(0.041)]		(0.079)]		(0.123)]
SBAE + Session ($\beta_1 + \beta_3$)		0.124***		0.025		0.135
		(0.039)		(0.081)		(0.115)
Lee bounds		[0.097***		[0.013		[0.102
		(0.034),		(0.080),		(0.108),
		0.106***		0.039		0.146
		(0.034)]		(0.082)]		(0.116)]
SBAE + Session + FFD ($\beta_1 + \beta_2 + \beta_3$)		0.096**		0.079		0.135
		(0.044)		(0.094)		(0.117)
Lee bounds		[0.073*		[0.010		[-0.044
		(0.044),		(0.092),		(0.094),
		0.119***		0.163*		0.188*
		(0.040)]		(0.088)]		(0.120)]
Summary Statistics for Control Schools						
Control Mean (without logs)	0.702	0.702	4692	4692	3407	3407
Control SD (without logs)	0.458	0.458	4664	4664	4491	4491
Observations	1488	1488	1118	1118	1488	1488
Number of clusters	155	155	152	152	155	155

Notes: This table reports intent-to-treat effects on students' income and savings in 2022, retrospectively reported in the endline survey. The top portion of each panel reports the estimates under each specification; the second portion then reports the combined effects of the SBAE program and informational treatments. These effects are computed using linear combinations of estimates. All regression specifications include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 6: Endline effects on students' income and savings *(continued)*

	Y=Non-farm income (Liberian dollars)					
	OLS extensive margin: Pr(Y>0)		OLS intensive margin: Log(Y)|Y>0		Poisson regression: Proportion change in Y	
	(7)	(8)	(9)	(10)	(11)	(12)
$SBAE_p$	-0.052**	-0.061*	-0.016	-0.067	-0.087	-0.101
	(0.024)	(0.036)	(0.087)	(0.155)	(0.074)	(0.139)
$SBAE_p \times FFD_s$		0.067		0.262		0.299
		(0.049)		(0.159)		(0.238)
$SBAE_p \times Session_s$		-0.051		-0.118		-0.185
		(0.040)		(0.150)		(0.104)
p-value: $(\beta_2 + \beta_3 = 0)$		0.82		0.59		0.82
Summary: Combined effects of SBAE and informational treatments						
SBAE only (β_1)	-0.052**	-0.061*	-0.016	-0.067	-0.087	-0.101
	(0.024)	(0.036)	(0.087)	(0.155)	(0.074)	(0.139)
Lee bounds	[-0.053**	[-0.070*	[-0.031	[-0.064	[-0.089	[-0.118
	(0.025),	(0.038),	(0.088),	(0.156),	(0.074),	(0.137),
	-0.052**	-0.050	-0.011	-0.017	-0.082	0.086
	(0.025)]	(0.038)]	(0.087)]	(0.150)]	(0.075)]	(0.165)]
SBAE + FFD ($\beta_1 + \beta_2$)		0.006		0.195**		0.168
		(0.038)		(0.096)		(0.139)
Lee bounds		[-0.011		[0.105		[0.125
		(0.041),		(0.098),		(0.134),
		0.020		0.263**		0.208
		(0.041)]		(0.104)]		(0.145)]
SBAE + Session ($\beta_1 + \beta_3$)		-0.112***		-0.185		-0.267**
		(0.034)		(0.128)		(0.091)
Lee bounds		[-0.118***		[-0.201		[-0.321***
		(0.035),		(0.127),		(0.083),
		-0.109***		-0.100		-0.261**
		(0.035)]		(0.104)]		(0.092)]
SBAE + Session + FFD ($\beta_1 + \beta_2 + \beta_3$)		-0.045		0.077		-0.048
		(0.045)		(0.159)		(0.135)
Lee bounds		[-0.075		[0.009		[-0.301**
		(0.048),		(0.154),		(0.098),
		-0.030		0.152		-0.001
		(0.047)]		(0.151)]		(0.142)]
	Summary Statistics for Control Schools					
Control Mean (without logs)	0.465	0.465	4009	4009	1860	1860
Control SD (without logs)	0.499	0.499	3818	3818	3272	3272
Observations	1488	1488	638	638	1488	1488
Number of clusters	155	155	137	137	155	155

Notes: This table reports intent-to-treat effects on students' income and savings in 2022, retrospectively reported in the endline survey. The top portion of each panel reports the estimates under each specification; the second portion then reports the combined effects of the SBAE program and informational treatments. These effects are computed using linear combinations of estimates. All regression specifications include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 6: Endline effects on students' income and savings *(continued)*

	Y=Total income (Liberian dollars)					
	OLS extensive margin: Pr(Y>0)		OLS intensive margin: Log(Y)|Y>0		Poisson regression: Proportion change in Y	
	(13)	(14)	(15)	(16)	(17)	(18)
$SBAE_p$	0.020	0.032	-0.052	-0.102	0.032	0.014
	(0.017)	(0.038)	(0.049)	(0.091)	(0.054)	(0.110)
$SBAE_p \times FFD_s$		-0.018		0.105		0.096
		(0.052)		(0.115)		(0.157)
$SBAE_p \times Session_s$		-0.000		0.019		-0.039
		(0.037)		(0.093)		(0.097)
p-value: $(\beta_2 + \beta_3 = 0)$		0.79		0.43		0.78
Summary: Combined effects of SBAE and informational treatments						
SBAE only (β_1)	0.020	0.032	-0.052	-0.102	0.032	0.014
	(0.017)	(0.038)	(0.049)	(0.091)	(0.054)	(0.110)
Lee bounds	[0.015	[0.005	[-0.057	[-0.183*	[0.029	[-0.012
	(0.015),	(0.035),	(0.049),	(0.093),	(0.054),	(0.108),
	0.017	0.029	-0.049	-0.057	0.037	0.107
	(0.015)]	(0.035)]	(0.049)]	(0.090)]	(0.054)]	(0.119)]
SBAE + FFD ($\beta_1 + \beta_2$)		0.014		0.003		0.111
		(0.032)		(0.086)		(0.112)
Lee bounds		[-0.029		[-0.111		[0.060
		(0.033),		(0.085),		(0.105),
		0.041		0.079		0.149
		(0.031)]		(0.074)]		(0.114)]
SBAE + Session ($\beta_1 + \beta_3$)		0.032		-0.083		-0.026
		(0.034)		(0.080)		(0.093)
Lee bounds		[0.022		[-0.099		[-0.063
		(0.029),		(0.079),		(0.088),
		0.030		-0.063		-0.015
		(0.030)]		(0.080)]		(0.094)]
SBAE + Session + FFD ($\beta_1 + \beta_2 + \beta_3$)		0.013		0.022		0.068
		(0.039)		(0.091)		(0.109)
Lee bounds		[0.005		[-0.055		[-0.077
		(0.037),		(0.087),		(0.094),
		0.054*		0.136*		0.120
		(0.030)]		(0.082)]		(0.108)]
	Summary Statistics for Control Schools					
Control Mean (without logs)	0.832	0.832	6280	6280	5367	5367
Control SD (without logs)	0.374	0.374	6341	6341	6266	6266
Observations	1488	1488	1277	1277	1488	1488
Number of clusters	155	155	153	153	155	155

Notes: This table reports intent-to-treat effects on students' income and savings in 2022, retrospectively reported in the endline survey. The top portion of each panel reports the estimates under each specification; the second portion then reports the combined effects of the SBAE program and informational treatments. These effects are computed using linear combinations of estimates. All regression specifications include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 6: Endline effects on students' income and savings *(continued)*

	Y=Total savings (Liberian dollars)					
	OLS extensive margin: Pr(Y>0)		OLS intensive margin: Log(Y)|Y>0		Poisson regression: Proportion change in Y	
	(19)	(20)	(21)	(22)	(23)	(24)
$SBAE_p$	0.050**	-0.009	-0.067	-0.108	-0.005	-0.181
	(0.022)	(0.040)	(0.050)	(0.103)	(0.066)	(0.099)
$SBAE_p \times FFD_s$		0.056		0.062		0.226*
		(0.050)		(0.091)		(0.139)
$SBAE_p \times Session_s$		0.061		0.030		0.203
		(0.056)		(0.103)		(0.148)
p-value: $(\beta_2 + \beta_3 = 0)$		0.07		0.58		0.04
Summary: Combined effects of SBAE and informational treatments						
SBAE only (β_1)	0.050**	-0.009	-0.067	-0.108	-0.005	-0.181
	(0.022)	(0.040)	(0.050)	(0.103)	(0.066)	(0.099)
Lee bounds	[0.050**	[-0.024	[-0.073	[-0.207**	[-0.006	[-0.199*
	(0.023),	(0.042),	(0.050),	(0.102),	(0.066),	(0.097),
	0.051**	-0.003	-0.063	-0.069	0.002	-0.068
	(0.023)]	(0.042)]	(0.050)]	(0.103)]	(0.066)]	(0.111)]
SBAE + FFD ($\beta_1 + \beta_2$)		0.046		-0.045		0.005
		(0.046)		(0.073)		(0.097)
Lee bounds		[0.025		[-0.141**		[-0.032
		(0.048),		(0.071),		(0.093),
		0.067		0.044		0.038
		(0.047)]		(0.069)]		(0.101)]
SBAE + Session ($\beta_1 + \beta_3$)		0.052		-0.078		-0.014
		(0.049)		(0.070)		(0.097)
Lee bounds		[0.050		[-0.089		[-0.039
		(0.050),		(0.071),		(0.092),
		0.059		-0.058		-0.001
		(0.050)]		(0.070)]		(0.097)]
SBAE + Session + FFD ($\beta_1 + \beta_2 + \beta_3$)		0.107***		-0.016		0.209*
		(0.038)		(0.091)		(0.130)
Lee bounds		[0.096**		[-0.065		[0.021
		(0.041),		(0.090),		(0.107),
		0.139***		0.046		0.270**
		(0.041)]		(0.085)]		(0.135)]
	Summary Statistics for Control Schools					
Control Mean (without logs)	0.552	0.552	5338	5338	2952	2952
Control SD (without logs)	0.498	0.498	4442	4442	4237	4237
Observations	1488	1488	833	833	1488	1488
Number of clusters	155	155	148	148	155	155

Notes: This table reports intent-to-treat effects on students' income and savings in 2022, retrospectively reported in the endline survey. The top portion of each panel reports the estimates under each specification; the second portion then reports the combined effects of the SBAE program and informational treatments. These effects are computed using linear combinations of estimates. All regression specifications include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 7: Midline and endline intent-to-treat effects on students' aspirations

	Midline Study agriculture as elective (0/1)		Midline Name ag jobs as top 3 (0/1)		Endline Save for university (0/1)	
	(1)	(2)	(3)	(4)	(5)	(6)
$SBAE_p$	0.101**	0.013	0.089***	0.078*	0.025*	-0.016
	(0.042)	(0.071)	(0.020)	(0.040)	(0.014)	(0.021)
$SBAE_p \times FFD_s$		0.055		-0.027		0.075**
		(0.058)		(0.037)		(0.029)
$SBAE_p \times Session_s$		0.117*		0.064		0.005
		(0.065)		(0.042)		(0.026)
p-value: $(\beta_2 + \beta_3 = 0)$		0.07		0.59		0.03
Summary: Combined effects of SBAE and informational treatments						
SBAE only (β_1)	0.101**	0.013	0.089***	0.078*	0.025*	-0.016
	(0.042)	(0.071)	(0.020)	(0.040)	(0.014)	(0.021)
Lee bounds	[0.094**	[0.003	[0.078***	[0.067*	[0.025*	[-0.020
	(0.042),	(0.071),	(0.020),	(0.040),	(0.014),	(0.022),
	0.112***	0.023	0.100***	0.088**	0.026*	0.004
	(0.041)]	(0.071)]	(0.020)]	(0.040)]	(0.014)]	(0.021)]
SBAE + FFD ($\beta_1 + \beta_2$)		0.067		0.051*		0.059**
		(0.064)		(0.028)		(0.025)
Lee bounds		[0.057		[0.036		[0.054**
		(0.064),		(0.028),		(0.025),
		0.081		0.067**		0.066***
		(0.063)]		(0.029)]		(0.025)]
SBAE + Session ($\beta_1 + \beta_3$)		0.129**		0.142***		-0.011
		(0.054)		(0.026)		(0.025)
Lee bounds		[0.121**		[0.130***		[-0.017
		(0.054),		(0.027),		(0.024),
		0.141***		0.157***		-0.010
		(0.053)]		(0.026)]		(0.025)]
SBAE + Session + FFD ($\beta_1 + \beta_2 + \beta_3$)		0.184***		0.115***		0.064***
		(0.055)		(0.038)		(0.024)
Lee bounds		[0.173***		[0.099**		[0.033
		(0.055),		(0.039),		(0.024),
		[0.203***		[0.134***		[0.074***
		(0.055)]		(0.038)]		(0.024)]
Summary Statistics for Control Schools						
Control Mean	0.50	0.50	0.43	0.43	0.17	0.17
Control SD	0.50	0.50	0.50	0.50	0.38	0.38
Observations	2343	2343	2425	2425	1970	1970
Number of clusters	155	155	155	155	155	155

Notes: This table reports intent-to-treat effects on students' aspirations in the midline and endline surveys. The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. Again, these effects are computed using linear combinations of estimates. The reference regression specifications include pair-of-clusters fixed effects and standard errors are clustered at the cluster-of-schools level. The main regression specification includes school-grade section-gender fixed effects and robust standard errors are presented. Significance: *** p<0.01, ** p<0.05, * p<0.10.

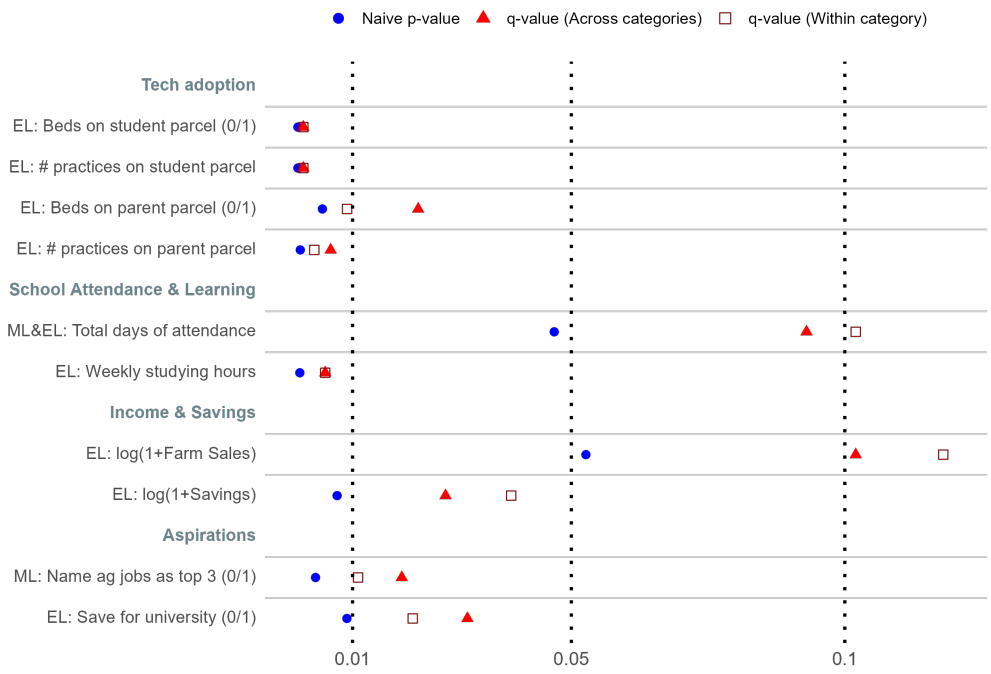


Figure 4: Robustness of augmented SBAE effects to testing multiple treatments and outcomes

Notes: This figure presents multiple-hypothesis-adjusted p-values for the treatment effects of the augmented SBAE program (i.e. summing the coefficients of the SBAE treatment, the program promotional session, and the annual farmer field day treatment). Naive p-values and q-values adjusted for multiple hypothesis correction, using the procedure of Anderson (2008), are presented for 10 main results. I present two types of adjusted q-values: (i) considering all outcomes together (testing 18 outcomes for 4 coefficients, thus 72 tests in total); and (ii) considering only outcomes within a category (testing 6 outcomes for 4 coefficients under "technology adoption"; testing 3 outcomes for 4 coefficients under "school attendance and learning"; testing 2 outcomes for 4 coefficients under "income and savings"; testing 4 outcomes for 4 coefficients under "spending and food security"; and testing 3 outcomes for 4 coefficients under "aspirations").

5. Mechanisms

5.1. What drives the pivotal effects of parental engagement?

5.1.1. Could increased responses to SBAE training explain?

Section 4 provides evidence that the SBAE program only succeeds in achieving its goals when parental engagement treatments are included. This section addresses the question of why. I begin by noting that sampled students have baseline farming experiences but may not have participated in the program. So a natural guess would be: parental engagement works by raising students' responses to training in the SBAE program and thus the ITT effects we observe in Section 4.

Table 8 examines this possibility by estimating midline and endline treatment effects on key intermediate outcomes associated with students' training in the SBAE program. Mirroring results in the first follow-up survey (Section 4.2), Columns (1), (3), (5), (7), (9), and (11) indicate that the overall SBAE effect is positive

as expected — 73.7% and 84.3% of sampled students in SBAE schools had participated in the program by the midline and endline surveys; SBAE school students scored 6.8% and 5.7% higher than control school students on tests of promoted techniques in the midline and endline; 5.8 and 7.9 percentage points more students in SBAE schools believed that they could harvest more cassava tubers per plant than a typical adult farmer in the community.

A closer inspection — in columns (2) and (8) — suggests that the majority of student participation in the program is driven by SBAE alone. Specifically, 67.7% of sampled students in SBAE-only schools had participated by the midline, and the program promotional session added only 13.2% of students to the program. By the endline, 80.9% of sampled students in SBAE-only schools had participated, whereas the program promotional session had no statistically significant effects on program participation. The evidence thus suggests that the program promotional session merely speeds up SBAE participation, and only by a small fraction. For increased SBAE participation to explain the dramatic effects of parental engagement — large differentials on technology adoption and reversed effects on education — we would expect marginal effects of parental engagement on SBAE participation that are large relative to pure SBAE effects.

Of course, parental engagement could have intensified students' absorption of SBAE training, even if participation remains unchanged. However, another closer inspection — in columns (4), (6), (10), and (12) — suggests that SBAE alone drives the improvements in students' agricultural knowledge and beliefs, whereas the parental engagement treatments have no effects. For increased students' absorption of SBAE training to explain the dramatic effects of parental engagement, we would expect that parental engagement treatments at least exert some effects on students' agricultural knowledge and beliefs.

So far we have examined measures that capture SBAE's impact within schools. But an important part of SBAE is for students to become agents of diffusion. Could it be that students only increased their intentions to look after household farms upon parental engagement? However, results in the first follow-up survey (Section 4.2) suggest that SBAE increased students' intentions to look after household farms before parental engagement. Column (14) in Table 8 further examines this from endline parent reports — whether students have asked to look after farms in the last two years. SBAE alone increases this measure by 12.2% (from 50% in control schools), whereas the program promotional session and the annual farmer field day have no statistically significant effects.

To sum up, the evidence suggests that the pivotal effects of parental engagement on primary outcomes cannot be driven by students' program participation, absorption of SBAE training, or intentions to look after household farms — SBAE alone accounts for these changes.

5.1.2. Parental engagement enables intra-household diffusion mechanisms otherwise muted

Having shown that SBAE alone is sufficient to equip students as agents of diffusion, we now turn to another possibility: parental engagement enables intra-household diffusion mechanisms otherwise muted, despite students' willingness to apply promoted techniques. I focus specifically on the dual diffusion mechanisms mentioned in the Introduction: (i) parents' visits to schools; (ii) students' home projects, intra-household communication, and parents' observations.

Parents' visits to schools. Table 9 shows results on parents' visits to schools (Panel A) and related outcomes (Panel B). Columns (1) and (3) indicate that, in line with a larger fraction of SBAE schools

(100%) having school farms than control schools (69%), SBAE overall has increased the fraction of parents who have visited school farms and the number of visits parents make in the second rainy season.

A closer inspection, however, suggests that parents' visits to schools farms are far from mechanical. First, only a small fraction of SBAE-only school parents visit school farms — 43.9%, as opposed to 33% in control schools (column (2)) — whereas including both types of parental engagement into the SBAE program increases this fraction to 65.3%. Second, and most surprisingly, SBAE alone does not increase the number of visits that parents make in the second rainy season (column (4)). Including both types of parental engagement increases this measure by 0.94, which means doubling visits from control schools (0.89 per parent). Third, and most importantly, including both types of parental engagement is crucial in enabling more parents to observe harvests on the school farm, from 37.9% in SBAE-only schools to 61.1% (column (6)).

Besides visits to school farms, Table 9 also examines parents' attendance of parent-teacher meetings, which could be crucial in encouraging parents' monitoring of students' academic studies and school attendance. In line with treatment effects on students' absenteeism and studying hours (Section 4.3.2), columns (8) and (10) indicate a reversal of effects on parents' attendance of parent-teacher meetings. With SBAE only, parents' interest in attendance surprisingly dropps — 10.2% less parents attend meetings (from 36% in control schools) and the number of meetings attended in 2022/23 drop by 38.7% (a drop of 0.35 from 0.91 in control schools). Both the program promotional session and the annual farmer field day contribute to a reversal in attendance — when included together in the SBAE program, 12.2% more parents attend (relative to 36% in control schools) and the number of meetings attended increase by 37%. These results indicate that, in line with potential concerns about SBAE, the program alone could distract parents' attention from students' academics — whereas the inclusion of parental engagement helps to reverse this distraction.

Is there further evidence that substantiates the impact of parents' visits to schools? Panel B of Table 9 examines if these visits increase parents' learning of promoted techniques. Specifically, I assess channels through which parents learn promoted techniques. In the endline survey, after probing parents' awareness of promoted techniques in open-ended questions about planting methods, enumerators follow up by asking when and how they first learnt the promoted techniques — in particular, whether they learnt it from the SBAE schools, from students specifically, or from other members of the community. Columns (11) to (14) examine treatment effects on the number of techniques parents have learnt, focusing on reports of learning within the past two years. Two interesting patterns emerge. First, SBAE alone increases parents' learning from both schools (11.4 percentage points; 190% rise from control schools; column (12)) and students (8.1 percentage points; 405% from control schools; column (14)). Second, both the program promotional session and the annual farmer field day substantially increase parents' learning from schools (column (12)), but not from students (column (14)). Specifically, including both types of parental engagement increases the number of techniques parents learn from schools from 0.174 to 0.714 — an increase by around fourfold. This pattern suggests that parental engagement is particularly important in enabling parents to learn the technical aspect of promoted techniques via visits to schools — a function that is less fulfilled by intra-household interactions.

Students' home projects, communication with parents, and parents' observations. Table 10 turns to the second diffusion mechanism of SBAE — intra-household interactions between students and parents. Panel A examines whether SBAE leads to increased student home projects, as manifested in various proxies about whether students manage household farms. Columns (1)-(8) focus on two measures that are reported by students in the midline and endline surveys: (i) whether students look after farms; (ii) whether students

choose planting methods on the student parcel. Two patterns emerge. First, while SBAE has some positive but imprecise effects on these proxies of student-led farms, the program promotional session significantly intensifies the process. Specifically, SBAE alone raises the fraction of students who look after farms in the midline and endline — only slightly — to 56.3% (from 53%) and 54.5% (from 48%); with the program promotional session, these measures increase to 63.6% and 58.8% respectively. The contrast is even sharper if we focus on the more stringent proxy about student-led farms — whether they choose planting methods on the student parcel. SBAE alone raises this fraction in the midline and endline to 23.9% (from 18%) and 27% (from 23%); with the program promotional session, these measures increase to 30.8% and 34.9%.

Besides student-led home projects, students could also influence household technology adoption and educational decisions through their communication with parents and parents' observations. Panel B turns to these other mechanisms of intra-household interactions. Columns (9)-(16) examine whether parental engagement has encouraged more active communication between students and parents. Columns (9)-(10) examine the fraction of students who have asked parents for permission to change farming practices on household farms, as reported by parents. Both the program promotional session and the annual farmer field day significantly increase this fraction. Their collective impact is substantial, increasing this fraction from 41.3% in SBAE-only schools (37% in control schools) to 56.3%.

Columns (11)-(16) examine three measures of communication that are reported in the midline student survey. These questions were asked as follow-ups when students have adopted promoted techniques on the student parcel. The three measures are: the number of techniques (i) that students have shown parents how to make; (ii) that students have shown parents the associated harvests; (iii) that students have advised parents to use on other parcels. Parental engagement substantially raises these measures (i) from 27% to 51.8% (27% in control schools); (ii) from 24.2% to 41.4% (21% in control schools); and (iii) from 23.1% to 32.9% (18% in control schools).

Students' home projects, requests to change practices on household farms, and communication with parents can all influence parents' technology adoption via their impact on parents' beliefs about the returns from adoption. Columns (17) and (18) capture this mechanism by asking endline parents whether they have observed improved root crop harvests on students' parcels in the last two years, where improved root crop harvests is defined as either an increase in the size or number of tubers per plant. Both the program promotional session and the annual farmer field day raise this fraction, though the former's effect is around three times larger. Their collective impact is substantial — raising the fraction from 60.5% (59% in control schools) to 78.5%.

5.1.3. Summary

I have presented evidence that parental engagement effects are unlikely due to increased student participation or responses to SBAE training in Section 5.1.1. This implies that the pivotal effects of parental engagement on the primary outcomes must be due to increased impact outside schools. Section 5.1.2 presents solid evidence that the dual intra-household mechanisms of SBAE — parents' visits to schools and intra-household interactions between students and parents — are enabled by parental engagement. Remarkably, the SBAE program on its own has only small and statistically insignificant effects on these mechanisms. Further, each mechanism has distinct effects on parents' exposure to promoted techniques — (i) parents' visits to

schools lead to a fourfold increase in parents' learning of promoted techniques from schools; (ii) students' home projects and increased interactions with parents leads to a 20-fold increase in parents' observations of improved yields on students' parcels. Both channels could contribute to improvements in technology adoption and school attendance via increasing parents' increased acceptance of promoted agricultural techniques.

Table A7 presents evidence that parental engagement indeed produces downstream effects on parents' agricultural knowledge and beliefs, as well as changes in attitudes towards education. Column (2) indicates that SBAE, on its own, increases parents' score on the test of promoted techniques by only 0.12 (from 5.23 out of 12 in control schools), which is statistically insignificant; whereas the treatment effect rises to 0.352 (6.7% increase from 5.23 out of 12 in control schools) when both types of parental engagement are included. Column (4) indicates that SBAE alone increases parents who believe that students would harvest more cassava tubers per plant than a typical adult farmer in the community by 4.4 percentage points (from 35% in control schools), which is statistically insignificant; this treatment effect increases to 10.5 percentage points with parental engagement. Lastly, mirroring the annual farmer field day's effects on absenteeism as well as studying and farming hours (Tables 4 and 5), column (6) shows that it reduces the fraction of students who report having missed school in the semester due to requests from household members to work by 7.5 percentage points (from 24% in control schools).

The evidence on mechanisms also contributes to our discussion of the relative importance of the program promotional session and the annual farmer field day, as in Section 4.3.4. Both types of parental engagement are quantitatively important in enabling the first diffusion mechanism — parents visiting schools and attending parent-teacher meetings, learning promoted techniques, and observing harvests on school farms. But only the program promotional session enables the second diffusion mechanism — students starting home projects and triggering parents' observation of improved yields on household farms. These differences likely explain why increased parents' technology adoption, increased students' farm sales, and reduced dropouts are only observed when the program promotional session is included. Meanwhile, the annual farmer field day supplements by reducing household requests of students to work, which is crucial in reducing absenteeism and overall child labor.

Table 8: Midline and endline intent-to-treat effects on students' training outcomes

	Panel A: Midline survey					
	Joined program (0/1)		Score on promoted techniques (0-10)		Beliefs: more cassava tubers than typical farmer (0/1)	
	(1)	(2)	(3)	(4)	(5)	(6)
$SBAE_p$	0.737***	0.677***	0.379***	0.392***	0.058***	0.092***
	(0.019)	(0.030)	(0.062)	(0.113)	(0.015)	(0.029)
$SBAE_p \times FFD_s$		-0.012		0.016		-0.042
		(0.035)		(0.099)		(0.031)
$SBAE_p \times Session_s$		0.132***		-0.071		-0.025
		(0.033)		(0.131)		(0.036)
p-value: $(\beta_2 + \beta_3 = 0)$		0.01		0.76		0.20
	Summary: Combined effects of SBAE and informational treatments					
SBAE only (β_1)	0.737***	0.677***	0.379***	0.392***	0.058***	0.092***
	(0.019)	(0.030)	(0.062)	(0.113)	(0.015)	(0.029)
Lee bounds	[0.732***	[0.677***	[0.307***	[0.292**	[0.047***	[0.081**
	(0.020),	(0.030),	(0.061),	(0.108),	(0.015),	(0.029),
	0.753***	0.676***	0.465***	0.489***	0.069***	0.106***
	(0.019)]	(0.000)]	(0.063)]	(0.000)]	(0.015)]	(0.000)]
SBAE + FFD ($\beta_1 + \beta_2$)		0.664***		0.408***		0.050*
		(0.035)		(0.099)		(0.027)
Lee bounds		[0.654***		[0.318***		[0.034
		(0.036),		(0.097),		(0.028),
		0.685***		0.497***		0.063**
		(0.034)]		(0.094)]		(0.028)]
SBAE + Session ($\beta_1 + \beta_3$)		0.808***		0.322***		0.067***
		(0.026)		(0.090)		(0.024)
Lee bounds		[0.803***		[0.234**		[0.053**
		(0.026),		(0.091),		(0.024),
		0.829***		0.432***		0.079***
		(0.025)]		(0.096)]		(0.024)]
SBAE + Session + FFD ($\beta_1 + \beta_2 + \beta_3$)		0.796***		0.337***		0.025
		(0.027)		(0.108)		(0.031)
Lee bounds		[0.789***		[0.226**		[0.004
		(0.028),		(0.105),		(0.031),
		[0.826***		[0.478***		[0.043
		(0.025)]		(0.101)]		(0.031)]
	Summary Statistics for Control Schools					
Control Mean	0.00	0.00	5.61	5.61	0.42	0.42
Control SD	0.00	0.00	1.81	1.81	0.49	0.49
Observations	2425	2425	2425	2425	2425	2425
Number of clusters	155	155	155	155	155	155

Notes: This table reports treatment effects on parents' visits to schools (Panel A) and channels through which parents report learning promoted techniques (Panel B) in the endline survey. The top portion of each panel reports the estimates under each specification; the second portion then reports the combined effects of the SBAE program and informational treatments. These effects are computed using linear combinations of estimates. All regression specifications include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 8: Midline and endline intent-to-treat effects on students' training outcomes *(continued)*

	Panel B: Endline survey							
	Joined program (0/1)		Score on promoted techniques (0-10)		Beliefs: more cassava tubers than typical farmer (0/1))		Parent Reports: student asked to look after farms (0/1))	
	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
$SBAE_p$	0.843***	0.809***	0.254***	0.151	0.079***	0.122***	0.111***	0.122***
	(0.012)	(0.026)	(0.074)	(0.107)	(0.017)	(0.030)	(0.026)	(0.044)
$SBAE_p \times FFD_s$		0.021		0.143		0.007		-0.049
		(0.028)		(0.129)		(0.036)		(0.048)
$SBAE_p \times Session_s$		0.042		0.047		-0.090**		0.022
		(0.031)		(0.132)		(0.035)		(0.041)
p-value: $(\beta_2 + \beta_3 = 0)$		0.13		0.34		0.11		0.73
	Summary: Combined effects of SBAE and informational treatments							
SBAE only (β_1)	0.843***	0.809***	0.254***	0.151	0.079***	0.122***	0.111***	0.122***
	(0.012)	(0.026)	(0.074)	(0.107)	(0.017)	(0.030)	(0.026)	(0.044)
Lee bounds	[0.843***	[0.809***	[0.248***	[0.077	[0.079***	[0.113***	[0.108***	[0.121**
	(0.012),	(0.026),	(0.075),	(0.109),	(0.017),	(0.031),	(0.026),	(0.044),
	0.843***	0.808***	0.259***	0.242**	0.080***	0.138***	0.118***	0.125**
	(0.012)]	(0.026)]	(0.074)]	(0.106)]	(0.017)]	(0.031)]	(0.026)]	(0.044)]
SBAE + FFD $(\beta_1 + \beta_2)$		0.829***		0.294***		0.129***		0.074**
		(0.027)		(0.109)		(0.027)		(0.029)
Lee bounds		[0.829***		[0.149		[0.115***		[0.067**
		(0.027),		(0.110),		(0.028),		(0.030),
		0.830***		0.446***		0.147***		0.080**
		(0.027)]		(0.112)]		(0.027)]		(0.029)]
SBAE + Session $(\beta_1 + \beta_3)$		0.851***		0.199*		0.032		0.144***
		(0.022)		(0.115)		(0.032)		(0.037)
Lee bounds		[0.849***		[0.176		[0.028		[0.140***
		(0.022),		(0.116),		(0.032),		(0.037),
		0.856***		0.236*		0.035		0.152***
		(0.022)]		(0.119)]		(0.032)]		(0.037)]
SBAE + Session + FFD $(\beta_1 + \beta_2 + \beta_3)$		0.871***		0.342**		0.039		0.095**
		(0.023)		(0.138)		(0.031)		(0.048)
Lee bounds		[0.865***		[0.216		[0.014		[0.087*
		(0.023),		(0.130),		(0.031),		(0.047),
		[0.908***		[0.475***		[0.056*		[0.109**
		(0.021)]		(0.130)]		(0.032)]		(0.047)]
	Summary Statistics for Control Schools							
Control Mean	0.00	0.00	4.43	4.43	0.39	0.39	0.50	0.50
Control SD	0.00	0.00	1.78	1.78	0.49	0.49	0.50	0.50
Observations	1970	1970	1970	1970	1970	1970	1543	1543
Number of clusters	155	155	155	155	155	155	155	155

Notes: This table reports treatment effects on parents' visits to schools (Panel A) and channels through which parents report learning promoted techniques (Panel B) in the endline survey. The top portion of each panel reports the estimates under each specification; the second portion then reports the combined effects of the SBAE program and informational treatments. These effects are computed using linear combinations of estimates. All regression specifications include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 9: Endline intent-to-treat effects on parents' visits to schools and related outcomes

	Panel A: Parents' visits to schools									
	Visited school farm (0/1)		# visits: rainy season 2022		Observed harvest (0/1)		Attended parent-teacher meetings (0/1)		# meetings: 2022/23	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$SBAE_p$	0.220***	0.109**	0.432***	-0.105	0.173***	0.059	0.014	-0.102**	0.008	-0.352**
	(0.028)	(0.047)	(0.108)	(0.131)	(0.030)	(0.045)	(0.026)	(0.045)	(0.093)	(0.146)
$SBAE_p \times FFD_s$		0.073		0.474***		0.150***		0.085**		0.269***
		(0.045)		(0.139)		(0.043)		(0.033)		(0.097)
$SBAE_p \times Session_s$		0.141***		0.569***		0.083*		0.138***		0.420***
		(0.052)		(0.145)		(0.048)		(0.047)		(0.140)
p-value: $(\beta_2 + \beta_3 = 0)$		0.01		0.00		0.00		0.00		0.00
	Summary: Combined effects of SBAE and informational treatments									
SBAE only (β_1)	0.220***	0.109**	0.432***	-0.105	0.173***	0.059	0.014	-0.102**	0.008	-0.352**
	(0.028)	(0.047)	(0.108)	(0.131)	(0.030)	(0.045)	(0.026)	(0.045)	(0.093)	(0.146)
Lee bounds	[0.216***	[0.107**	[0.376***	[-0.107	[0.008	[-0.102**	[-0.053	[-0.354**	[0.168***	[0.059
	(0.028),	(0.047),	(0.105),	(0.131),	(0.026),	(0.045),	(0.090),	(0.146),	(0.030),	(0.045),
	0.226***	0.112**	0.447***	-0.083	0.018	-0.099**	0.017	-0.311**	0.178***	0.062
	(0.029)]	(0.047)]	(0.108)]	(0.130)]	(0.026)]	(0.045)]	(0.094)]	(0.129)]	(0.030)]	(0.045)]
SBAE+FFD ($\beta_1 + \beta_2$)		0.182***		0.369***		0.208***		-0.016		-0.083
		(0.038)		(0.113)		(0.041)		(0.034)		(0.125)
Lee bounds		[0.178***		[0.358***		[-0.020		[-0.095		[0.206***
		(0.038),		(0.113),		(0.035),		(0.126),		(0.041),
		0.191***		0.403***		-0.006		-0.015		0.218***
		(0.038)]		(0.107)]		(0.034)]		(0.111)]		(0.041)]
SBAE+Session ($\beta_1 + \beta_3$)		0.250***		0.464***		0.142***		0.037		0.068
		(0.040)		(0.152)		(0.039)		(0.036)		(0.117)
Lee bounds		[0.245***		[0.413**		[0.030		[0.024		[0.137***
		(0.040),		(0.156),		(0.036),		(0.117),		(0.039),
		0.257***		0.478***		0.042		0.080		0.147***
		(0.040)]		(0.152)]		(0.036)]		(0.117)]		(0.039)]
SBAE+Session+FFD ($\beta_1 + \beta_2 + \beta_3$)		0.323***		0.938***		0.291***		0.122***		0.337***
		(0.049)		(0.159)		(0.048)		(0.039)		(0.122)
Lee bounds		[0.315***		[0.819***		[0.108**		[0.221*		[0.281***
		(0.048),		(0.145),		(0.039),		(0.120),		(0.048),
		[0.339***		[0.982***		[0.132***		[0.360***		[0.303***
		(0.048)]		(0.160)]		(0.039)]		(0.122)]		(0.049)]
	Summary Statistics for Control Schools									
Control Mean	0.33	0.33	0.89	0.89	0.32	0.32	0.36	0.36	0.91	0.91
Control SD	0.47	0.47	1.66	1.66	0.47	0.47	0.48	0.48	1.46	1.46
Observations	1543	1543	1543	1543	1543	1543	1543	1543	1543	1543
Number of clusters	155	155	155	155	155	155	155	155	155	155

Notes: This table reports intent-to-treat effects on parents' visits to schools (Panel A) and related outcomes (Panel B) reported in the endline survey. Columns (1)-(4) refer to parents' visits to school farms — (1) and (2) a binary indicator of whether parents have visited by endline; (3) and (4) the number of visits in the second rainy season. Columns (5)-(6) refer to a binary indicator of whether parents have observed harvest on school farms. Columns (7)-(10) refer to parents' attendance of parent-teacher meetings — (7)-(8) a binary indicator of whether parents have attended any meetings in 2022/23; (9)-(10) the number of meetings attended. All regressions include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level.

Table 9: Endline intent-to-treat effects on parents' visits to schools and related outcomes *(continued)*

	Panel B: Channels through which parents learn promoted techniques			
	Learnt from schools: # techniques		Learnt from students: # techniques	
	(11)	(12)	(13)	(14)
$SBAE_p$	0.387***	0.114*	0.079***	0.081***
	(0.036)	(0.068)	(0.013)	(0.027)
$SBAE_p \times FFD_s$		0.261***		0.000
		(0.065)		(0.027)
$SBAE_p \times Session_s$		0.279***		-0.002
		(0.072)		(0.030)
p-value: $(\beta_2 + \beta_3 = 0)$		0.00		0.97
	Summary: Combined effects of SBAE and informational treatments			
SBAE only (β_1)	0.387***	0.114*	0.079***	0.081***
	(0.036)	(0.068)	(0.013)	(0.027)
Lee bounds	[0.366***	[0.114	[0.053***	[0.081***
	(0.034),	(0.068),	(0.011),	(0.027),
	0.393***	0.121*	0.080***	0.086***
	(0.036)]	(0.068)]	(0.013)]	(0.026)]
SBAE+FFD ($\beta_1 + \beta_2$)		0.375***		0.081***
		(0.045)		(0.021)
Lee bounds		[0.375***		[0.081***
		(0.046),		(0.021),
		0.400***		0.092***
		(0.046)]		(0.022)]
SBAE+Session ($\beta_1 + \beta_3$)		0.393***		0.079***
		(0.045)		(0.020)
Lee bounds		[0.366***		[0.048**
		(0.043),		(0.022),
		0.399***		0.080***
		(0.046)]		(0.020)]
SBAE+Session+FFD ($\beta_1 + \beta_2 + \beta_3$)		0.654***		0.079***
		(0.072)		(0.027)
Lee bounds		[0.619***		[0.027
		(0.067),		(0.020),
		[0.669***		[0.082***
		(0.073)]		(0.027)]
	Summary Statistics for Control Schools			
Control Mean	0.06	0.06	0.02	0.02
Control SD	0.30	0.30	0.14	0.14
Observations	1543	1543	1543	1543
Number of clusters	155	155	155	155

Notes: This table reports intent-to-treat effects on parents' visits to schools (Panel A) and related outcomes (Panel B) reported in the endline survey. Columns (11)-(14) refer to the number of promoted agricultural techniques parents learn — (11)-(12) from schools; (13)-(14) from specific students. All regressions include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level.

Table 10: Midline and endline intent-to-treat effects on students' management of household farms and related outcomes

	Panel A: Students' management of household farms									
	Midline Student Reports				Endline Student Reports				Endline Parent Reports	
	Look after farms (0/1)		Choose methods: student parcel (0/1)		Look after farms (0/1)		Choose methods: student parcel (0/1)		Student asked to change practices (0/1)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$SBAE_p$	0.079***	0.033	0.116***	0.059	0.087***	0.065	0.101***	0.040	0.120***	0.043
	(0.021)	(0.042)	(0.023)	(0.058)	(0.025)	(0.043)	(0.021)	(0.044)	(0.023)	(0.034)
$SBAE_p \times FFD_s$		0.030		0.040		-0.002		0.034		0.071*
		(0.040)		(0.057)		(0.053)		(0.042)		(0.040)
$SBAE_p \times Session_s$		0.073*		0.069		0.043		0.079*		0.079**
		(0.043)		(0.055)		(0.049)		(0.047)		(0.033)
p-value: $(\beta_2 + \beta_3 = 0)$		0.12		0.29		0.55		0.09		0.01
	Summary: Combined effects of SBAE and informational treatments									
SBAE only (β_1)	0.079***	0.033	0.116***	0.059	0.087***	0.065	0.101***	0.040	0.120***	0.043
	(0.021)	(0.042)	(0.023)	(0.058)	(0.025)	(0.043)	(0.021)	(0.044)	(0.023)	(0.034)
Lee bounds	[0.071***	[0.020	[0.101***	[0.053	[0.087***	[0.052	[0.101***	[0.034	[0.115***	[0.043
	(0.021),	(0.041),	(0.023),	(0.059),	(0.025),	(0.043),	(0.021),	(0.044),	(0.023),	(0.034),
	0.092***	0.044	0.122***	0.077	0.088***	0.078*	0.102***	0.058	0.125***	0.045
	(0.020)]	(0.290)]	(0.023)]	(0.188)]	(0.025)]	(0.043)]	(0.021)]	(0.043)]	(0.023)]	(0.034)]
SBAE+FFD ($\beta_1 + \beta_2$)		0.063*		0.099***		0.063		0.074*		0.114***
		(0.035)		(0.030)		(0.044)		(0.038)		(0.035)
Lee bounds		[0.050		[0.074**		[0.047		[0.067*		[0.110***
		(0.035),		(0.031),		(0.044),		(0.039),		(0.035),
		0.079**		0.105***		0.081*		0.101**		0.124***
		(0.035)]		(0.031)]		(0.044)]		(0.037)]		(0.035)]
SBAE+Session ($\beta_1 + \beta_3$)		0.106***		0.128***		0.108**		0.119***		0.123***
		(0.030)		(0.031)		(0.046)		(0.035)		(0.031)
Lee bounds		[0.096***		[0.108***		[0.104**		[0.112***		[0.118***
		(0.030),		(0.030),		(0.046),		(0.035),		(0.031),
		0.122***		0.136***		0.113**		0.123***		0.129***
		(0.029)]		(0.031)]		(0.046)]		(0.035)]		(0.031)]
SBAE+Session+FFD ($\beta_1 + \beta_2 + \beta_3$)		0.136***		0.168***		0.106**		0.154***		0.193***
		(0.035)		(0.054)		(0.042)		(0.035)		(0.038)
Lee bounds		[0.125***		[0.145**		[0.089**		[0.128***		[0.182***
		(0.036),		(0.054),		(0.043),		(0.034),		(0.037),
		[0.160***		[0.182***		[0.130***		[0.169***		[0.206***
		(0.036)]		(0.055)]		(0.041)]		(0.036)]		(0.038)]
	Summary Statistics for Control Schools									
Control Mean	0.53	0.53	0.18	0.18	0.48	0.48	0.23	0.23	0.37	0.37
Control SD	0.50	0.50	0.39	0.39	0.50	0.50	0.42	0.42	0.48	0.48
Observations	2425	2425	1384	1384	1488	1488	1488	1488	1543	1543
Number of clusters	155	155	155	155	155	155	155	155	155	155

Notes: This table reports intent-to-treat effects on students' management of household farms (Panel A) and related outcomes (Panel B) reported in the midline and endline surveys. All regressions include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level.

Table 10: Midline and endline intent-to-treat effects on students' management of farms and related outcomes *(continued)*

	Panel B: Outcomes related to students' management of household farms							
	Midline Student Reports						**Endline Parent Reports**	
	Shown parents how to make: # techniques		Shown parents associated harvests: # techniques		Advised parents to use on other parcels: # techniques		Observed: increased size or number of tubers on student parcels (0/1)	
	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
$SBAE_p$	0.125***	-0.007	0.121***	0.032	0.103***	0.051	0.113***	0.015
	(0.038)	(0.069)	(0.034)	(0.058)	(0.037)	(0.053)	(0.028)	(0.043)
$SBAE_p \times FFD_s$		0.067		0.083*		0.008		0.043
		(0.060)		(0.050)		(0.043)		(0.053)
$SBAE_p \times Session_s$		0.188**		0.088*		0.090*		0.137***
		(0.088)		(0.047)		(0.047)		(0.048)
p-value: $(\beta_2 + \beta_3 = 0)$		0.03		0.03		0.19		0.01
	Summary: Combined effects of SBAE and informational treatments							
SBAE only (β_1)	0.125***	-0.007	0.121***	0.032	0.103***	0.051	0.113***	0.015
	(0.038)	(0.069)	(0.034)	(0.058)	(0.037)	(0.053)	(0.028)	(0.043)
Lee bounds	[0.075**	[-0.015	[0.066**	[0.025	[0.054*	[0.047	[0.091***	[0.019
	(0.037),	(0.070),	(0.030),	(0.059),	(0.031),	(0.054),	(0.028),	(0.040),
	0.132***	0.036	0.129***	0.087	0.109***	0.099*	0.101***	0.022
	(0.038)]	(0.597)]	(0.034)]	(0.099)]	(0.038)]	(0.051)]	(0.028)]	(0.041)]
SBAE+FFD ($\beta_1 + \beta_2$)		0.060		0.115***		0.059		0.058
		(0.062)		(0.043)		(0.046)		(0.046)
Lee bounds		[0.001		[0.039		[-0.020		[0.028
		(0.062),		(0.047),		(0.044),		(0.044),
		0.069		0.127***		0.069		0.043
		(0.063)]		(0.044)]		(0.048)]		(0.045)]
SBAE+Session ($\beta_1 + \beta_3$)		0.181***		0.120***		0.141***		0.152***
		(0.057)		(0.044)		(0.041)		(0.045)
Lee bounds		[0.126**		[0.055		[0.079*		[0.131***
		(0.051),		(0.037),		(0.041),		(0.043),
		0.195***		0.130***		0.152***		0.143***
		(0.057)]		(0.044)]		(0.041)]		(0.043)]
SBAE+Session+FFD ($\beta_1 + \beta_2 + \beta_3$)		0.248***		0.204***		0.149***		0.195***
		(0.070)		(0.042)		(0.050)		(0.048)
Lee bounds		[0.189**		[0.137***		[0.078*		[0.145***
		(0.070),		(0.045),		(0.042),		(0.046),
		[0.264***		[0.220***		[0.164***		[0.173***
		(0.071)]		(0.042)]		(0.051)]		(0.045)]
	Summary Statistics for Control Schools							
Control Mean	0.27	0.27	0.21	0.21	0.18	0.18	0.59	0.59
Control SD	0.59	0.59	0.51	0.51	0.45	0.45	0.49	0.49
Observations	1458	1458	1458	1458	1458	1458	1543	1543
Number of clusters	155	155	155	155	155	155	155	155

Notes: This table reports intent-to-treat effects on students' management of household farms (Panel A) and related outcomes (Panel B) reported in the midline and endline surveys. All regressions include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level.

5.2. The video plays a pivotal role by addressing a key information friction

5.2.1. Contrasting the marginal effects of the video and the farmer field day

Contrasting the marginal effects of the farmer field day and the promotional video sheds light on what drives households' technology adoption and the reversal of SBAE effects on students' school attendance. Both interventions provide information to parents, but of different types. The farmer field day offers hands-on technical training on promoted soil management techniques, which parents can employ while retaining students as assistants on family farms. The promotional video, by contrast, highlights how SBAE creates complementarities between students' agricultural work and education—emphasizing that students can manage farms independently and pursue further education to strengthen their skills as farmers. If the key constraint to technology adoption is parents' technical knowledge or program awareness, the farmer field day should have qualitatively similar effects as the video. If instead the constraint is parents' beliefs about the value of delegating farms to students and, more broadly, the returns to supporting students' education, the video should prove more effective—particularly for activating intrahousehold technology transfer—the second diffusion mechanism of SBAE.

Figure 5 presents the marginal effects on SBAE's first diffusion mechanism: parents' visits to school farms. Both the farmer field day and the promotional video play important roles in activating this channel. The farmer field day increases the fraction of parents who visit school farms by 7.3 percentage points and the number of visits in the second season by 0.47 (59% from the SBAE mean of 0.8); the video produces stronger, but qualitatively similar effects of 14.1 percentage points and 0.57, respectively. Similarly, both treatments increase the fraction of parents who observe school farm harvests (by 15 and 8.3 percentage points, respectively), the number of techniques parents learn from school (by 0.26 and 0.28, 153% and 165% from the SBAE mean of 0.17), and attendance at parent-teacher meetings (by 0.27 and 0.42, 47% and 74% from the SBAE mean of 0.57) in the third school year. Overall, therefore, the evidence suggests that both treatments are effective in encouraging parents to visit schools, learn promoted techniques, and monitor students' academic progress.

Figure 6 presents evidence on the second diffusion mechanism—intra-household technology diffusion from students to parents. Here, the video's dominance is stark. Only the video, not the farmer field day, increases the fraction of students who look after household farms at both midline and endline. The video increases the fraction of students who look after farms by 7.3 percentage points at midline and 4.3 percentage points at endline, while the corresponding farmer field day effects are negligible (3 percentage points and -0.2 percentage points, respectively). More strikingly, students whose parents receive the video are approximately three times more likely to demonstrate promoted techniques to their parents, and over ten times more likely to advise parents to adopt these techniques on their own parcels, compared to students whose parents attend only the farmer field day. In turn, the increase in the fraction of parents who report observing improved cassava yields on students' parcels is substantially higher with the video (13.7 percentage points) than with the farmer field day (4.3 percentage points)—a ratio of more than three to one. Overall, therefore, the evidence suggests that the video, not the farmer field day, plays a dominant role in activating the second diffusion mechanism of SBAE.

As highlighted in the introduction, since farmers prefer learning from socioeconomic peers (BenYishay & Mobarak, 2019), observing technologies under familiar conditions (Munshi, 2004), and gathering multiple

observations before adopting (Beaman et al., 2021), students' home projects—SBAE's second diffusion mechanism—could prove pivotal in overcoming barriers for technology adoption. Consistent with this, Figure 7 shows that while both treatments contribute to technology adoption on students' parcels, only the video produces effects strong enough to increase adoption on parents' parcels. This has significant implications for households' production, as parents' production tends to be bigger in scale and generate far greater farm sales than students' parcels. Similarly, consistent with the video reshaping parents' aspirations for their children's education, the video accounts for the entire reduction in student dropouts, and, correspondingly, the entire increase in the total number of days students attend school.

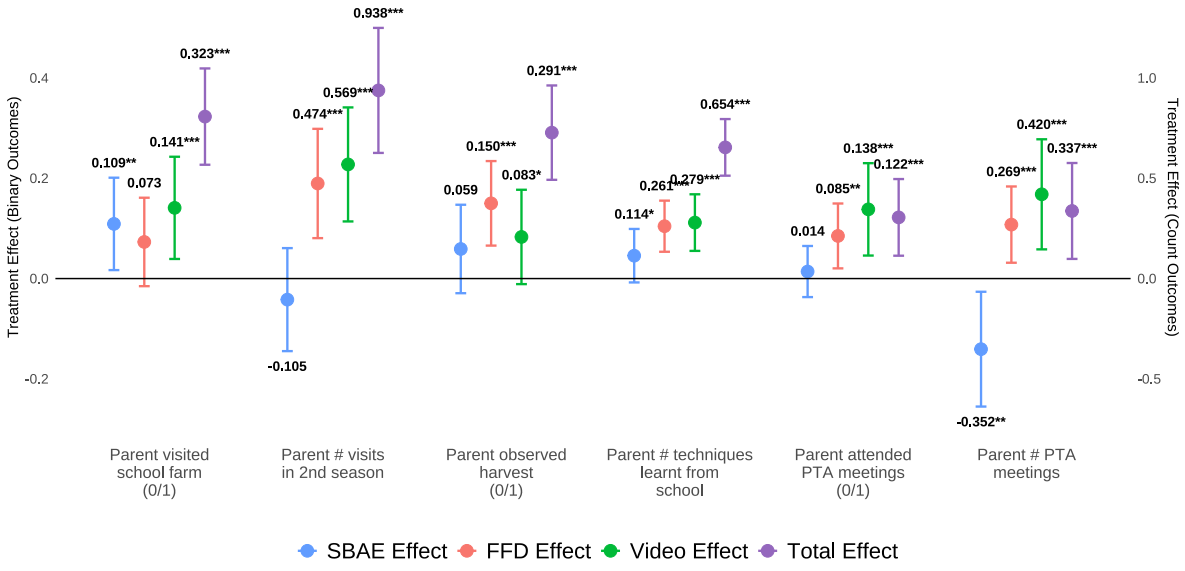


Figure 5: Contrasting marginal treatment effects on parents' visits to schools

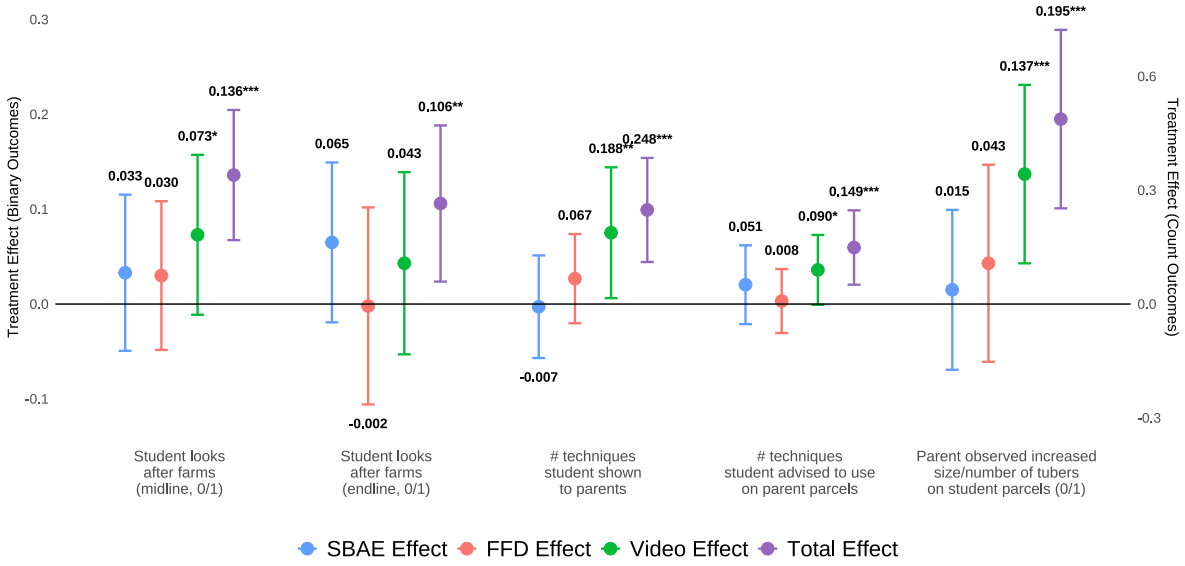


Figure 6: Contrasting marginal treatment effects on intrahousehold technology diffusion

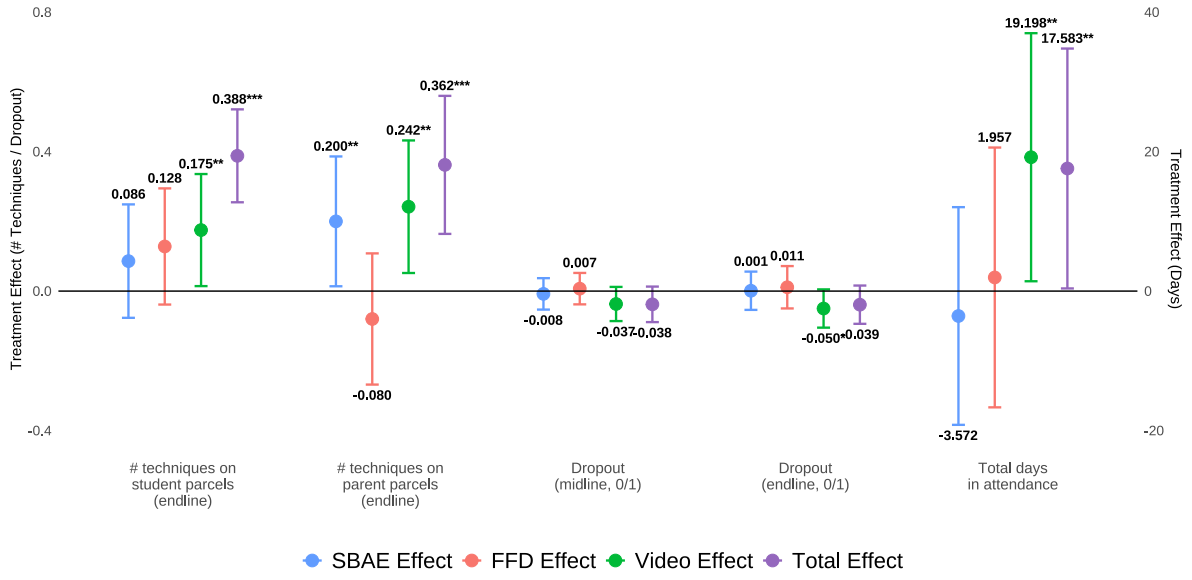


Figure 7: Contrasting marginal treatment effects on technology adoption and school attendance

5.2.2. Evidence that the video is tackling a key information friction

So far, we have shown that the promotional video plays a dominant role in activating SBAE's second mechanism—intra-household technology diffusion—and that only the video increases parents' technology adoption and reduces student dropouts. We now discuss evidence that this is due to the design of treatments: the video differs from the farmer field day in that it clarifies the student-centered pedagogy underlying SBAE, leading parents to expect stronger growth in students' agricultural skills and greater returns from keeping them in school. My companion paper (JMP2, Lee, 2025) provides strong evidence that the video addresses this key information friction.

Specifically, 97% of parents who watch the video anticipate growth in students' farm management abilities in one year, including their cassava yields, knowledge about farm management, and knowledge about promoted techniques. Consistent with the video providing crucial information that parents otherwise miss, their beliefs about students' cassava yields remain significantly more optimistic at both midline and endline compared to parents in schools without video sessions. Moreover, parents' beliefs about the difference between students' cassava yields and those of typical adult farmers—measured immediately after watching the video—strongly predict their midline responses to SBAE. One standard-deviation increase in these beliefs is associated with a 14% increase in visits to school farms, a 22% increase in the likelihood of observing school farm harvests, a 31% increase in the likelihood of working on school farms, and a 12% increase in the likelihood of meeting with the responsible teacher. These associations do not merely reflect social desirability bias: one standard-deviation increase is also associated with a 2.7 percentage point increase in parents correctly naming field officers (from a base of 11%) and 9% higher attendance at parent-teacher meetings.

6. Cost-benefit and cost-effectiveness analyses

The previous sections show that SBAE, when augmented with both parental engagement treatments, is highly effective in increasing students' savings, school attendance, and parents' adoption of agricultural technologies. But do these benefits justify the costs and is the augmented SBAE program cost-effective?

6.1. Assumptions

Table 11 lists all key assumptions in the cost-benefit and cost-effectiveness analyses. I first define the program beneficiaries. Corresponding to the randomized trial, I consider a three-year period in which three cohorts of students in 100 SBAE schools are targeted. Since sampled students represent a random 30% of students in targeted grades, the program reaches roughly 65 students per school—a total of 19,500 students across all cohorts and schools. For the analysis of technology adoption, which may include spillovers of farming practices to non-parent households, it is important to note that approximately 410,000 people reside within 2 kilometers of the SBAE schools (Liberia Institute of Statistics and Geo-Information Services, 2008).

The total cost over this three-year period is $801,139 (2023 US dollars), or about $40 per student. This covers two components: program implementation ($621,139), including training extension officers and teachers, monthly school monitoring, setting up school farms, and hosting national events for extracurricular clubs; and parental engagement ($180,000), including a promotional session for one parent per student and an annual farmer field day.

An important consideration for program sustainability is the potential for drastically lower costs after the initial three years. Setup costs are not repeated, and I assume monitoring visits are sharply curtailed to once a year, reducing three-year implementation costs from $621,139 to $120,000—a 63% drop in total costs.

Turning to the side of program benefits, the first important component is the increase in students' savings — Section 4.3 shows that in the second year of the program, there is a 21% increase in annual savings from a base level of $20 (2022 US dollars) for students who have not dropped out from schools. I assume that this increase carries over to the third year.

The second important component is the increase in students' school attendance — due to both reduced dropouts and reduced absenteeism. To combine effects on both dropouts and absenteeism, I focus on the treatment effect on the imputed total number of days in attendance, which is 17.6 according to Table 4.

For the cost-benefit analysis, we must monetize this increase in school attendance. Adopting a conservative approach, we assume an 8% annual return to schooling, implying a return of 0.04% per school day (based on approximately 200 school days per year).[22] Applying a 10% discount rate and assuming that students work for 30 years, we estimate that each additional school day generates $1.89 in discounted lifetime earnings.

[22] The global average return to an additional year of schooling is approximately 8.8% (Psacharopoulos & Patrinos, 2018). We use 8% as a conservative estimate. Returns in sub-Saharan Africa, especially at the secondary level, are typically higher, ranging from 10–13% (Montenegro & Patrinos, 2014). A World Bank document (World Bank, 2019) also reports that based on household survey data in 2016, individuals who completed senior high school earned on average 18 percent more than those who completed only junior high school, though this descriptive finding reflects correlation rather than causal returns. Downward revision from these optimistic estimates is reasonable, as standard Mincerian estimates may overstate returns for rural populations who are more likely to remain in the agricultural or informal sector, where credential-based wage premiums are less pronounced.

The third component, which is also the most uncertain component, is parents' benefits from technology adoption. To calculate these benefits as conservatively as possible, I consider only adoption of bed preparation in one season (only in verified third year) and in 1/4 of the typical cultivated area of roots and vegetables. This tends to understate the benefits as (i) SBAE also increases joint adoption of multiple techniques (as we discuss in Section 4.3) tend further raise yields; (ii) parents who experience improved yields tend to continue using the technologies. Further, as yields in the general population are often found to be lower than in agronomic trials, I assume only half of the yield gains in agronomic trials. I use cassava and sweet potatoes as the representative crops (each comprising half of the area of adoption) — these crops are widely grown by Liberian farmers as their flexible use helps satisfy subsistence needs, but have lower prices in the market than other root crops and vegetables. I also consider the potential increase in labor costs due to constructing beds — 150 hours per hectare. Taking all factors into account, I calculate that each adoption of bed preparation creates a net production value of $64.6.

Table 11: Key assumptions and parameter values

Item	Value	Justification
Scope & reach		
Period of consideration	3 years	RCT period
Number of schools	100	Schools in SBAE arm
Direct beneficiaries (households)	19,500	3 cohorts × 6,500 students
Community population (within 2 km of schools)	410,000	2008 census data
Cost assumptions (Years 1–3)		
Total cost	$801,139	
Program cost	$621,139	Training of extension officers and teachers, monitoring costs, setup costs
Parental engagement cost	$180,000	Program promotional session to every cohort and annual farmer field day
Cost assumptions (Years 4–6 and beyond)		
Total cost	$300,000	
Program cost	$120,000	Dropping monitoring costs and setup costs
Parental engagement cost	$180,000	Same as before
Students' savings		
Base annual savings per student	$21	Average in control schools (translated into 2023 US dollars)
Growth rate of savings	21%	RCT finding on Year 2: Table 6
Savings increase (Year 2) for non-dropouts	4.41	92% increase
Savings increase (Year 3) for non-dropouts	4.41	Assuming saving increase in Year 2 carries over
Students' discounted lifetime earnings		
Number of additional school days per student	17.6	Table 4
Base annual income	$500	Typical annual income in rural Liberia
Return on additional school day	8/200%	Implying annualized return of 8%
Discount rate	10%	Standard rate in program evaluation
Present value per extra school day	$1.89	Working for 30 years after finishing school
Parents' benefits from adopting bed preparation		
Number of parents adopting	4,251	RCT finding: 19,500 households × 2 parents × 10.9%
Cassava base yield (kg/ha)	8,280	FAOSTAT database
Cassava yield increase	15%	1/2 of the 30% yield gain in agronomic trial
Sweet potato base yield (kg/ha)	1,283.4	FAOSTAT database
Sweet potato yield increase	35%	1/2 of the 70% yield gain in agronomic trial
Adoption area per household (ha)	0.15	1/4 of typical cultivated area for roots and vegetables
Labor increase (hrs/ha)	150	Estimates from FAO guidebooks
Cassava price (USD/kg)	$0.21	Average market price in 2023 from FAO
Sweet potato price (USD/kg)	$0.30	Average market price in 2023 from FAO
Labor cost (USD/hour)	$0.43	Agricultural wage rate in rural Liberia
Net increase in production value per adoption	$64.60	Increased production value net of increased labor cost

Notes: This table lists all essential assumptions used for cost-benefit and cost-effectiveness analyses in Section 6. All monetary values are expressed in 2023 US dollars.

6.2. Cost-benefit analysis

We distinguish between two types of analysis: first, whether benefits justify costs in the first three years, with known data and assumptions from Section 6.1; second, whether SBAE is a good long-term investment, viewing from the perspective of a policymaker who commits a stream of spending on SBAE.

Figure 8 presents the benefits and costs over the first three years, revealing two key observations. First, benefits exceed costs during this period, yielding a benefit-cost ratio of 1.33. Second, even when excluding

benefits from parents' technology adoption—a conservative approach for those skeptical about the magnitude of yield gains—the benefit-cost ratio remains at 0.98. In other words, the benefits accruing to students alone nearly justify the program, even under the most pessimistic assumptions regarding parents' gains from technology adoption and potential spillovers of improved farming practices to non-parent farmers.

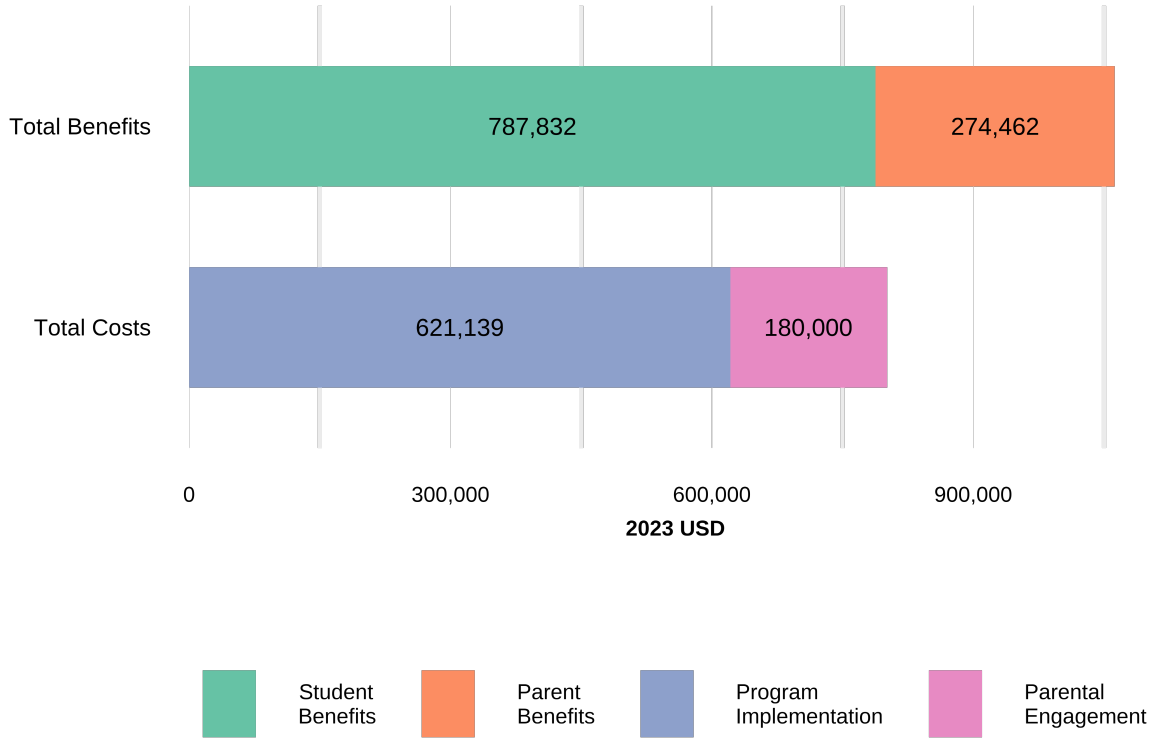


Figure 8: Benefits and costs in the first 3 years of the augmented SBAE program

Notes: This figure plots all components of benefits and costs of the augmented SBAE program in the three-year RCT period. All monetary values are expressed in 2023 US dollars.

Figure 8 demonstrates that the augmented SBAE program is economically justified in the short term. However, this analysis overlooks a key strength of SBAE: its emphasis on capacity building means the program can be sustained through periodic teacher training alone, without ongoing monitoring visits from extension officers. This raises a critical question: what are the long-term returns on SBAE? The answer hinges on two competing considerations. On the one hand, costs drop substantially after the initial three years—setup costs are not repeated, and monitoring visits are curtailed to once a year while teacher training is retained, reducing implementation costs from \$621,139 to \$120,000, a 63% reduction (inclusive of parental engagement). On the other hand, the key uncertainty is whether the program remains equally effective once external support is scaled back.

Figure 9 presents the 10-year cost-benefit analysis under two scenarios. During the initial three-year implementation period, the benefit-cost ratio is 1.33. In Panel A, which assumes program benefits remain

unchanged after the first three years, the benefit-cost ratio for Years 4–10 rises to 3.54, yielding a cumulative benefit-cost ratio of 2.36 over the ten-year horizon. In Panel B, which assumes a more conservative scenario where program benefits decline to 50% after the first three years, the benefit-cost ratio for Years 4–10 is 1.77, with a cumulative benefit-cost ratio of 1.53 over the ten-year period. While Panel A highlights SBAE's potential to deliver even higher returns over time, the results in Panel B provide a robustness check that strongly justifies long-term policy commitment. By capitalizing on the institutional capacity and teacher training established during the initial implementation period, sustained returns can be achieved at a fraction of the initial costs.

6.3. Cost-effectiveness analysis

We now compare SBAE's cost-effectiveness in addressing the dual challenges of diffusing agricultural technologies and increasing school attendance against major alternatives in sub-Saharan Africa, drawing on randomized trial evidence from Innovations for Poverty Action. The Online Appendix provides further details on these programs and cost-effectiveness calculations.

Figure 10 compares the impact per $100 of the augmented SBAE program with other African programs evaluated using randomized trials. I first assess SBAE's cost-effectiveness in increasing technology adoption by calculating the number of households that adopt promoted techniques per $100 invested. For SBAE, I measure adoption of bed preparation on parents' parcels at endline. A key unknown—which I will track in a 2026 follow-up survey—is the adoption rate among randomly sampled non-parent farmers in nearby communities. I assume this rate (for communities within 2 kilometers of schools) ranges from 0% to 5%, in line with estimates of positive spillovers in farming practices from the social learning literature.

The prevailing mode of agricultural extension in developing countries is the contact farmer model, in which extension agents train selected village farmers who are then expected to disseminate practices to their peers. As Suri & Udry (2022) summarize, this approach has generally proven ineffective at triggering adoption beyond direct trainees. Efforts to improve extension have largely focused on enhancing this existing model—by tweaking how contact farmers are trained, leveraging information technologies to deliver messages to farmers, or increasing social learning by choosing contact farmers more carefully. I compare SBAE, which represents a departure from the contact farmer approach, against these RCT-evaluated programs. Despite differences in the specific techniques promoted, all programs target soil management—a pervasive agronomic problem—making them reasonably comparable.

Figure 10 shows that, with the exception of BenYishay & Mobarak (2019), SBAE's impact per $100—0.46 to 0.94 households adopting bed preparation, or 0.55 to 1.02 households adopting bed preparation *and* a complementary soil management technique—far exceeds that of extension programs using the contact farmer model. Kondylis et al. (2017) attempt to address imperfect implementation and accountability among extension agents by adding a centralized training workshop for contact farmers. While this increases adoption among contact farmers, it fails to reach untrained farmers, yielding an impact of just 0.11 households adopting pit planting per $100. Two other notable studies evaluate the use of farmer field days and video messaging. These interventions are similar in cost to the annual farmer field day and promotional video I introduce as parental engagement treatments in SBAE, but differ in that agricultural training targets adults rather than students. Fabregas et al. (2023) study the effects of farmer field days in promoting Mavuno

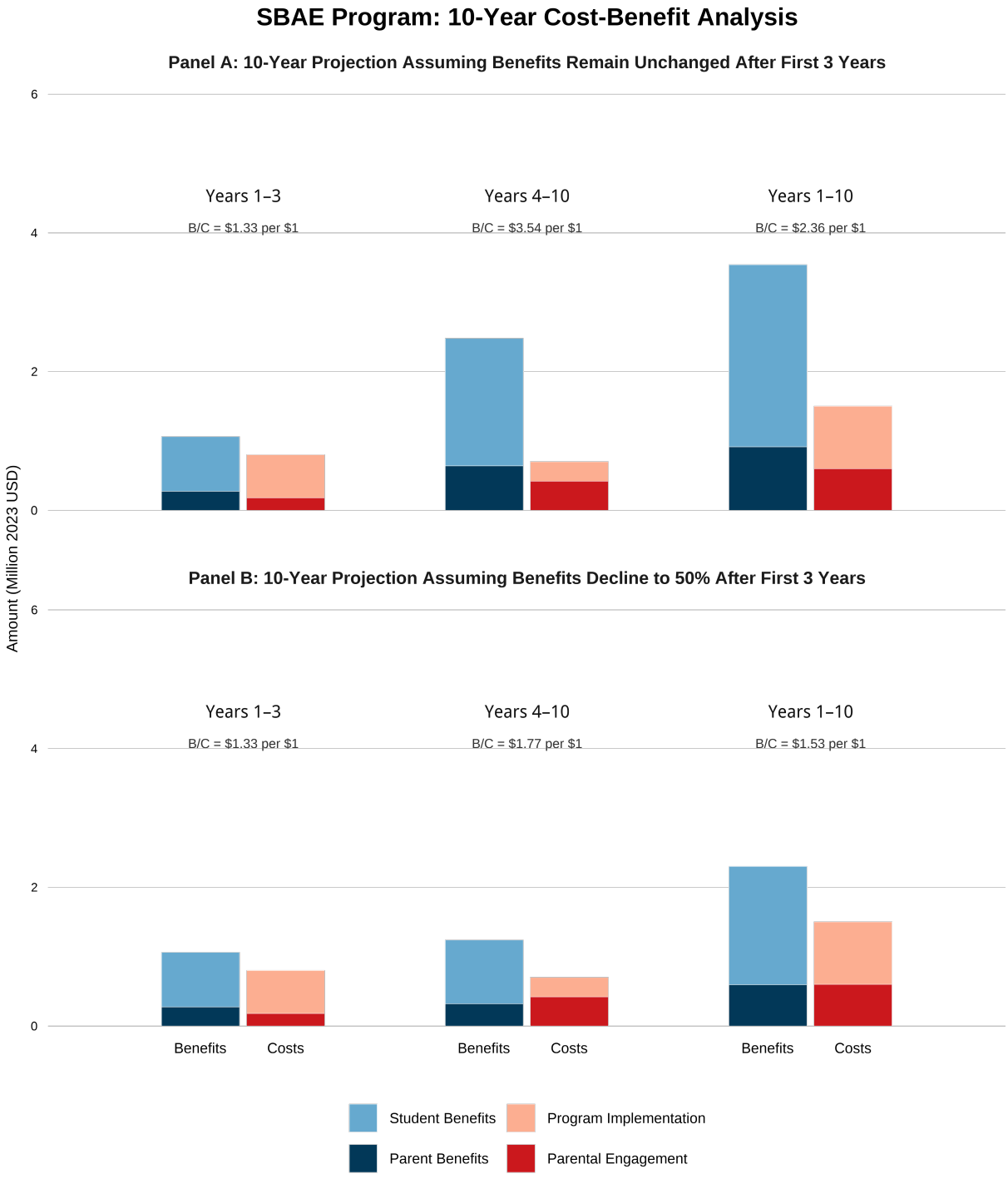


Figure 9: SBAE's benefit-cost ratios under different assumptions about post-RCT program effectiveness

fertilizer adoption. With an intent-to-treat effect of 4 percentage points and 100 to 300 farmers reached per field day, the impact is 0.15 to 0.46 households per $100. Van Campenhout et al. (2021) evaluate video-mediated extension messages designed to teach optimal crop spacing to maize farmers in Uganda. This technique is comparable to bed preparation in the present study, as both are relatively easy to adopt. Despite encouraging effects on knowledge, practices, and yields, the program induces only 0.62 households to adopt per $100 spent, placing it at the lower end of SBAE's cost-effectiveness range. The only program that outperforms SBAE is village-based extension in Malawi augmented with performance incentives for contact farmers (BenYishay & Mobarak, 2019)—and only when the outcome is composting adoption (1.53 households per $100). When considering pit planting, a technique more comparable to bed preparation given similar labor intensity and complexity, the impact falls to 0.6 households per $100, below SBAE's range.

Second, I assess SBAE's cost-effectiveness in increasing school attendance by calculating additional school days per student per $100 invested. I benchmark against conditional cash transfers (CCTs), one of the most popular approaches to boosting school attendance in sub-Saharan Africa and among the most rigorously evaluated. Drawing on Baird et al. (2011)'s RCT in Malawi, Figure 10 shows that the augmented SBAE program (43.46 days per $100) is more than twice as cost-effective as CCTs (17.06–24.98 days per $100). My companion paper (JMP2, Lee, 2025) shows that this advantage widens further when SBAE is augmented with both parental engagement and information nudges signaling parents' openness to delegate farms to students: as students shift toward sales-oriented crops to finance school fees, cost-effectiveness rises to 59.51 days per $100—nearly 3.5 times that of CCTs.

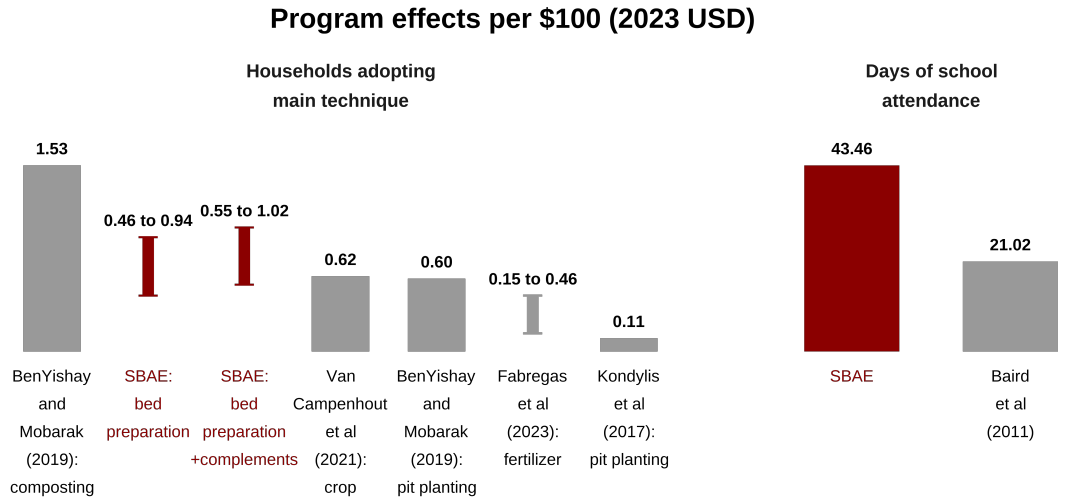


Figure 10: Comparing the augmented SBAE program with alternatives using randomized trial evidence

Notes: This figure plots the cost-effectiveness of various programs, in terms of impact per $100 invested. All monetary values are expressed in 2023 US dollars. Relevant calculations can be found In the Online Appendix.

6.4. Summary

Overall, these back-of-the-envelope calculations provide strong support of the merits of the augmented SBAE program, at least in the context of Liberia. First, SBAE is highly sustainable — it offers a stream of benefits that increasingly outweigh costs after external support for schools are dropped. Second, in terms of

cost-effectiveness, SBAE outperforms major alternatives in diffusing agricultural technologies and increasing school attendance.

How generalizable are these results? I speculate that there would be a shift of balance between short-term and long-term benefits as SBAE is extended to countries that are higher up the ladder of economic development. The short-term benefits might become smaller for three reasons. First, there is less room to promote agricultural technologies that involve low monetary costs, can easily be used by students, and yet improve yields by a large extent. This would result in less dramatic increases in students' savings. Second, there is less room to improve school attendance as the dropout rates are much lower. Third, informational barriers are less likely to be the main constraints for parents' adoption of off-the-shelf agricultural technologies.

In the long term, however, SBAE's framework could generate more benefits for countries higher up the economic ladder, especially if SBAE could partner or integrate with complementary programs. First, the returns from promoting more advanced techniques, such as improved crop varieties and proper use of chemical fertilizers, would far exceed the returns from the promoted techniques in Liberia. While factors such as credit constraints would limit the use of these techniques, the SBAE framework remains powerful in alleviating information barriers about the extent of yield gains. Second, rather than encouraging school attendance per se, education systems in these countries are more well-equipped to deliver the potential of student-centered learning under SBAE. In the very first step, SBAE could provide an entry point to train and motivate teachers to use more effective pedagogies in teaching science outside the few structured lessons related to agriculture, thereby improving students' understanding of and performance in science and related subjects. More importantly, the development of an educational culture that is centered on scientific understanding and open inquiries would facilitate bottom-up experimentation of farming techniques by students and their households, thereby sustaining the SBAE program without external extension support.

7. Concluding remarks

Finding an effective delivery model for agricultural extension stands out as one of the foremost challenges in African rural development. Using a randomized experiment with 197 schools, this paper provides proof of concept that schools can be a promising entry point for diffusing yield-improving agricultural technologies. This promise lies in two parts. First, I show that — contrary to concerns about pitfalls — with appropriate implementation, school-based agricultural extension can in fact encourage school attendance alongside increasing households' technology adoption. Second, I show that due to scale economies in leveraging the school infrastructure, the program offers both scope of impact and cost-effectiveness. Specifically, I show that the program increases households' technology adoption, students' savings, and school attendance. It is on par with the best village-based extension programs in increasing technology adoption; more cost-effective than non-school-based agricultural training in increasing youth's savings; and more cost-effective than conditional cash transfers in increasing school attendance.

Alongside the promise of school-based agricultural extension, it should be noted that its large, positive effects are achieved only with parental engagement and sub-treatments discussed in Lee (2025). To the extent that school-based development programs have strategic advantages in logistics (e.g. scalability and sustainability), this result offers two general lessons. First, we should conduct joint evaluations of school-based development

programs and household interventions, as this paper reveals that the former could have hidden potentials that are only unlocked with the latter. Second, we should investigate ways to enhance students' agency in amplifying the impact of school programs, not just as a matter of youth empowerment, but as ways to increase the cost-effectiveness of school programs.

Beyond immediate impacts, this paper demonstrates that schools can serve as sustainable rural development hubs when coupled with low-cost parental engagement events. The first step is to motivate students to attend schools, triggering a virtuous cycle in which schools are strengthened through improved incomes and provide higher-quality education. However, while the current RCT demonstrates positive effects on technology adoption and school attendance, it shows no improvements in students' academic performance or agribusiness knowledge. The second step, therefore, is to synergize SBAE with complementary interventions. One direction that I am taking is to integrate the highly effective, student-centered "Learning to Learn (LTL)" pedagogy into SBAE, which is shown in a randomized trial in Uganda by Nourani et al. (2025) to be amongst the top 5% education interventions in terms of learning-adjusted years of schooling. While complementary with the student-centered, experiential instruction promoted by SBAE, this intervention does not prescribe fixed teaching methods. Instead LTL prompts teachers to reflect on how they learn, emphasizing active learning, critical thinking, and problem-solving skills essential for students to effectively apply agricultural knowledge in real-world contexts. Another direction is to link SBAE to home-grown school-feeding programs, creating direct market demand for students' agricultural produce while expanding the scope of agribusiness education. A third direction is to leverage the high-frequency student data collection system developed during the RCT to conduct iterated randomized trials on different agricultural technologies. This approach is particularly promising when combined with algorithmic farming advice, a partnership currently under development.

As of writing, the current randomized trial has significant policy impact in sub-Saharan Africa, scaling to all rural schools in Liberia by 2035, with expansion to Côte d'Ivoire and Kenya beginning in 2026. In the long term, this holistic approach of SBAE can contribute to our understanding of macroeconomic development in agrarian economies. First, it addresses whether schools can simultaneously promote rural livelihoods while equipping students with adaptable skills for structural transformation toward higher-value sectors. Second, it contributes to our ability to understand and shape the direction of technical change through a system of high-frequency feedback and enhanced data collection.

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Appendix

Contents

Additional Tables A-1

Additional Tables

Table A1: External validity of sampled schools and students

	Sampled schools	Eligible schools	Targeted schools	Targeted counties
Panel A: Sampled schools vs. other schools in targeted counties				
# students	444.47	419.17	423.33	279.72
	(244.96)	(228.14)	(258.93)	(208.56)
# elementary students	182.48	174.76	170.46	126.43
	(118.07)	(112.08)	(120.34)	(115.91)
# junior high students	91.93	78.26	86.55	25.66
	(107.25)	(94.65)	(109.64)	(71.14)
# teachers	14.52	13.78	15.53	8.75
	(6.33)	(6.60)	(9.38)	(7.29)
Student-teacher ratio	31.83	32.48	29.34	39.29
	(13.02)	(15.29)	(14.14)	(25.00)
Functional PTA according to WASH (0/1)	0.84	0.78	0.74	
	(0.37)	(0.41)	(0.44)	
Water source according to WASH (0/1)	0.61	0.53	0.51	
	(0.49)	(0.50)	(0.50)	
# schools	197	304	447	1794

	Sampled students	All targeted students
Panel B: Sampled vs. all students in targeted grades		
% farming	100.00 (0.00)	98.65 (6.32)
% farming before 2021	99.38 (1.82)	94.96 (11.08)
% joined cooperative farming groups (kuu)	93.91 (18.39)	83.71 (18.89)
% joined kuu before 2021	82.12 (17.09)	77.14 (22.32)
% students sampled in each school	30.33 (17.33)	

Notes: This table compares the average characteristics of sampled schools and students to population-level statistics in Liberia. Panel A compares the characteristics of sampled schools to eligible schools (after screening by 4-H Liberia), targeted schools using the WASH database, and all schools listed in the EMIS database 2015-16 and in targeted counties. Panel B uses responses to the grade-level screening survey to compare the average characteristics of sampled students and all students in targeted grades. Standard errors are in parentheses.

Table A2: Balance in baseline characteristics across treatment arms

Baseline variable	Overall Mean	Control Mean	$SBAE_p$	$SBAE_p \times Session_s$	$SBAE_p \times FFD_s$
Panel A: School characteristics					
# students	299.42	300.23	3.570	17.393	-17.880
	(221.95)	(234.26)	(57.129)	(54.413)	(51.472)
# elementary students	124.67	122.65	-5.869	16.165	8.301
	(85.85)	(73.75)	(23.489)	(25.653)	(26.038)
# junior high students	67.32	68.75	4.266	-5.607	-3.687
	(84.96)	(96.75)	(18.731)	(18.080)	(17.843)
# teachers	13.75	13.65	1.164	-1.796	-0.623
	(5.87)	(5.74)	(1.634)	(1.753)	(1.311)
Student-teacher ratio	23.54	23.37	0.257	3.074	-2.051
	(14.25)	(13.51)	(2.826)	(3.047)	(3.066)
# classrooms	9.94	9.75	0.773	-0.471	-0.717
	(3.00)	(3.11)	(0.821)	(0.790)	(0.728)
Water source (0/1)	0.73	0.77	-0.144	0.143	0.003
	(0.44)	(0.42)	(0.126)	(0.155)	(0.127)
Provides breakfast (0/1)	0.10	0.11	0.031	-0.045	-0.016
	(0.30)	(0.31)	(0.058)	(0.093)	(0.066)
Provides lunch (0/1)	0.24	0.23	0.100	-0.068	-0.126
	(0.43)	(0.42)	(0.088)	(0.104)	(0.089)
# district education officer visits	3.14	3.10	-0.211	-0.049	0.559
	(1.37)	(1.39)	(0.321)	(0.345)	(0.340)
Kept enrollment list (0/1)	0.68	0.68	-0.089	0.197	-0.046
	(0.47)	(0.47)	(0.116)	(0.122)	(0.113)
# parent-teacher association meetings	1.95	1.97	-0.215	0.289	-0.013
	(0.89)	(0.94)	(0.234)	(0.225)	(0.255)
# school management committee meetings	1.01	1.18	-0.313	-0.202	-0.020
	(1.17)	(1.24)	(0.297)	(0.371)	(0.309)
Public school (0/1)	0.72	0.75	-0.029	0.062	-0.037
	(0.45)	(0.43)	(0.098)	(0.114)	(0.119)
Private or faith-based school (0/1)	0.22	0.20	0.010	-0.004	-0.029
	(0.42)	(0.40)	(0.096)	(0.103)	(0.096)
# students sampled	19.14	18.68	0.943	0.464	-0.717
	(3.64)	(3.74)	(1.277)	(1.277)	(1.109)
	F-test p-values		0.382	0.865	0.646

Notes: This table shows a test of balance for the baseline characteristics of schools (Panel A), students (Panel B), and parents (Panel C) in the randomized experiment. Column 1 provides covariate means and standard deviations in the entire sample. Column 2 provides covariate means and standard deviations in control schools. Column 3 shows the marginal effects of SBAE schools over control schools. Column 4 shows the marginal effects of program promotional sessions over SBAE only schools. Column 5 shows the marginal effects of the annual farmer field day over SBAE only schools. P-values from Wald tests of joint significance of treatment effects over all covariates are presented at the bottom of each panel.

Table A2: Balance in baseline characteristics *(continued)*

Baseline variable	Overall Mean	Control Mean	$SBAE_p$	$SBAE_p \times Session_s$	$SBAE_p \times FFD_s$
Panel B: Student characteristics					
Age	15.70	15.74	0.065	-0.143	-0.110
	(1.97)	(1.86)	(0.154)	(0.174)	(0.202)
Grade	5.69	5.69	0.046	-0.230	0.146
	(1.47)	(1.43)	(0.128)	(0.142)	(0.124)
Male (0/1)	0.52	0.52	0.013	-0.013	-0.016
	(0.50)	(0.50)	(0.016)	(0.017)	(0.017)
Years of farming	7.85	7.84	0.028	-0.035	0.024
	(3.32)	(3.36)	(0.196)	(0.229)	(0.226)
	F-test p-values		0.924	0.447	0.561
Panel C: Parent characteristics					
Age	43.15	43.14	-0.183	0.004	0.453
	(6.95)	(3.33)	(0.161)	(0.174)	(0.274)
Male (0/1)	0.47	0.48	0.015	-0.034	-0.018
	(0.50)	(0.50)	(0.024)	(0.027)	(0.030)
Parent or not (0/1)	0.74	0.75	-0.040	0.002	-0.010
	(0.44)	(0.43)	(0.033)	(0.043)	(0.036)
Household head (0/1)	0.63	0.61	0.001	-0.034	0.058
	(0.48)	(0.49)	(0.033)	(0.040)	(0.039)
Invited to surveys (0/1)	0.98	0.99	0.009	-0.016	-0.012
	(0.13)	(0.12)	(0.007)	(0.011)	(0.011)
Farmer (0/1)	1.00	1.00	0.001	-0.005*	-0.000
	(0.03)	(0.00)	(0.002)	(0.002)	(0.002)
Read and write English (0/1)	0.59	0.59	0.016	0.009	0.006
	(0.49)	(0.49)	(0.041)	(0.042)	(0.048)
Finished elementary school (0/1)	0.54	0.55	0.001	-0.017	0.035
	(0.50)	(0.50)	(0.040)	(0.042)	(0.045)
Finished junior high school (0/1)	0.39	0.38	-0.008	-0.013	0.046
	(0.49)	(0.49)	(0.042)	(0.040)	(0.050)
	F-test p-values		0.400	0.445	0.205

Notes: This table shows a test of balance for the baseline characteristics of schools (Panel A), students (Panel B), and parents (Panel C) in the randomized experiment. Column 1 provides covariate means and standard deviations in the entire sample. Column 2 provides covariate means and standard deviations in control schools. Column 3 shows the marginal effects of SBAE schools over control schools. Column 4 shows the marginal effects of program promotional sessions over SBAE only schools. Column 5 shows the marginal effects of the annual farmer field day over SBAE only schools. P-values from Wald tests of joint significance of treatment effects over all covariates are presented at the bottom of each panel.

Table A3: Balance in attrition rates across treatment arms

	Panel A: Balance in attrition		
	Midline Students Attrited (0/1)	Endline Students Attrited (0/1)	Endline Parents Attrited (0/1)
$SBAE_p$	0.020	0.028	0.042**
Standard errors	(0.035)	(0.034)	(0.020)
$SBAE_p \times Session_s$	-0.073**	-0.092***	0.017
Standard errors	(0.034)	(0.030)	(0.020)
$SBAE_p \times FFD_s$	0.001	0.046	-0.040**
Standard errors	(0.036)	(0.037)	(0.020)
Control Mean	0.36	0.48	0.09
Overall Mean	0.35	0.48	0.12
Observations	3757	3757	1689
Number of clusters	155	155	155
	Panel B: Reasons for attrition		
	Dropped out or switched schools	Dropped out or switched schools	Travelled
	34.9%	45.5%	4.6%
	— Confirmed dropped out	— Confirmed dropped out	Relocated
	18.3%	23.9%	4.3%
	— Confirmed switched schools	— Confirmed switched schools	Working
	16.3%	20.9%	1.1%
	Travelled	Travelled	Sick
	0.2%	0.6%	0.9%
	Moved to foreign countries	Moved to foreign countries	Moved to foreign countries
	0.1%	0.5%	0.2%
	Passed away	Passed away	Passed away
	0.1%	0.2%	0.9%
	Other reasons	Other reasons	Other reasons
	0.4%	0.9%	0.4%
	Panel C: Balance in major attrition reasons		
	Dropped out or switched schools (0/1)	Dropped out or switched schools (0/1)	Travelled or relocated (0/1)
$SBAE_p$	-0.000	0.024	0.038*
Standard errors	(0.033)	(0.033)	(0.021)
$SBAE_p \times Session_s$	-0.061*	-0.099***	0.014
Standard errors	(0.034)	(0.031)	(0.023)
$SBAE_p \times FFD_s$	0.014	0.046	-0.047**
Standard errors	(0.035)	(0.037)	(0.020)

Notes: This table shows a test of balance for the attrition rates among students and parents in the midline and endline surveys. Panel A regresses a binary indicator of attrition on a binary indicator of the SBAE treatment, an interaction term between the SBAE treatment and the program promotional session treatment, and an interaction term between the SBAE treatment and the annual farmer field day treatment. Panel B shows the distributions of reasons for attrition in each survey. Panel C repeats the balance test in Panel A, but using only attrition due to major reasons. All regressions include pair-of-clusters-of-schools fixed effects and a binary indicator of faith-based schools. Standard errors are clustered at the cluster-of-schools level.

Table A4: Selection of students and parents into the endline sample

Baseline variable	Change in attrition probability in percentages (%)	E(Attritors) - E(Non-attritors) in standard deviations	F-test p-value: Differential selection by treatment arms
	Panel A: Student characteristics		
Age	2.623***	0.214***	0.572
	(0.394)	(0.032)	
Grade	1.962***	0.119***	0.786
	(0.605)	(0.037)	
Male (0/1)	-1.403	-0.030	0.598
	(1.737)	(0.037)	
Years of farming	0.371	0.054	0.684
	(0.261)	(0.038)	
	Panel B: Parent characteristics		
Age	-0.231*	-0.174*	0.725
	(0.117)	(0.091)	
Male (0/1)	2.798	0.127	0.066
	(1.912)	(0.086)	
Parent or not (0/1)	-4.268**	-0.183**	0.791
	(2.000)	(0.085)	
Household head (0/1)	1.702	0.075	0.117
	(1.696)	(0.075)	
Invited to surveys (0/1)	-8.560	-0.113	0.108
	(5.870)	(0.077)	
Read and write English (0/1)	0.719	0.030	0.954
	(1.776)	(0.075)	
Finished elementary school (0/1)	1.468	0.062	0.967
	(1.918)	(0.080)	
Finished junior high school (0/1)	-1.439	-0.063	0.785
	(2.112)	(0.092)	

Notes: This table summarizes how the endline samples of students (Panel A) and parents (Panel B) differ from the original samples. Column 1 regresses a binary indicator on each subject characteristic, controlling for treatment status. Column 2 regresses each subject characteristic on a binary indicator of attrition, controlling for treatment status. The marginal effects of attrition are then divided by the control group standard deviations. Column 3 regresses a binary indicator of attrition on all treatments, fully interacted with each subject characteristic. P-values from Wald test of joint significance of all treatment interactions are reported. All regressions include pair-of-clusters-of-schools fixed effects and a binary indicator of faith-based schools. Standard errors are clustered at the cluster-of-schools level.

Table A5: Midline and endline intent-to-treat effects on joint adoption of bed preparation with other techniques

	Dependent variables							
	Panel A: Student parcel							
	Midline		Midline		Endline		Endline	
	Other promoted techniques (0/1)		Deviation from traditional (0/1)		Other promoted techniques (0/1)		Deviation from traditional (0/1)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$SBAE_p$	0.110***	-0.000	0.059*	-0.017	0.129***	0.088**	0.102***	0.002
Standard errors	(0.031)	(0.053)	(0.034)	(0.052)	(0.024)	(0.037)	(0.031)	(0.048)
$SBAE_p \times Session_s$		0.132***		0.101**		0.043		0.121**
Standard errors		(0.046)		(0.041)		(0.033)		(0.049)
$SBAE_p \times FFD_s$		0.090*		0.055		0.027		0.056
Standard errors		(0.047)		(0.050)		(0.036)		(0.049)
p-value: $(\beta_2 + \beta_3 = 0)$		0.01		0.04		0.17		0.01
Linear Combinations of Estimates								
SBAE only (β_1)	0.110***	-0.000	0.059*	-0.017	0.129***	0.088**	0.102***	0.002
Standard errors	(0.031)	(0.053)	(0.034)	(0.052)	(0.024)	(0.037)	(0.031)	(0.048)
Lee bounds	[0.109***	[0.008	[0.048	[-0.019	[0.129***	[0.083**	[0.102***	[-0.004
	(0.031),	(0.057),	(0.033),	(0.053),	(0.024),	(0.038),	(0.031),	(0.049),
	0.129***	0.033	0.067**	0.007	0.130***	0.114***	0.103***	0.023
	(0.030)]	(0.555)]	(0.033)]	(0.889)]	(0.023)]	(0.035)]	(0.031)]	(0.048)]
SBAE + Session ($\beta_1 + \beta_2$)		0.132***		0.084**		0.131***		0.124**
Standard errors		(0.037)		(0.042)		(0.036)		(0.049)
Lee bounds		[0.117***		[0.069		[0.126***		[0.118**
		(0.038),		(0.043),		(0.036),		(0.048),
		0.145***		0.096**		0.134***		0.127**
		(0.038)]		(0.042)]		(0.036)]		(0.049)]
SBAE + FFD ($\beta_1 + \beta_3$)		0.089***		0.038		0.114***		0.058
Standard errors		(0.034)		(0.043)		(0.027)		(0.037)
Lee bounds		[0.095**		[0.021		[0.108***		[0.049
		(0.036),		(0.042),		(0.027),		(0.038),
		0.124***		0.051		0.146***		0.085**
		(0.035)]		(0.042)]		(0.025)]		(0.036)]
SBAE + Session + FFD ($\beta_1 + \beta_2 + \beta_3$)		0.222***		0.139***		0.157***		0.180***
Standard errors		(0.047)		(0.047)		(0.028)		(0.038)
Lee bounds		[0.202***		[0.106**		[0.131***		[0.153***
		(0.054),		(0.047),		(0.028),		(0.038),
		[0.243***		[0.141***		[0.170***		[0.196***
		(0.053)]		(0.046)]		(0.028)]		(0.039)]
Control Mean	0.32	0.32	0.43	0.43	0.17	0.17	0.25	0.25
Control SD	0.47	0.47	0.50	0.50	0.38	0.38	0.44	0.44
Observations	1384	1384	1384	1384	1488	1488	1488	1488
Number of clusters	155	155	155	155	155	155	155	155

Notes: This table reports intent-to-treat effects on the joint adoption of bed preparation with other techniques on the student parcel (Panel A) and the parent parcel (Panel B) in the midline and endline surveys. Columns (1), (2), (5), (6), (9), and (10) refer to a binary indicator of whether bed preparation is used along with other promoted techniques, including composting, regular spacing, or row planting. Columns (3), (4), (7), (8), (11), and (12) refer to a binary indicator of whether bed preparation is used along with at least one deviation from traditional cultivation practices, including fallowing, rice cultivation, and mixedcropping of roots or vegetables with rice. All regressions include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level.

Table A5: Midline and endline intent-to-treat effects on joint adoption of bed preparation with other techniques *(continued)*

	Panel B: Parent parcel			
	Endline Other promoted techniques (0/1)		Endline Deviation from traditional (0/1)	
	(9)	(10)	(11)	(12)
$SBAE_p$	0.109***	0.076*	0.025	-0.023
Standard errors	(0.026)	(0.045)	(0.030)	(0.045)
$SBAE_p \times Session_s$		0.072*		0.098*
Standard errors		(0.042)		(0.051)
$SBAE_p \times FFD_s$		-0.012		-0.011
Standard errors		(0.042)		(0.037)
p-value: $(\beta_2 + \beta_3 = 0)$		0.40		0.23
Linear Combinations of Estimates				
SBAE only (β_1)	0.109***	0.076*	0.025	-0.023
Standard errors	(0.026)	(0.045)	(0.030)	(0.045)
Lee bounds	[0.104***	[0.075	[0.021	[-0.025
	(0.026),	(0.045),	(0.030),	(0.045),
	0.114***	0.077*	0.030	-0.021
	(0.026)]	(0.045)]	(0.030)]	(0.045)]
SBAE + Session ($\beta_1 + \beta_2$)		0.148***		0.075*
Standard errors		(0.034)		(0.039)
Lee bounds		[0.142***		[0.070*
		(0.034),		(0.039),
		0.154***		0.082**
		(0.034)]		(0.039)]
SBAE + FFD ($\beta_1 + \beta_3$)		0.064**		-0.035
Standard errors		(0.032)		(0.040)
Lee bounds		[0.058*		[-0.043
		(0.032),		(0.040),
		0.072**		-0.026
		(0.032)]		(0.040)]
SBAE + Session + FFD ($\beta_1 + \beta_2 + \beta_3$)		0.136***		0.063
Standard errors		(0.042)		(0.048)
Lee bounds		[0.124***		[0.053
		(0.042),		(0.048),
		[0.147***		[0.076
		(0.043)]		(0.049)]
Control Mean	0.34	0.34	0.52	0.52
Control SD	0.48	0.48	0.50	0.50
Observations	1363	1363	1363	1363
Number of clusters	155	155	155	155

Notes: This table reports intent-to-treat effects on the joint adoption of bed preparation with other techniques on the student parcel (Panel A) and the parent parcel (Panel B) in the midline and endline surveys. Columns (1), (2), (5), (6), (9), and (10) refer to a binary indicator of whether bed preparation is used along with other promoted techniques, including composting, regular spacing, or row planting. Columns (3), (4), (7), (8), (11), and (12) refer to a binary indicator of whether bed preparation is used along with at least one deviation from traditional cultivation practices, including fallowing, rice cultivation, and mixedcropping of roots or vegetables with rice. All regressions include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level.

Table A6: Midline and endline effects on students' spending and meals missed

	Panel A: Midline (2021)							
	Y=Total spending (Liberian dollars)						Y=Meals missed (0-14)	
	OLS extensive margin: Pr(Y>0)		OLS intensive margin: Log(Y)|Y>0		Poisson regression: Proportion change in Y		OLS	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$SBAE_p$	-0.004	-0.029	0.005	0.166*	0.050	0.118	0.005	0.074
	(0.012)	(0.021)	(0.054)	(0.097)	(0.042)	(0.105)	(0.099)	(0.178)
$SBAE_p \times FFD_s$		0.048*		-0.071		-0.018		0.004
		(0.028)		(0.101)		(0.103)		(0.225)
$SBAE_p \times Session_s$		0.010		-0.207**		-0.065		-0.100
		(0.021)		(0.102)		(0.101)		(0.186)
p-value: $(\beta_2 + \beta_3 = 0)$		0.12		0.09		0.60		0.76
Summary: Combined effects of SBAE and informational treatments								
SBAE only (β_1)	-0.004	-0.029	0.005	0.166*	0.050	0.118	0.005	0.074
	(0.012)	(0.021)	(0.054)	(0.097)	(0.042)	(0.105)	(0.099)	(0.178)
Lee bounds	[-0.007	[-0.053**	[-0.034	[0.013	[-0.034	[0.088	[-0.178*	[0.001
	(0.012),	(0.021),	(0.054),	(0.095),	(0.038),	(0.102),	(0.101),	(0.178),
	0.015	-0.023	0.121**	0.200**	0.073*	0.186*	0.060	0.249
	(0.012)]	(0.022)]	(0.048)]	(0.098)]	(0.043)]	(0.115)]	(0.099)]	(0.167)]
SBAE + FFD ($\beta_1 + \beta_2$)		0.019		0.095		0.098		0.078
		(0.021)		(0.077)		(0.092)		(0.172)
Lee bounds		[0.014		[0.023		[-0.044		[-0.103
		(0.020),		(0.074),		(0.069),		(0.156),
		0.045**		0.193**		0.133		0.160
		(0.018)]		(0.076)]		(0.094)]		(0.174)]
SBAE + Session ($\beta_1 + \beta_3$)		-0.019		-0.041		0.045		-0.026
		(0.019)		(0.083)		(0.079)		(0.164)
Lee bounds		[-0.024		[-0.090		[-0.057		[-0.242
		(0.019),		(0.082),		(0.068),		(0.154),
		0.001		0.077		0.072		0.037
		(0.018)]		(0.075)]		(0.080)]		(0.167)]
SBAE + FFD + Session ($\beta_1 + \beta_2 + \beta_3$)		0.030		-0.112		0.026		-0.022
		(0.023)		(0.093)		(0.089)		(0.186)
Lee bounds		[0.022		[-0.193**		[-0.105		[-0.303
		(0.021),		(0.094),		(0.077),		(0.194),
		0.061***		0.068		0.071		0.075
		(0.019)]		(0.094)]		(0.088)]		(0.185)]
Summary Statistics for Control Schools								
Control Mean (without logs)	0.826	0.826	3490	3490	2903	2903	2.62	2.62
Control SD (without logs)	0.379	0.379	3182	3182	3182	3182	2.93	2.93
Observations	2425	2425	2011	2011	2425	2425	2423	2423
Number of clusters	155	155	155	155	155	155	155	155

Notes: This table reports intent-to-treat effects on students' spending in the last six months and consumption of meals, reported during the midline and endline surveys. Columns (1) and (2) refer to students' log-transformed total spending, which is a sum of spending on agricultural inputs, school fees and supplies, and consumption goods. Columns (3) and (4) refer to the number of meals (specifically, breakfasts and lunches) students miss in the last seven days. All monetary values are expressed in 2022 Liberian dollars and winsorized at the 2.5 and 97.5 percentiles. All regressions include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level.

Table A6: Midline and endline effects on students' spending and meals missed *(continued)*

	Panel B: Endline (2022)							
	Y=Total spending (Liberian dollars)						Y=Meals missed (0-14)	
	OLS extensive margin: Pr(Y>0)		OLS intensive margin: Log(Y)|Y>0		Poisson regression: Proportion change in Y		OLS	
	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
$SBAE_p$	-0.022	-0.001	0.083	0.210	0.033	0.113	-0.032	0.185
	(0.016)	(0.031)	(0.050)	(0.141)	(0.041)	(0.101)	(0.145)	(0.192)
$SBAE_p \times FFD_s$		0.009		-0.217		-0.067		-0.235
		(0.037)		(0.146)		(0.079)		(0.238)
$SBAE_p \times Session_s$		-0.053*		-0.000		-0.065		-0.210
		(0.029)		(0.156)		(0.094)		(0.252)
p-value: $(\beta_2 + \beta_3 = 0)$		0.43		0.39		0.35		0.17
Summary: Combined effects of SBAE and informational treatments								
SBAE only (β_1)	-0.022	-0.001	0.083	0.210	0.033	0.113	-0.032	0.185
	(0.016)	(0.031)	(0.050)	(0.141)	(0.041)	(0.101)	(0.145)	(0.192)
Lee bounds	[-0.018	[-0.028	[-0.072	[-0.057	[0.031	[0.083	[-0.037	[0.135
	(0.011),	(0.024),	(0.103),	(0.243),	(0.041),	(0.098),	(0.145),	(0.191),
	0.053***	0.000	0.029	0.221	0.111*	0.189*	-0.052**	0.297
	(0.020)]	(0.024)]	(0.048)]	(0.247)]	(0.061)]	(0.108)]	(0.025)]	(0.191)]
SBAE + FFD ($\beta_1 + \beta_2$)		0.008		-0.006		0.039		-0.049
		(0.020)		(0.103)		(0.077)		(0.207)
Lee bounds		[-0.035**		[-0.286		[-0.002		[-0.135
		(0.014),		(0.187),		(0.074),		(0.209),
		0.039**		0.310		0.075		0.019
		(0.016)]		(0.202)]		(0.080)]		(0.212)]
SBAE + Session ($\beta_1 + \beta_3$)		-0.055**		0.210**		0.041		-0.024
		(0.028)		(0.092)		(0.070)		(0.258)
Lee bounds		[-0.038*		[-0.147		[0.022		[-0.071
		(0.020),		(0.206),		(0.069),		(0.244),
		-0.028		-0.034		301.826		-0.002
		(0.020)]		(0.195)]		(396.235)]		(0.257)]
SBAE + FFD + Session ($\beta_1 + \beta_2 + \beta_3$)		-0.046		-0.006		-0.029		-0.259
		(0.034)		(0.128)		(0.074)		(0.234)
Lee bounds		[-0.033		[-0.322		[-0.115*		[-0.531**
		(0.025),		(0.242),		(0.065),		(0.218),
		0.011		0.073		0.010		-0.181
		(0.015)]		(0.183)]		(0.074)]		(0.235)]
Summary Statistics for Control Schools								
Control Mean (without logs)	0.918	0.918	5847	5847	5520	5520	2.18	2.18
Control SD (without logs)	0.274	0.274	4686	4686	4747	4747	2.56	2.56
Observations	1488	1488	1390	1390	1488	1488	1487	1487
Number of clusters	155	155	154	154	155	155	155	155

Notes: This table reports intent-to-treat effects on students' spending in the last six months and consumption of meals, reported during the midline and endline surveys. Columns (1) and (2) refer to students' log-transformed total spending, which is a sum of spending on agricultural inputs, school fees and supplies, and consumption goods. Columns (3) and (4) refer to the number of meals (specifically, breakfasts and lunches) students miss in the last seven days. All monetary values are expressed in 2022 Liberian dollars and winsorized at the 2.5 and 97.5 percentiles. All regressions include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level.

Table A7: Endline intent-to-treat effects on downstream outcomes associated with parents

	Parent Reports				**Student Reports**	
	Score on: techniques (0-12)		Beliefs: student more cassava tubers than typical farmer (0/1)		Student missed school because asked to work (0/1))	
	(1)	(2)	(3)	(4)	(5)	(6)
$SBAE_p$	0.253***	0.122	0.073**	0.044	0.007	0.033
	(0.085)	(0.127)	(0.029)	(0.044)	(0.017)	(0.027)
$SBAE_p \times FFD_s$		-0.000		0.039		-0.075**
		(0.121)		(0.048)		(0.032)
$SBAE_p \times Session_s$		0.229		0.022		0.032
		(0.152)		(0.042)		(0.027)
p-value: $(\beta_2 + \beta_3 = 0)$		0.25		0.31		0.35
	Summary: Combined effects of SBAE and informational treatments					
SBAE only (β_1)	0.253***	0.122	0.073**	0.044	0.007	0.033
	(0.085)	(0.127)	(0.029)	(0.044)	(0.017)	(0.027)
Lee bounds	[0.220**	[0.111	[0.067**	[0.043	[0.007	[0.028
	(0.085),	(0.128),	(0.029),	(0.044),	(0.017),	(0.028),
	0.303***	0.113	0.077**	0.047	0.008	0.050*
	(0.086)]	(0.127)]	(0.029)]	(0.044)]	(0.017)]	(0.028)]
SBAE+FFD ($\beta_1 + \beta_2$)		0.122		0.083*		-0.042
		(0.131)		(0.045)		(0.027)
Lee bounds		[0.084		[0.078*		[-0.050*
		(0.133),		(0.045),		(0.027),
		0.170		0.093**		-0.015
		(0.132)]		(0.045)]		(0.026)]
SBAE+Session ($\beta_1 + \beta_3$)		0.352***		0.066		0.065***
		(0.124)		(0.043)		(0.021)
Lee bounds		[0.308**		[0.059		[0.059**
		(0.127),		(0.043),		(0.021),
		0.403***		0.072		0.067***
		(0.124)]		(0.043)]		(0.021)]
SBAE+Session+FFD ($\beta_1 + \beta_2 + \beta_3$)		0.352**		0.105***		-0.010
		(0.138)		(0.038)		(0.027)
Lee bounds		[0.277*		[0.095**		[-0.048*
		(0.139),		(0.039),		(0.025),
		[0.434***		[0.116***		[-0.001
		(0.130)]		(0.038)]		(0.028)]
Control Mean	5.23	5.23	0.35	0.35	0.24	0.24
Control SD	1.66	1.66	0.48	0.48	0.43	0.43
Observations	1543	1543	1543	1543	1894	1894
Number of clusters	155	155	155	155	155	155

Notes: This table reports endline intent-to-treat effects on downstream effects associated with parents. Columns (1)-(4) refer to parents' knowledge and beliefs — (1)-(2) refer to parents' scores on a test on promoted agricultural techniques; (3)-(4) refer to a binary indicator of whether parents believe that students could harvest more tubers from 4 cassava plants than a typical adult in the community. (5)-(6) refer to a binary indicator of whether students report having missed school in 2022/23 because they were asked to work for households. All regressions include pair-of-clusters-of-schools fixed effects and if a binary indicator of charity schools when appropriate. Standard errors are clustered at the cluster-of-schools level.